

2

THE DERIVATIVE

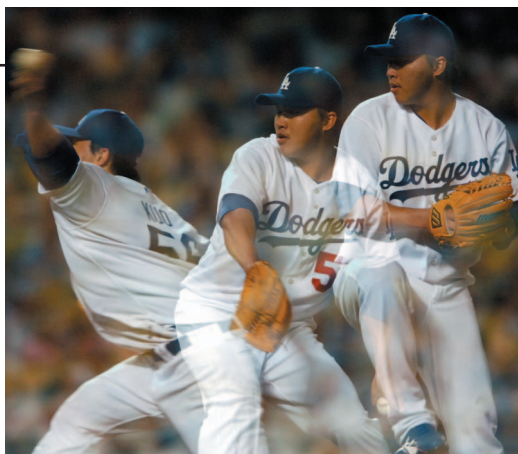


Photo by Kirby Lee/WireImage/Getty Images

One of the crowning achievements of calculus is its ability to capture continuous motion mathematically, allowing that motion to be analyzed instant by instant.

Many real-world phenomena involve changing quantities—the speed of a rocket, the inflation of currency, the number of bacteria in a culture, the shock intensity of an earthquake, the voltage of an electrical signal, and so forth. In this chapter we will develop the concept of a “derivative,” which is the mathematical tool for studying the rate at which one quantity changes relative to another. The study of rates of change is closely related to the geometric concept of a tangent line to a curve, so we will also be discussing the general definition of a tangent line and methods for finding its slope and equation. Later in the chapter, we will consider some applications of the derivative. These will include ways in which different rates of change can be related as well as the use of linear functions to approximate nonlinear functions.

2.1 TANGENT LINES AND RATES OF CHANGE

In this section we will discuss three ideas: tangent lines to curves, the velocity of an object moving along a line, and the rate at which one variable changes relative to another. Our goal is to show how these seemingly unrelated ideas are, in actuality, closely linked.

TANGENT LINES

In Example 1 of Section 1.1, we showed how the notion of a limit could be used to find an equation of a tangent line to a curve. At that stage in the text we did not have precise definitions of tangent lines and limits to work with, so the argument was intuitive and informal. However, now that limits have been defined precisely, we are in a position to give a mathematical definition of the tangent line to a curve $y = f(x)$ at a point $P(x_0, f(x_0))$ on the curve. As illustrated in Figure 2.1.1, consider a point $Q(x, f(x))$ on the curve that is distinct from P , and compute the slope m_{PQ} of the secant line through P and Q :

$$m_{PQ} = \frac{f(x) - f(x_0)}{x - x_0}$$

If we let x approach x_0 , then the point Q will move along the curve and approach the point P . If the secant line through P and Q approaches a limiting position as $x \rightarrow x_0$, then we will regard that position to be the position of the tangent line at P . Stated another way, if the slope m_{PQ} of the secant line through P and Q approaches a limit as $x \rightarrow x_0$, then we regard that limit to be the slope m_{tan} of the tangent line at P . Thus, we make the following definition.

✓ QUICK CHECK ANSWERS 2.4

1. (a) $x^2 f'(x) + 2x f(x)$ (b) $\frac{(x^2 + 1)f'(x) - 2x f(x)}{(x^2 + 1)^2}$ (c) $\frac{2x f(x) - (x^2 + 1)f'(x)}{[f(x)^2]}$ 2. (a) 7 (b) -4 (c) 7 (d) $\frac{5}{9}$

2.5 DERIVATIVES OF TRIGONOMETRIC FUNCTIONS

The main objective of this section is to obtain formulas for the derivatives of the six basic trigonometric functions. If needed, you will find a review of trigonometric functions in Appendix B.

We will assume in this section that the variable x in the trigonometric functions $\sin x$, $\cos x$, $\tan x$, $\cot x$, $\sec x$, and $\csc x$ is measured in radians. Also, we will need the limits in Theorem 1.6.3, but restated as follows using h rather than x as the variable:

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1 \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{1 - \cos h}{h} = 0 \quad (1-2)$$

Let us start with the problem of differentiating $f(x) = \sin x$. Using the definition of the derivative we obtain

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} && \text{By the addition formula for sine} \\ &= \lim_{h \rightarrow 0} \left[\sin x \left(\frac{\cos h - 1}{h} \right) + \cos x \left(\frac{\sin h}{h} \right) \right] \\ &= \lim_{h \rightarrow 0} \left[\cos x \left(\frac{\sin h}{h} \right) - \sin x \left(\frac{1 - \cos h}{h} \right) \right] && \text{Algebraic reorganization} \\ &= \lim_{h \rightarrow 0} \cos x \cdot \lim_{h \rightarrow 0} \frac{\sin h}{h} - \lim_{h \rightarrow 0} \sin x \cdot \lim_{h \rightarrow 0} \frac{1 - \cos h}{h} \\ &= \left(\lim_{h \rightarrow 0} \cos x \right) (1) - \left(\lim_{h \rightarrow 0} \sin x \right) (0) && \text{Formulas (1) and (2)} \\ &= \lim_{h \rightarrow 0} \cos x = \cos x && \begin{array}{l} \cos x \text{ does not involve the variable } h \text{ and hence} \\ \text{is treated as a constant in the limit computation.} \end{array} \end{aligned}$$

Formulas (1) and (2) and the derivation of Formulas (3) and (4) are only valid if h and x are in radians. See Exercise 49 for how Formulas (3) and (4) change when x is measured in degrees.

Thus, we have shown that

$$\frac{d}{dx}[\sin x] = \cos x \quad (3)$$

In the exercises we will ask you to use the same method to derive the following formula for the derivative of $\cos x$:

$$\frac{d}{dx}[\cos x] = -\sin x \quad (4)$$

► **Example 1** Find dy/dx if $y = x \sin x$.

Solution. Using Formula (3) and the product rule we obtain

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}[x \sin x] \\ &= x \frac{d}{dx}[\sin x] + \sin x \frac{d}{dx}[x] \\ &= x \cos x + \sin x \quad \blacktriangleleft\end{aligned}$$

► **Example 2** Find dy/dx if $y = \frac{\sin x}{1 + \cos x}$.

Solution. Using the quotient rule together with Formulas (3) and (4) we obtain

$$\begin{aligned}\frac{dy}{dx} &= \frac{(1 + \cos x) \cdot \frac{d}{dx}[\sin x] - \sin x \cdot \frac{d}{dx}[1 + \cos x]}{(1 + \cos x)^2} \\ &= \frac{(1 + \cos x)(\cos x) - (\sin x)(-\sin x)}{(1 + \cos x)^2} \\ &= \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2} = \frac{\cos x + 1}{(1 + \cos x)^2} = \frac{1}{1 + \cos x} \quad \blacktriangleleft\end{aligned}$$

Since Formulas (3) and (4) are only valid if x is in radians, the same is true for Formulas (5)–(8).

The derivatives of the remaining trigonometric functions are

$$\frac{d}{dx}[\tan x] = \sec^2 x \qquad \frac{d}{dx}[\sec x] = \sec x \tan x \qquad (5-6)$$

$$\frac{d}{dx}[\cot x] = -\csc^2 x \qquad \frac{d}{dx}[\csc x] = -\csc x \cot x \qquad (7-8)$$

These can all be obtained using the definition of the derivative, but it is easier to use Formulas (3) and (4) and apply the quotient rule to the relationships

$$\tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}, \quad \sec x = \frac{1}{\cos x}, \quad \csc x = \frac{1}{\sin x}$$

For example,

$$\begin{aligned}\frac{d}{dx}[\tan x] &= \frac{d}{dx}\left[\frac{\sin x}{\cos x}\right] = \frac{\cos x \cdot \frac{d}{dx}[\sin x] - \sin x \cdot \frac{d}{dx}[\cos x]}{\cos^2 x} \\ &= \frac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x\end{aligned}$$

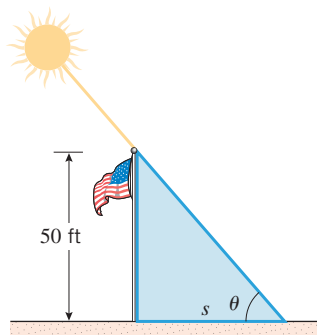
When finding the value of a derivative at a specific point $x = x_0$, it is important to substitute x_0 *after* the derivative is obtained. Thus, in Example 3 we made the substitution $x = \pi/4$ after f'' was calculated. What would have happened had we *incorrectly* substituted $x = \pi/4$ into $f'(x)$ before calculating f'' ?

► **Example 3** Find $f''(\pi/4)$ if $f(x) = \sec x$.

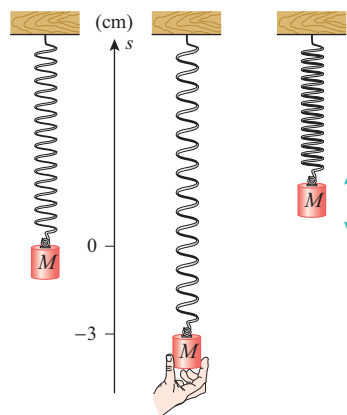
$$\begin{aligned}f'(x) &= \sec x \tan x \\ f''(x) &= \sec x \cdot \frac{d}{dx}[\tan x] + \tan x \cdot \frac{d}{dx}[\sec x] \\ &= \sec x \cdot \sec^2 x + \tan x \cdot \sec x \tan x \\ &= \sec^3 x + \sec x \tan^2 x\end{aligned}$$

Thus,

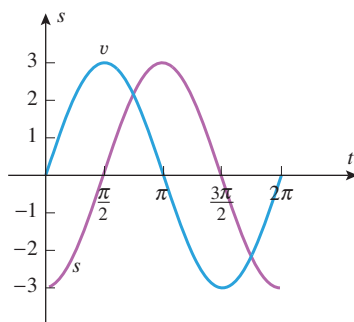
$$\begin{aligned} f''(\pi/4) &= \sec^3(\pi/4) + \sec(\pi/4) \tan^2(\pi/4) \\ &= (\sqrt{2})^3 + (\sqrt{2})(1)^2 = 3\sqrt{2} \quad \blacktriangleleft \end{aligned}$$



▲ Figure 2.5.1



▲ Figure 2.5.2



▲ Figure 2.5.3

In Example 5, the top of the mass has its maximum speed when it passes through its rest position. Why? What is that maximum speed?

► **Example 4** On a sunny day, a 50 ft flagpole casts a shadow that changes with the angle of elevation of the Sun. Let s be the length of the shadow and θ the angle of elevation of the Sun (Figure 2.5.1). Find the rate at which the length of the shadow is changing with respect to θ when $\theta = 45^\circ$. Express your answer in units of feet/degree.

Solution. The variables s and θ are related by $\tan \theta = 50/s$ or, equivalently,

$$s = 50 \cot \theta \quad (9)$$

If θ is measured in radians, then Formula (7) is applicable, which yields

$$\frac{ds}{d\theta} = -50 \csc^2 \theta$$

which is the rate of change of shadow length with respect to the elevation angle θ in units of feet/radian. When $\theta = 45^\circ$ (or equivalently $\theta = \pi/4$ radians), we obtain

$$\left. \frac{ds}{d\theta} \right|_{\theta=\pi/4} = -50 \csc^2(\pi/4) = -100 \text{ feet/radian}$$

Converting radians (rad) to degrees (deg) yields

$$-100 \frac{\text{ft}}{\text{rad}} \cdot \frac{\pi \text{ rad}}{180 \text{ deg}} = -\frac{5}{9} \pi \frac{\text{ft}}{\text{deg}} \approx -1.75 \text{ ft/deg}$$

Thus, when $\theta = 45^\circ$, the shadow length is decreasing (because of the minus sign) at an approximate rate of 1.75 ft/deg increase in the angle of elevation. ◀

► **Example 5** As illustrated in Figure 2.5.2, suppose that a spring with an attached mass is stretched 3 cm beyond its rest position and released at time $t = 0$. Assuming that the position function of the top of the attached mass is

$$s = -3 \cos t \quad (10)$$

where s is in centimeters and t is in seconds, find the velocity function and discuss the motion of the attached mass.

Solution. The velocity function is

$$v = \frac{ds}{dt} = \frac{d}{dt}[-3 \cos t] = 3 \sin t$$

Figure 2.5.3 shows the graphs of the position and velocity functions. The position function tells us that the top of the mass oscillates between a low point of $s = -3$ and a high point of $s = 3$ with one complete oscillation occurring every 2π seconds [the period of (10)]. The top of the mass is moving up (the positive s -direction) when v is positive, is moving down when v is negative, and is at a high or low point when $v = 0$. Thus, for example, the top of the mass moves up from time $t = 0$ to time $t = \pi$, at which time it reaches the high point $s = 3$ and then moves down until time $t = 2\pi$, at which time it reaches the low point of $s = -3$. The motion then repeats periodically. ◀

QUICK CHECK EXERCISES 2.5 (See page 153 for answers.)

- Find dy/dx .
 - $y = \sin x$
 - $y = \cos x$
 - $y = \tan x$
 - $y = \sec x$
- Find $f'(x)$ and $f'(\pi/3)$ if $f(x) = \sin x \cos x$.
- Use a derivative to evaluate each limit.
 - $\lim_{h \rightarrow 0} \frac{\sin(\frac{\pi}{2} + h) - 1}{h}$
 - $\lim_{h \rightarrow 0} \frac{\csc(x + h) - \csc x}{h}$

EXERCISE SET 2.5 Graphing Utility

1–18 Find $f'(x)$. ■

- $f(x) = 4 \cos x + 2 \sin x$
- $f(x) = \frac{5}{x^2} + \sin x$
- $f(x) = -4x^2 \cos x$
- $f(x) = 2 \sin^2 x$
- $f(x) = \frac{5 - \cos x}{5 + \sin x}$
- $f(x) = \frac{\sin x}{x^2 + \sin x}$
- $f(x) = \sec x - \sqrt{2} \tan x$
- $f(x) = (x^2 + 1) \sec x$
- $f(x) = 4 \csc x - \cot x$
- $f(x) = \cos x - x \csc x$
- $f(x) = \sec x \tan x$
- $f(x) = \csc x \cot x$
- $f(x) = \frac{\cot x}{1 + \csc x}$
- $f(x) = \frac{\sec x}{1 + \tan x}$
- $f(x) = \sin^2 x + \cos^2 x$
- $f(x) = \sec^2 x - \tan^2 x$
- $f(x) = \frac{\sin x \sec x}{1 + x \tan x}$
- $f(x) = \frac{(x^2 + 1) \cot x}{3 - \cos x \csc x}$

19–24 Find d^2y/dx^2 . ■

- $y = x \cos x$
- $y = \csc x$
- $y = x \sin x - 3 \cos x$
- $y = x^2 \cos x + 4 \sin x$
- $y = \sin x \cos x$
- $y = \tan x$
- Find the equation of the line tangent to the graph of $\tan x$ at
 - $x = 0$
 - $x = \pi/4$
 - $x = -\pi/4$
- Find the equation of the line tangent to the graph of $\sin x$ at
 - $x = 0$
 - $x = \pi$
 - $x = \pi/4$
- Show that $y = x \sin x$ is a solution to $y'' + y = 2 \cos x$.
 - Show that $y = x \sin x$ is a solution of the equation $y^{(4)} + y'' = -2 \cos x$.
- Show that $y = \cos x$ and $y = \sin x$ are solutions of the equation $y'' + y = 0$.
 - Show that $y = A \sin x + B \cos x$ is a solution of the equation $y'' + y = 0$ for all constants A and B .
- Find all values in the interval $[-2\pi, 2\pi]$ at which the graph of f has a horizontal tangent line.
 - $f(x) = \sin x$
 - $f(x) = x + \cos x$
 - $f(x) = \tan x$
 - $f(x) = \sec x$
- Use a graphing utility to make rough estimates of the values in the interval $[0, 2\pi]$ at which the graph of $y = \sin x \cos x$ has a horizontal tangent line.
 - Find the exact locations of the points where the graph has a horizontal tangent line.

- A 10 ft ladder leans against a wall at an angle θ with the horizontal, as shown in the accompanying figure. The top of the ladder is x feet above the ground. If the bottom of the ladder is pushed toward the wall, find the rate at which x changes with respect to θ when $\theta = 60^\circ$. Express the answer in units of feet/degree.

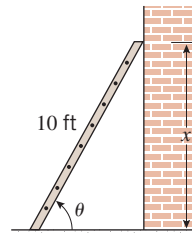


Figure Ex-31

- An airplane is flying on a horizontal path at a height of 3800 ft, as shown in the accompanying figure. At what rate is the distance s between the airplane and the fixed point P changing with respect to θ when $\theta = 30^\circ$? Express the answer in units of feet/degree.

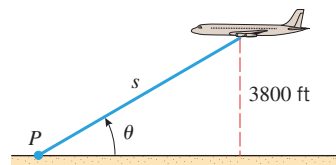


Figure Ex-32

- A searchlight is trained on the side of a tall building. As the light rotates, the spot it illuminates moves up and down the side of the building. That is, the distance D between ground level and the illuminated spot on the side of the building is a function of the angle θ formed by the light beam and the horizontal (see the accompanying figure). If the searchlight is located 50 m from the building, find the rate at which D is changing with respect to θ when $\theta = 45^\circ$. Express your answer in units of meters/degree.

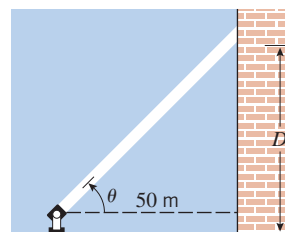


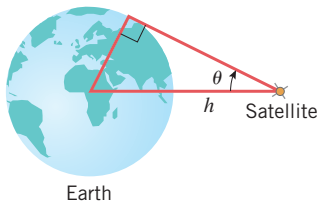
Figure Ex-33

34. An Earth-observing satellite can see only a portion of the Earth's surface. The satellite has horizon sensors that can detect the angle θ shown in the accompanying figure. Let r be the radius of the Earth (assumed spherical) and h the distance of the satellite from the Earth's surface.

(a) Show that $h = r(\csc \theta - 1)$.

(b) Using $r = 6378$ km, find the rate at which h is changing with respect to θ when $\theta = 30^\circ$. Express the answer in units of kilometers/degree.

Source: Adapted from *Space Mathematics*, NASA, 1985.



◀ Figure Ex-34

35–38 True–False Determine whether the statement is true or false. Explain your answer. ■

35. If $g(x) = f(x) \sin x$, then $g'(x) = f'(x) \cos x$.
 36. If $g(x) = f(x) \sin x$, then $g'(0) = f(0)$.
 37. If $f(x) \cos x = \sin x$, then $f'(x) = \sec^2 x$.
 38. Suppose that $g(x) = f(x) \sec x$, where $f(0) = 8$ and $f'(0) = -2$. Then

$$\begin{aligned} g'(0) &= \lim_{h \rightarrow 0} \frac{f(h) \sec h - f(0)}{h} = \lim_{h \rightarrow 0} \frac{8(\sec h - 1)}{h} \\ &= 8 \cdot \frac{d}{dx} [\sec x] \Big|_{x=0} = 8 \sec 0 \tan 0 = 0 \end{aligned}$$

39–40 Make a conjecture about the derivative by calculating the first few derivatives and observing the resulting pattern. ■

39. $\frac{d^{87}}{dx^{87}} [\sin x]$

40. $\frac{d^{100}}{dx^{100}} [\cos x]$

41. Let $f(x) = \cos x$. Find all positive integers n for which $f^{(n)}(x) = \sin x$.
 42. Let $f(x) = \sin x$. Find all positive integers n for which $f^{(n)}(x) = \sin x$.

FOCUS ON CONCEPTS

43. In each part, determine where f is differentiable.

(a) $f(x) = \sin x$

(b) $f(x) = \cos x$

(c) $f(x) = \tan x$

(d) $f(x) = \cot x$

(e) $f(x) = \sec x$

(f) $f(x) = \csc x$

(g) $f(x) = \frac{1}{1 + \cos x}$

(h) $f(x) = \frac{1}{\sin x \cos x}$

(i) $f(x) = \frac{\cos x}{2 - \sin x}$

44. (a) Derive Formula (4) using the definition of a derivative.
 (b) Use Formulas (3) and (4) to obtain (7).
 (c) Use Formula (4) to obtain (6).
 (d) Use Formula (3) to obtain (8).

45. Use Formula (1), the alternative form for the definition of derivative given in Formula (13) of Section 2.2, that is,

$$f'(x) = \lim_{w \rightarrow x} \frac{f(w) - f(x)}{w - x}$$

and the difference identity

$$\sin \alpha - \sin \beta = 2 \sin \left(\frac{\alpha - \beta}{2} \right) \cos \left(\frac{\alpha + \beta}{2} \right)$$

to show that $\frac{d}{dx} [\sin x] = \cos x$.

46. Follow the directions of Exercise 45 using the difference identity

$$\cos \alpha - \cos \beta = -2 \sin \left(\frac{\alpha - \beta}{2} \right) \sin \left(\frac{\alpha + \beta}{2} \right)$$

to show that $\frac{d}{dx} [\cos x] = -\sin x$.

47. (a) Show that $\lim_{h \rightarrow 0} \frac{\tan h}{h} = 1$.
 (b) Use the result in part (a) to help derive the formula for the derivative of $\tan x$ directly from the definition of a derivative.

48. Without using any trigonometric identities, find

$$\lim_{x \rightarrow 0} \frac{\tan(x + y) - \tan y}{x}$$

[Hint: Relate the given limit to the definition of the derivative of an appropriate function of y .]

49. The derivative formulas for $\sin x$, $\cos x$, $\tan x$, $\cot x$, $\sec x$, and $\csc x$ were obtained under the assumption that x is measured in radians. If x is measured in degrees, then

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{\pi}{180}$$

(See Exercise 39 of Section 1.6). Use this result to prove that if x is measured in degrees, then

(a) $\frac{d}{dx} [\sin x] = \frac{\pi}{180} \cos x$

(b) $\frac{d}{dx} [\cos x] = -\frac{\pi}{180} \sin x$

50. **Writing** Suppose that f is a function that is differentiable everywhere. Explain the relationship, if any, between the periodicity of f and that of f' . That is, if f is periodic, must f' also be periodic? If f' is periodic, must f also be periodic?

✓ QUICK CHECK ANSWERS 2.5

1. (a) $\cos x$ (b) $-\sin x$ (c) $\sec^2 x$ (d) $\sec x \tan x$ 2. $f'(x) = \cos^2 x - \sin^2 x$, $f'(\pi/3) = -\frac{1}{2}$
 3. (a) $\frac{d}{dx}[\sin x] \Big|_{x=\pi/2} = 0$ (b) $\frac{d}{dx}[\csc x] = -\csc x \cot x$

2.6 THE CHAIN RULE

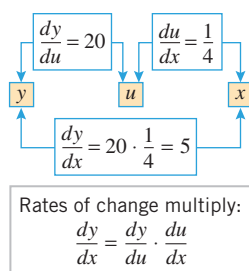
In this section we will derive a formula that expresses the derivative of a composition $f \circ g$ in terms of the derivatives of f and g . This formula will enable us to differentiate complicated functions using known derivatives of simpler functions.

DERIVATIVES OF COMPOSITIONS



Mike Brinson/Getty Images

The cost of a car trip is a combination of fuel efficiency and the cost of gasoline.



▲ Figure 2.6.1

Suppose you are traveling to school in your car, which gets 20 miles per gallon of gasoline. The number of miles you can travel in your car without refueling is a function of the number of gallons of gas you have in the gas tank. In symbols, if y is the number of miles you can travel and u is the number of gallons of gas you have initially, then y is a function of u , or $y = f(u)$. As you continue your travels, you note that your local service station is selling gasoline for \$4 per gallon. The number of gallons of gas you have initially is a function of the amount of money you spend for that gas. If x is the number of dollars you spend on gas, then $u = g(x)$. Now 20 miles per gallon is the rate at which your mileage changes with respect to the amount of gasoline you use, so

$$f'(u) = \frac{dy}{du} = 20 \text{ miles per gallon}$$

Similarly, since gasoline costs \$4 per gallon, each dollar you spend will give you $1/4$ of a gallon of gas, and

$$g'(x) = \frac{du}{dx} = \frac{1}{4} \text{ gallons per dollar}$$

Notice that the number of miles you can travel is also a function of the number of dollars you spend on gasoline. This fact is expressible as the composition of functions

$$y = f(u) = f(g(x))$$

You might be interested in how many miles you can travel per dollar, which is dy/dx . Intuition suggests that rates of change multiply in this case (see Figure 2.6.1), so

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{20 \text{ miles}}{1 \text{ gallon}} \cdot \frac{1 \text{ gallon}}{4 \text{ dollars}} = \frac{20 \text{ miles}}{4 \text{ dollars}} = 5 \text{ miles per dollar}$$

The following theorem, the proof of which is given in Appendix D, formalizes the preceding ideas.

2.6.1 THEOREM (The Chain Rule) If g is differentiable at x and f is differentiable at $g(x)$, then the composition $f \circ g$ is differentiable at x . Moreover, if

$$y = f(g(x)) \quad \text{and} \quad u = g(x)$$

then $y = f(u)$ and

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad (1)$$

The name “chain rule” is appropriate because the desired derivative is obtained by a two-link “chain” of simpler derivatives.

Formula (1) is easy to remember because the left side is exactly what results if we “cancel” the du ’s on the right side. This “canceling” device provides a good way of deducing the correct form of the chain rule when different variables are used. For example, if w is a function of x and x is a function of t , then the chain rule takes the form

$$\frac{dw}{dt} = \frac{dw}{dx} \cdot \frac{dx}{dt}$$

► **Example 1** Find dy/dx if $y = \cos(x^3)$.

Solution. Let $u = x^3$ and express y as $y = \cos u$. Applying Formula (1) yields

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= \frac{d}{du}[\cos u] \cdot \frac{d}{dx}[x^3] \\ &= (-\sin u) \cdot (3x^2) \\ &= (-\sin(x^3)) \cdot (3x^2) = -3x^2 \sin(x^3) \quad \blacktriangleleft\end{aligned}$$

► **Example 2** Find dw/dt if $w = \tan x$ and $x = 4t^3 + t$.

Solution. In this case the chain rule computations take the form

$$\begin{aligned}\frac{dw}{dt} &= \frac{dw}{dx} \cdot \frac{dx}{dt} \\ &= \frac{d}{dx}[\tan x] \cdot \frac{d}{dt}[4t^3 + t] \\ &= (\sec^2 x) \cdot (12t^2 + 1) \\ &= [\sec^2(4t^3 + t)] \cdot (12t^2 + 1) = (12t^2 + 1) \sec^2(4t^3 + t) \quad \blacktriangleleft\end{aligned}$$

■ AN ALTERNATIVE VERSION OF THE CHAIN RULE

Formula (1) for the chain rule can be unwieldy in some problems because it involves so many variables. As you become more comfortable with the chain rule, you may want to dispense with writing out the dependent variables by expressing (1) in the form

Confirm that (2) is an alternative version of (1) by letting $y = f(g(x))$ and $u = g(x)$.

$$\frac{d}{dx}[f(g(x))] = (f \circ g)'(x) = f'(g(x))g'(x) \quad (2)$$

A convenient way to remember this formula is to call f the “outside function” and g the “inside function” in the composition $f(g(x))$ and then express (2) in words as:

The derivative of $f(g(x))$ is the derivative of the outside function evaluated at the inside function times the derivative of the inside function.

$$\frac{d}{dx}[f(g(x))] = \underbrace{f'(g(x))}_{\text{Derivative of the outside function evaluated at the inside function}} \cdot \underbrace{g'(x)}_{\text{Derivative of the inside function}}$$

Derivative of the outside function evaluated at the inside function

Derivative of the inside function

► **Example 3** (Example 1 revisited) Find $h'(x)$ if $h(x) = \cos(x^3)$.

Solution. We can think of h as a composition $f(g(x))$ in which $g(x) = x^3$ is the inside function and $f(x) = \cos x$ is the outside function. Thus, Formula (2) yields

$$\begin{aligned}
 h'(x) &= \underbrace{f'(g(x))}_{\substack{\text{Derivative of the outside} \\ \text{function evaluated at the} \\ \text{inside function}}} \cdot \underbrace{g'(x)}_{\substack{\text{Derivative of the} \\ \text{inside function}}} \\
 &= f'(x^3) \cdot 3x^2 \\
 &= -\sin(x^3) \cdot 3x^2 = -3x^2 \sin(x^3)
 \end{aligned}$$

which agrees with the result obtained in Example 1. ◀

► **Example 4**

$$\begin{aligned}
 \frac{d}{dx} [\tan^2 x] &= \frac{d}{dx} [(\tan x)^2] = \underbrace{(2 \tan x)}_{\substack{\text{Derivative of the outside} \\ \text{function evaluated at the} \\ \text{inside function}}} \cdot \underbrace{(\sec^2 x)}_{\substack{\text{Derivative of the} \\ \text{inside function}}} = 2 \tan x \sec^2 x \\
 \\
 \frac{d}{dx} [\sqrt{x^2 + 1}] &= \frac{1}{\underbrace{2\sqrt{x^2 + 1}}_{\substack{\text{Derivative of the outside} \\ \text{function evaluated at the} \\ \text{inside function}}}} \cdot \underbrace{2x}_{\substack{\text{Derivative of the} \\ \text{inside function}}} = \frac{x}{\sqrt{x^2 + 1}} \quad \text{See Formula (6) of Section 2.3.} \quad \blacktriangleleft
 \end{aligned}$$

GENERALIZED DERIVATIVE FORMULAS

There is a useful third variation of the chain rule that strikes a middle ground between Formulas (1) and (2). If we let $u = g(x)$ in (2), then we can rewrite that formula as

$$\frac{d}{dx} [f(u)] = f'(u) \frac{du}{dx} \quad (3)$$

This result, called the **generalized derivative formula** for f , provides a way of using the derivative of $f(x)$ to produce the derivative of $f(u)$, where u is a function of x . Table 2.6.1 gives some examples of this formula.

Table 2.6.1
GENERALIZED DERIVATIVE FORMULAS

$\frac{d}{dx} [u^r] = ru^{r-1} \frac{du}{dx}$	
$\frac{d}{dx} [\sin u] = \cos u \frac{du}{dx}$	$\frac{d}{dx} [\cos u] = -\sin u \frac{du}{dx}$
$\frac{d}{dx} [\tan u] = \sec^2 u \frac{du}{dx}$	$\frac{d}{dx} [\cot u] = -\csc^2 u \frac{du}{dx}$
$\frac{d}{dx} [\sec u] = \sec u \tan u \frac{du}{dx}$	$\frac{d}{dx} [\csc u] = -\csc u \cot u \frac{du}{dx}$

► **Example 5** Find

$$\begin{array}{lll} \text{(a)} \quad \frac{d}{dx}[\sin(2x)] & \text{(b)} \quad \frac{d}{dx}[\tan(x^2 + 1)] & \text{(c)} \quad \frac{d}{dx}[\sqrt{x^3 + \csc x}] \\ \text{(d)} \quad \frac{d}{dx}[x^2 - x + 2]^{3/4} & \text{(e)} \quad \frac{d}{dx}[(1 + x^5 \cot x)^{-8}] \end{array}$$

Solution (a). Taking $u = 2x$ in the generalized derivative formula for $\sin u$ yields

$$\frac{d}{dx}[\sin(2x)] = \frac{d}{dx}[\sin u] = \cos u \frac{du}{dx} = \cos 2x \cdot \frac{d}{dx}[2x] = \cos 2x \cdot 2 = 2 \cos 2x$$

Solution (b). Taking $u = x^2 + 1$ in the generalized derivative formula for $\tan u$ yields

$$\begin{aligned} \frac{d}{dx}[\tan(x^2 + 1)] &= \frac{d}{dx}[\tan u] = \sec^2 u \frac{du}{dx} \\ &= \sec^2(x^2 + 1) \cdot \frac{d}{dx}[x^2 + 1] = \sec^2(x^2 + 1) \cdot 2x \\ &= 2x \sec^2(x^2 + 1) \end{aligned}$$

Solution (c). Taking $u = x^3 + \csc x$ in the generalized derivative formula for $\sqrt{u}$ yields

$$\begin{aligned} \frac{d}{dx}[\sqrt{x^3 + \csc x}] &= \frac{d}{dx}[\sqrt{u}] = \frac{1}{2\sqrt{u}} \frac{du}{dx} = \frac{1}{2\sqrt{x^3 + \csc x}} \cdot \frac{d}{dx}[x^3 + \csc x] \\ &= \frac{1}{2\sqrt{x^3 + \csc x}} \cdot (3x^2 - \csc x \cot x) = \frac{3x^2 - \csc x \cot x}{2\sqrt{x^3 + \csc x}} \end{aligned}$$

Solution (d). Taking $u = x^2 - x + 2$ in the generalized derivative formula for $u^{3/4}$ yields

$$\begin{aligned} \frac{d}{dx}[x^2 - x + 2]^{3/4} &= \frac{d}{dx}[u^{3/4}] = \frac{3}{4}u^{-1/4} \frac{du}{dx} \\ &= \frac{3}{4}(x^2 - x + 2)^{-1/4} \cdot \frac{d}{dx}[x^2 - x + 2] \\ &= \frac{3}{4}(x^2 - x + 2)^{-1/4}(2x - 1) \end{aligned}$$

Solution (e). Taking $u = 1 + x^5 \cot x$ in the generalized derivative formula for u^{-8} yields

$$\begin{aligned} \frac{d}{dx}[(1 + x^5 \cot x)^{-8}] &= \frac{d}{dx}[u^{-8}] = -8u^{-9} \frac{du}{dx} \\ &= -8(1 + x^5 \cot x)^{-9} \cdot \frac{d}{dx}[1 + x^5 \cot x] \\ &= -8(1 + x^5 \cot x)^{-9} \cdot [x^5(-\csc^2 x) + 5x^4 \cot x] \\ &= (8x^5 \csc^2 x - 40x^4 \cot x)(1 + x^5 \cot x)^{-9} \quad \blacktriangleleft \end{aligned}$$

Sometimes you will have to make adjustments in notation or apply the chain rule more than once to calculate a derivative.

► **Example 6** Find

$$\begin{array}{ll} \text{(a)} \quad \frac{d}{dx}[\sin(\sqrt{1 + \cos x})] & \text{(b)} \quad \frac{d\mu}{dt} \text{ if } \mu = \sec \sqrt{\omega t} \quad (\omega \text{ constant}) \end{array}$$

Solution (a). Taking $u = \sqrt{1 + \cos x}$ in the generalized derivative formula for $\sin u$ yields

$$\begin{aligned}\frac{d}{dx} [\sin(\sqrt{1 + \cos x})] &= \frac{d}{dx} [\sin u] = \cos u \frac{du}{dx} \\ &= \cos(\sqrt{1 + \cos x}) \cdot \frac{d}{dx} [\sqrt{1 + \cos x}] \\ &= \cos(\sqrt{1 + \cos x}) \cdot \frac{-\sin x}{2\sqrt{1 + \cos x}} \\ &= -\frac{\sin x \cos(\sqrt{1 + \cos x})}{2\sqrt{1 + \cos x}}\end{aligned}$$

We used the generalized derivative formula for $\sqrt{u}$ with $u = 1 + \cos x$.

Solution (b).

$$\begin{aligned}\frac{d\mu}{dt} &= \frac{d}{dt} [\sec \sqrt{\omega t}] = \sec \sqrt{\omega t} \tan \sqrt{\omega t} \frac{d}{dt} [\sqrt{\omega t}] \\ &= \sec \sqrt{\omega t} \tan \sqrt{\omega t} \frac{\omega}{2\sqrt{\omega t}}\end{aligned}$$

We used the generalized derivative formula for $\sec u$ with $u = \sqrt{\omega t}$.

We used the generalized derivative formula for $\sqrt{u}$ with $u = \omega t$.

DIFFERENTIATING USING COMPUTER ALGEBRA SYSTEMS

Even with the chain rule and other differentiation rules, some derivative computations can be tedious to perform. For complicated derivatives, engineers and scientists often use computer algebra systems such as *Mathematica*, *Maple*, or *Sage*. For example, although we have all the mathematical tools to compute

$$\frac{d}{dx} \left[\frac{(x^2 + 1)^{10} \sin^3(\sqrt{x})}{\sqrt{1 + \csc x}} \right] \quad (4)$$

by hand, the computation is sufficiently involved that it may be more efficient (and less error-prone) to use a computer algebra system.

TECHNOLOGY MASTERY

If you have a CAS, use it to perform the differentiation in (4).

QUICK CHECK EXERCISES 2.6 (See page 160 for answers.)

- The chain rule states that the derivative of the composition of two functions is the derivative of the _____ function evaluated at the _____ function times the derivative of the _____ function.
- If y is a differentiable function of u , and u is a differentiable function of x , then

$$\frac{dy}{dx} = \text{_____} \cdot \text{_____}$$

- Find dy/dx .
(a) $y = (x^2 + 5)^{10}$ (b) $y = \sqrt{1 + 6x}$
- Find dy/dx .
(a) $y = \sin(3x + 2)$ (b) $y = (x^2 \tan x)^4$
- Suppose that $f(2) = 3$, $f'(2) = 4$, $g(3) = 6$, and $g'(3) = -5$. Evaluate
(a) $h'(2)$, where $h(x) = g(f(x))$
(b) $k'(3)$, where $k(x) = f(\frac{1}{3}g(x))$.

EXERCISE SET 2.6



Graphing Utility



CAS

- Given that
 $f'(0) = 2$, $g(0) = 0$ and $g'(0) = 3$
find $(f \circ g)'(0)$.
- Given that
 $f'(9) = 5$, $g(2) = 9$ and $g'(2) = -3$
find $(f \circ g)'(2)$.

- Let $f(x) = x^5$ and $g(x) = 2x - 3$.
(a) Find $(f \circ g)(x)$ and $(f \circ g)'(x)$.
(b) Find $(g \circ f)(x)$ and $(g \circ f)'(x)$.
- Let $f(x) = 5\sqrt{x}$ and $g(x) = 4 + \cos x$.
(a) Find $(f \circ g)(x)$ and $(f \circ g)'(x)$.
(b) Find $(g \circ f)(x)$ and $(g \circ f)'(x)$.

FOCUS ON CONCEPTS

5. Given the following table of values, find the indicated derivatives in parts (a) and (b).

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
3	5	-2	5	7
5	3	-1	12	4

(a) $F'(3)$, where $F(x) = f(g(x))$

(b) $G'(3)$, where $G(x) = g(f(x))$

6. Given the following table of values, find the indicated derivatives in parts (a) and (b).

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
-1	2	3	2	-3
2	0	4	1	-5

(a) $F'(-1)$, where $F(x) = f(g(x))$

(b) $G'(-1)$, where $G(x) = g(f(x))$

7-26 Find $f'(x)$. ■

7. $f(x) = (x^3 + 2x)^{37}$ 8. $f(x) = (3x^2 + 2x - 1)^6$

9. $f(x) = \left(x^3 - \frac{7}{x}\right)^{-2}$ 10. $f(x) = \frac{1}{(x^5 - x + 1)^9}$

11. $f(x) = \frac{4}{(3x^2 - 2x + 1)^3}$ 12. $f(x) = \sqrt{x^3 - 2x + 5}$

13. $f(x) = \sqrt{4 + \sqrt{3x}}$ 14. $f(x) = \sqrt[3]{12 + \sqrt{x}}$

15. $f(x) = \sin\left(\frac{1}{x^2}\right)$ 16. $f(x) = \tan \sqrt{x}$

17. $f(x) = 4\cos^5 x$ 18. $f(x) = 4x + 5\sin^4 x$

19. $f(x) = \cos^2(3\sqrt{x})$ 20. $f(x) = \tan^4(x^3)$

21. $f(x) = 2\sec^2(x^7)$ 22. $f(x) = \cos^3\left(\frac{x}{x+1}\right)$

23. $f(x) = \sqrt{\cos(5x)}$ 24. $f(x) = \sqrt{3x - \sin^2(4x)}$

25. $f(x) = [x + \csc(x^3 + 3)]^{-3}$

26. $f(x) = [x^4 - \sec(4x^2 - 2)]^{-4}$

27-40 Find dy/dx . ■

27. $y = x^3 \sin^2(5x)$ 28. $y = \sqrt{x} \tan^3(\sqrt{x})$

29. $y = x^5 \sec(1/x)$ 30. $y = \frac{\sin x}{\sec(3x + 1)}$

31. $y = \cos(\cos x)$ 32. $y = \sin(\tan 3x)$

33. $y = \cos^3(\sin 2x)$ 34. $y = \frac{1 + \csc(x^2)}{1 - \cot(x^2)}$

35. $y = (5x + 8)^7 (1 - \sqrt{x})^6$ 36. $y = (x^2 + x)^5 \sin^8 x$

37. $y = \left(\frac{x-5}{2x+1}\right)^3$ 38. $y = \left(\frac{1+x^2}{1-x^2}\right)^{17}$

39. $y = \frac{(2x+3)^3}{(4x^2-1)^8}$ 40. $y = [1 + \sin^3(x^5)]^{12}$

C 41-42 Use a CAS to find dy/dx . ■

41. $y = [x \sin 2x + \tan^4(x^7)]^5$

42. $y = \tan^4\left(2 + \frac{(7-x)\sqrt{3x^2+5}}{x^3 + \sin x}\right)$

43-50 Find an equation for the tangent line to the graph at the specified value of x . ■

43. $y = x \cos 3x$, $x = \pi$

44. $y = \sin(1 + x^3)$, $x = -3$

45. $y = \sec^3\left(\frac{\pi}{2} - x\right)$, $x = -\frac{\pi}{2}$

46. $y = \left(x - \frac{1}{x}\right)^3$, $x = 2$ 47. $y = \tan(4x^2)$, $x = \sqrt{\pi}$

48. $y = 3 \cot^4 x$, $x = \frac{\pi}{4}$ 49. $y = x^2 \sqrt{5 - x^2}$, $x = 1$

50. $y = \frac{x}{\sqrt{1-x^2}}$, $x = 0$

51-54 Find d^2y/dx^2 . ■

51. $y = x \cos(5x) - \sin^2 x$ 52. $y = \sin(3x^2)$

53. $y = \frac{1+x}{1-x}$ 54. $y = x \tan\left(\frac{1}{x}\right)$

55-58 Find the indicated derivative. ■

55. $y = \cot^3(\pi - \theta)$; find $\frac{dy}{d\theta}$.

56. $\lambda = \left(\frac{au+b}{cu+d}\right)^6$; find $\frac{d\lambda}{du}$ (a, b, c, d constants).

57. $\frac{d}{d\omega}[a \cos^2 \pi\omega + b \sin^2 \pi\omega]$ (a, b constants)

58. $x = \csc^2\left(\frac{\pi}{3} - y\right)$; find $\frac{dx}{dy}$.

59. (a) Use a graphing utility to obtain the graph of the function $f(x) = x\sqrt{4-x^2}$.

(b) Use the graph in part (a) to make a rough sketch of the graph of f' .

(c) Find $f'(x)$, and then check your work in part (b) by using the graphing utility to obtain the graph of f' .

(d) Find the equation of the tangent line to the graph of f at $x = 1$, and graph f and the tangent line together.

60. (a) Use a graphing utility to obtain the graph of the function $f(x) = \sin x^2 \cos x$ over the interval $[-\pi/2, \pi/2]$.

(b) Use the graph in part (a) to make a rough sketch of the graph of f' over the interval.

(c) Find $f'(x)$, and then check your work in part (b) by using the graphing utility to obtain the graph of f' over the interval.

(d) Find the equation of the tangent line to the graph of f at $x = 1$, and graph f and the tangent line together over the interval.

61–64 True–False Determine whether the statement is true or false. Explain your answer. ■

61. If $y = f(x)$, then $\frac{d}{dx}[\sqrt{y}] = \sqrt{f'(x)}$.
62. If $y = f(u)$ and $u = g(x)$, then $dy/dx = f'(x) \cdot g'(x)$.
63. If $y = \cos[g(x)]$, then $dy/dx = -\sin[g'(x)]$.
64. If $y = \sin^3(3x^3)$, then $dy/dx = 27x^2 \sin^2(3x^3) \cos(3x^3)$.
65. If an object suspended from a spring is displaced vertically from its equilibrium position by a small amount and released, and if the air resistance and the mass of the spring are ignored, then the resulting oscillation of the object is called **simple harmonic motion**. Under appropriate conditions the displacement y from equilibrium in terms of time t is given by

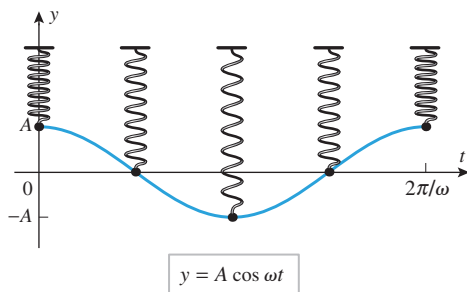
$$y = A \cos \omega t$$

where A is the initial displacement at time $t = 0$, and ω is a constant that depends on the mass of the object and the stiffness of the spring (see the accompanying figure). The constant $|A|$ is called the **amplitude** of the motion and ω the **angular frequency**.

(a) Show that

$$\frac{d^2 y}{dt^2} = -\omega^2 y$$

- (b) The **period** T is the time required to make one complete oscillation. Show that $T = 2\pi/\omega$.
- (c) The **frequency** f of the vibration is the number of oscillations per unit time. Find f in terms of the period T .
- (d) Find the amplitude, period, and frequency of an object that is executing simple harmonic motion given by $y = 0.6 \cos 15t$, where t is in seconds and y is in centimeters.



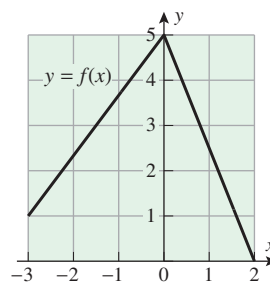
▲ Figure Ex-65

66. Find the value of the constant A so that $y = A \sin 3t$ satisfies the equation
- $$\frac{d^2 y}{dt^2} + 2y = 4 \sin 3t$$

FOCUS ON CONCEPTS

67. Use the graph of the function f in the accompanying figure to evaluate

$$\left. \frac{d}{dx} [\sqrt{x + f(x)}] \right|_{x=-1}$$



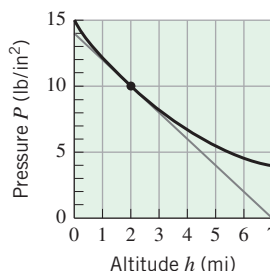
◀ Figure Ex-67

68. Using the function f in Exercise 67, evaluate

$$\left. \frac{d}{dx} [f(2 \sin x)] \right|_{x=\pi/6}$$

69. The accompanying figure shows the graph of atmospheric pressure p (lb/in²) versus the altitude h (mi) above sea level.

- (a) From the graph and the tangent line at $h = 2$ shown on the graph, estimate the values of p and dp/dh at an altitude of 2 mi.
- (b) If the altitude of a space vehicle is increasing at the rate of 0.3 mi/s at the instant when it is 2 mi above sea level, how fast is the pressure changing with time at this instant?



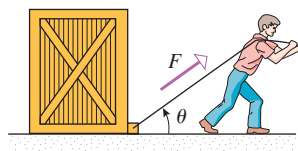
◀ Figure Ex-69

70. The force F (in pounds) acting at an angle θ with the horizontal that is needed to drag a crate weighing W pounds along a horizontal surface at a constant velocity is given by

$$F = \frac{\mu W}{\cos \theta + \mu \sin \theta}$$

where μ is a constant called the **coefficient of sliding friction** between the crate and the surface (see the accompanying figure). Suppose that the crate weighs 150 lb and that $\mu = 0.3$.

- (a) Find $dF/d\theta$ when $\theta = 30^\circ$. Express the answer in units of pounds/degree.
- (b) Find dF/dt when $\theta = 30^\circ$ if θ is decreasing at the rate of $0.5^\circ/\text{s}$ at this instant.



◀ Figure Ex-70

71. Recall that

$$\frac{d}{dx}(|x|) = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$

Use this result and the chain rule to find

$$\frac{d}{dx}(|\sin x|)$$

for nonzero x in the interval $(-\pi, \pi)$.

72. Use the derivative formula for $\sin x$ and the identity

$$\cos x = \sin\left(\frac{\pi}{2} - x\right)$$

to obtain the derivative formula for $\cos x$.

73. Let

$$f(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

- Show that f is continuous at $x = 0$.
- Use Definition 2.2.1 to show that $f'(0)$ does not exist.
- Find $f'(x)$ for $x \neq 0$.
- Determine whether $\lim_{x \rightarrow 0} f'(x)$ exists.

74. Let

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

- Show that f is continuous at $x = 0$.
- Use Definition 2.2.1 to find $f'(0)$.
- Find $f'(x)$ for $x \neq 0$.
- Show that f' is not continuous at $x = 0$.

75. Given the following table of values, find the indicated derivatives in parts (a) and (b).

x	$f(x)$	$f'(x)$
2	1	7
8	5	-3

- $g'(2)$, where $g(x) = [f(x)]^3$
- $h'(2)$, where $h(x) = f(x^3)$

76. Given that $f'(x) = \sqrt{3x+4}$ and $g(x) = x^2 - 1$, find $F'(x)$ if $F(x) = f(g(x))$.

77. Given that $f'(x) = \frac{x}{x^2+1}$ and $g(x) = \sqrt{3x-1}$, find $F'(x)$ if $F(x) = f(g(x))$.

78. Find $f'(x^2)$ if $\frac{d}{dx}[f(x^2)] = x^2$.

79. Find $\frac{d}{dx}[f(x)]$ if $\frac{d}{dx}[f(3x)] = 6x$.

80. Recall that a function f is **even** if $f(-x) = f(x)$ and **odd** if $f(-x) = -f(x)$, for all x in the domain of f . Assuming that f is differentiable, prove:

- f' is odd if f is even
- f' is even if f is odd.

81. Draw some pictures to illustrate the results in Exercise 80, and write a paragraph that gives an informal explanation of why the results are true.

82. Let $y = f_1(u)$, $u = f_2(v)$, $v = f_3(w)$, and $w = f_4(x)$. Express dy/dx in terms of dy/du , dw/dx , du/dv , and dv/dw .

83. Find a formula for

$$\frac{d}{dx}[f(g(h(x)))]$$

84. **Writing** The “co” in “cosine” comes from “complementary,” since the cosine of an angle is the sine of the complementary angle, and vice versa:

$$\cos x = \sin\left(\frac{\pi}{2} - x\right) \quad \text{and} \quad \sin x = \cos\left(\frac{\pi}{2} - x\right)$$

Suppose that we define a function g to be a *cofunction* of a function f if

$$g(x) = f\left(\frac{\pi}{2} - x\right) \quad \text{for all } x$$

Thus, cosine and sine are cofunctions of each other, as are cotangent and tangent, and also cosecant and secant. If g is the cofunction of f , state a formula that relates g' and the cofunction of f' . Discuss how this relationship is exhibited by the derivatives of the cosine, cotangent, and cosecant functions.

✓ QUICK CHECK ANSWERS 2.6

- outside; inside; inside 2. $\frac{dy}{du} \cdot \frac{du}{dx}$ 3. (a) $10(x^2 + 5)^9 \cdot 2x = 20x(x^2 + 5)^9$ (b) $\frac{1}{2\sqrt{1+6x}} \cdot 6 = \frac{3}{\sqrt{1+6x}}$
- (a) $3 \cos(3x + 2)$ (b) $4(x^2 \tan x)^3(2x \tan x + x^2 \sec^2 x)$ 5. (a) $g'(f(2))f'(2) = -20$ (b) $f'\left(\frac{1}{3}g(3)\right) \cdot \frac{1}{3}g'(3) = -\frac{20}{3}$

3

THE DERIVATIVE IN GRAPHING AND APPLICATIONS



Stone/Getty Images

Derivatives can help to find the most cost-effective location for an offshore oil-drilling rig.

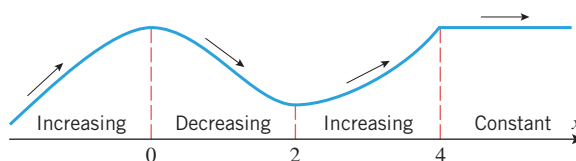
In this chapter we will study various applications of the derivative. For example, we will use methods of calculus to analyze functions and their graphs. In the process, we will show how calculus and graphing utilities, working together, can provide most of the important information about the behavior of functions. Another important application of the derivative will be in the solution of optimization problems. For example, if time is the main consideration in a problem, we might be interested in finding the quickest way to perform a task, and if cost is the main consideration, we might be interested in finding the least expensive way to perform a task. Mathematically, optimization problems can be reduced to finding the largest or smallest value of a function on some interval, and determining where the largest or smallest value occurs. Using the derivative, we will develop the mathematical tools necessary for solving such problems. We will also use the derivative to study the motion of a particle moving along a line, and we will show how the derivative can help us to approximate solutions of equations.

3.1 ANALYSIS OF FUNCTIONS I: INCREASE, DECREASE, AND CONCAVITY

Although graphing utilities are useful for determining the general shape of a graph, many problems require more precision than graphing utilities are capable of producing. The purpose of this section is to develop mathematical tools that can be used to determine the exact shape of a graph and the precise locations of its key features.

■ INCREASING AND DECREASING FUNCTIONS

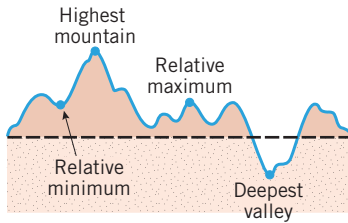
The terms *increasing*, *decreasing*, and *constant* are used to describe the behavior of a function as we travel left to right along its graph. For example, the function graphed in Figure 3.1.1 can be described as increasing to the left of $x = 0$, decreasing from $x = 0$ to $x = 2$, increasing from $x = 2$ to $x = 4$, and constant to the right of $x = 4$.



► Figure 3.1.1

3.2 ANALYSIS OF FUNCTIONS II: RELATIVE EXTREMA; GRAPHING POLYNOMIALS

In this section we will develop methods for finding the high and low points on the graph of a function and we will discuss procedures for analyzing the graphs of polynomials.



▲ Figure 3.2.1

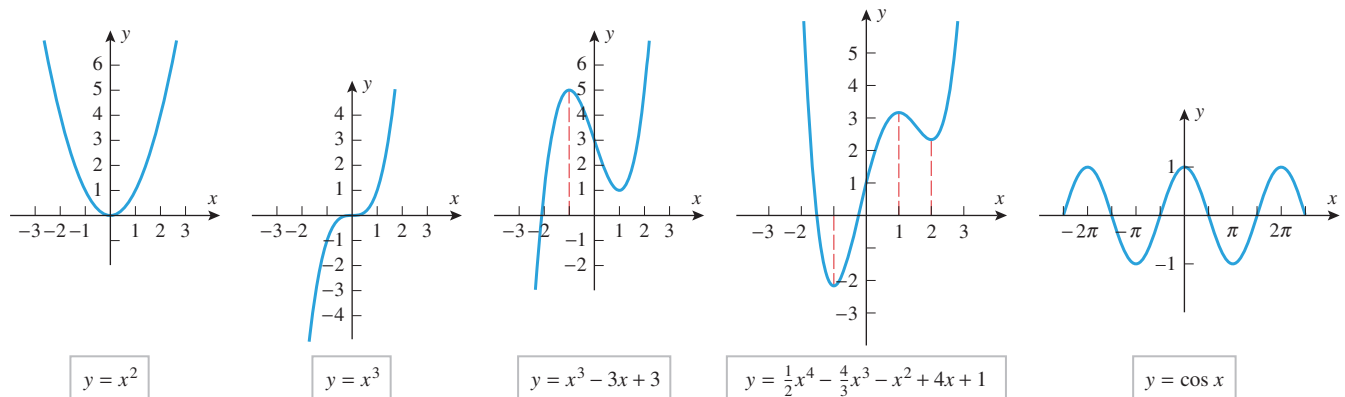
■ RELATIVE MAXIMA AND MINIMA

If we imagine the graph of a function f to be a two-dimensional mountain range with hills and valleys, then the tops of the hills are called “relative maxima,” and the bottoms of the valleys are called “relative minima” (Figure 3.2.1). The relative maxima are the high points in their *immediate vicinity*, and the relative minima are the low points. A relative maximum need not be the highest point in the entire mountain range, and a relative minimum need not be the lowest point—they are just high and low points *relative* to the nearby terrain. These ideas are captured in the following definition.

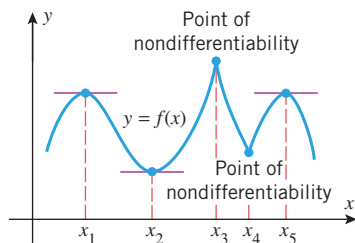
3.2.1 DEFINITION A function f is said to have a **relative maximum** at x_0 if there is an open interval containing x_0 on which $f(x_0)$ is the largest value, that is, $f(x_0) \geq f(x)$ for all x in the interval. Similarly, f is said to have a **relative minimum** at x_0 if there is an open interval containing x_0 on which $f(x_0)$ is the smallest value, that is, $f(x_0) \leq f(x)$ for all x in the interval. If f has either a relative maximum or a relative minimum at x_0 , then f is said to have a **relative extremum** at x_0 .

► **Example 1** We can see from Figure 3.2.2 that:

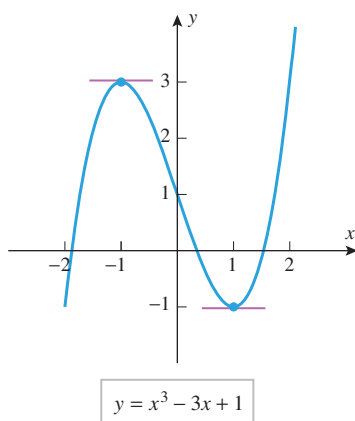
- $f(x) = x^2$ has a relative minimum at $x = 0$ but no relative maxima.
- $f(x) = x^3$ has no relative extrema.
- $f(x) = x^3 - 3x + 3$ has a relative maximum at $x = -1$ and a relative minimum at $x = 1$.
- $f(x) = \frac{1}{2}x^4 - \frac{4}{3}x^3 - x^2 + 4x + 1$ has relative minima at $x = -1$ and $x = 2$ and a relative maximum at $x = 1$.
- $f(x) = \cos x$ has relative maxima at all even multiples of π and relative minima at all odd multiples of π . ◀



▲ Figure 3.2.2



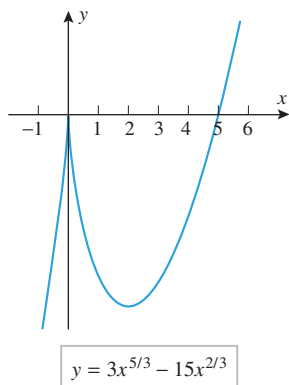
▲ **Figure 3.2.3** The points x_1, x_2, x_3, x_4 , and x_5 are critical points. Of these, x_1, x_2 , and x_5 are stationary points.



▲ **Figure 3.2.4**

What is the maximum number of critical points that a polynomial of degree n can have? Why?

TECHNOLOGY MASTERY



▲ **Figure 3.2.5**

The relative extrema for the five functions in Example 1 occur at points where the graphs of the functions have horizontal tangent lines. Figure 3.2.3 illustrates that a relative extremum can also occur at a point where a function is not differentiable. In general, we define a **critical point** for a function f to be a point in the domain of f at which either the graph of f has a horizontal tangent line or f is not differentiable. To distinguish between the two types of critical points we call x a **stationary point** of f if $f'(x) = 0$. The following theorem, which is proved in Appendix D, states that the critical points for a function form a complete set of candidates for relative extrema on the interior of the domain of the function.

3.2.2 THEOREM Suppose that f is a function defined on an open interval containing the point x_0 . If f has a relative extremum at $x = x_0$, then $x = x_0$ is a critical point of f ; that is, either $f'(x_0) = 0$ or f is not differentiable at x_0 .

► **Example 2** Find all critical points of $f(x) = x^3 - 3x + 1$.

Solution. The function f , being a polynomial, is differentiable everywhere, so its critical points are all stationary points. To find these points we must solve the equation $f'(x) = 0$. Since

$$f'(x) = 3x^2 - 3 = 3(x + 1)(x - 1)$$

we conclude that the critical points occur at $x = -1$ and $x = 1$. This is consistent with the graph of f in Figure 3.2.4. ◀

► **Example 3** Find all critical points of $f(x) = 3x^{5/3} - 15x^{2/3}$.

Solution. The function f is continuous everywhere and its derivative is

$$f'(x) = 5x^{2/3} - 10x^{-1/3} = 5x^{-1/3}(x - 2) = \frac{5(x - 2)}{x^{1/3}}$$

We see from this that $f'(x) = 0$ if $x = 2$ and $f'(x)$ is undefined if $x = 0$. Thus $x = 0$ and $x = 2$ are critical points and $x = 2$ is a stationary point. This is consistent with the graph of f shown in Figure 3.2.5. ◀

Your graphing utility may have trouble producing portions of the graph in Figure 3.2.5 because of the fractional exponents. If this is the case for you, graph the function

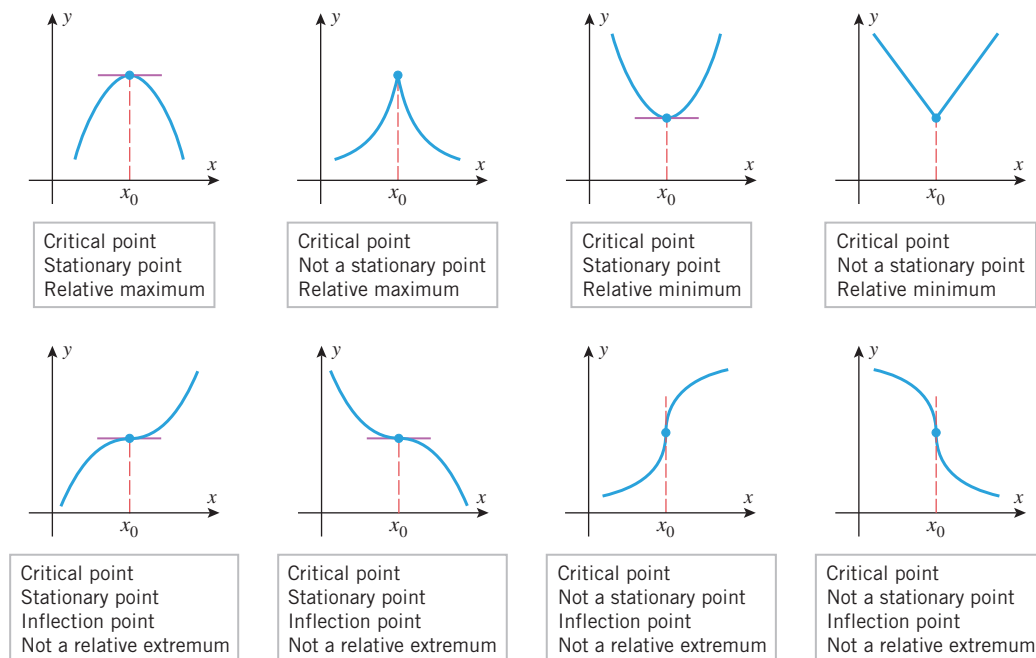
$$y = 3(|x|/x)|x|^{5/3} - 15|x|^{2/3}$$

which is equivalent to $f(x)$ for $x \neq 0$. Appendix A explores the method suggested here in more detail.

FIRST DERIVATIVE TEST

Theorem 3.2.2 asserts that the relative extrema must occur at critical points, but it does *not* say that a relative extremum occurs at *every* critical point. For example, for the eight critical points in Figure 3.2.6, relative extrema occur at each x_0 in the top row but not at any x_0 in the bottom row. Moreover, at the critical points in the first row the derivatives have opposite signs on the two sides of x_0 , whereas at the critical points in the second row the signs of the derivatives are the same on both sides. This suggests:

A function f has a relative extremum at those critical points where f' changes sign.



▲ Figure 3.2.6

We can actually take this a step further. At the two relative maxima in Figure 3.2.6 the derivative is positive on the left side and negative on the right side, and at the two relative minima the derivative is negative on the left side and positive on the right side. All of this is summarized more precisely in the following theorem.

3.2.3 THEOREM (First Derivative Test) Suppose that f is continuous at a critical point x_0 .

- (a) If $f'(x) > 0$ on an open interval extending left from x_0 and $f'(x) < 0$ on an open interval extending right from x_0 , then f has a relative maximum at x_0 .
- (b) If $f'(x) < 0$ on an open interval extending left from x_0 and $f'(x) > 0$ on an open interval extending right from x_0 , then f has a relative minimum at x_0 .
- (c) If $f'(x)$ has the same sign on an open interval extending left from x_0 as it does on an open interval extending right from x_0 , then f does not have a relative extremum at x_0 .

Informally stated, parts (a) and (b) of Theorem 3.2.3 tell us that for a continuous function, relative maxima occur at critical points where the derivative changes from $+$ to $-$ and relative minima where it changes from $-$ to $+$.

Use the first derivative test to confirm the behavior at x_0 of each graph in Figure 3.2.6.

PROOF We will prove part (a) and leave parts (b) and (c) as exercises. We are assuming that $f'(x) > 0$ on the interval (a, x_0) and that $f'(x) < 0$ on the interval (x_0, b) , and we want to show that

$$f(x_0) \geq f(x)$$

for all x in the interval (a, b) . However, the two hypotheses, together with Theorem 3.1.2 and its associated marginal note imply that f is increasing on the interval $(a, x_0]$ and decreasing on the interval $[x_0, b)$. Thus, $f(x_0) \geq f(x)$ for all x in (a, b) with equality only at x_0 . ■

► **Example 4** We showed in Example 3 that the function $f(x) = 3x^{5/3} - 15x^{2/3}$ has critical points at $x = 0$ and $x = 2$. Figure 3.2.5 suggests that f has a relative maximum at $x = 0$ and a relative minimum at $x = 2$. Confirm this using the first derivative test.

Table 3.2.1

INTERVAL	$5(x-2)/x^{1/3}$	$f'(x)$
$x < 0$	$(-)/(-)$	$+$
$0 < x < 2$	$(-)/(+)$	$-$
$x > 2$	$(+)/(+)$	$+$

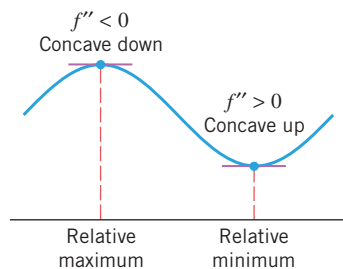
Solution. We showed in Example 3 that

$$f'(x) = \frac{5(x-2)}{x^{1/3}}$$

A sign analysis of this derivative is shown in Table 3.2.1. The sign of f' changes from $+$ to $-$ at $x = 0$, so there is a relative maximum at that point. The sign changes from $-$ to $+$ at $x = 2$, so there is a relative minimum at that point. ◀

■ SECOND DERIVATIVE TEST

There is another test for relative extrema that is based on the following geometric observation: A function f has a relative maximum at a stationary point if the graph of f is concave down on an open interval containing that point, and it has a relative minimum if it is concave up (Figure 3.2.7).



► Figure 3.2.7

3.2.4 THEOREM (Second Derivative Test) Suppose that f is twice differentiable at the point x_0 .

- (a) If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a relative minimum at x_0 .
- (b) If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a relative maximum at x_0 .
- (c) If $f'(x_0) = 0$ and $f''(x_0) = 0$, then the test is inconclusive; that is, f may have a relative maximum, a relative minimum, or neither at x_0 .

The second derivative test is often easier to apply than the first derivative test. However, the first derivative test can be used at any critical point of a continuous function, while the second derivative test applies only at stationary points where the second derivative exists.

We will prove parts (a) and (c) and leave part (b) as an exercise.

PROOF (a) We are given that $f'(x_0) = 0$ and $f''(x_0) > 0$, and we want to show that f has a relative minimum at x_0 . Expressing $f''(x_0)$ as a limit and using the two given conditions we obtain

$$f''(x_0) = \lim_{x \rightarrow x_0} \frac{f'(x) - f'(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} \frac{f'(x)}{x - x_0} > 0$$

This implies that for x sufficiently close to but different from x_0 we have

$$\frac{f'(x)}{x - x_0} > 0 \quad (1)$$

Thus, there is an open interval extending left from x_0 and an open interval extending right from x_0 on which (1) holds. On the open interval extending left the denominator in (1) is

negative, so $f'(x) < 0$, and on the open interval extending right the denominator is positive, so $f'(x) > 0$. It now follows from part (b) of the first derivative test (Theorem 3.2.3) that f has a relative minimum at x_0 .

PROOF (c) To prove this part of the theorem we need only provide functions for which $f'(x_0) = 0$ and $f''(x_0) = 0$ at some point x_0 , but with one having a relative minimum at x_0 , one having a relative maximum at x_0 , and one having neither at x_0 . We leave it as an exercise for you to show that three such functions are $f(x) = x^4$ (relative minimum at $x = 0$), $f(x) = -x^4$ (relative maximum at $x = 0$), and $f(x) = x^3$ (neither a relative maximum nor a relative minimum at x_0). ■

► **Example 5** Find the relative extrema of $f(x) = 3x^5 - 5x^3$.

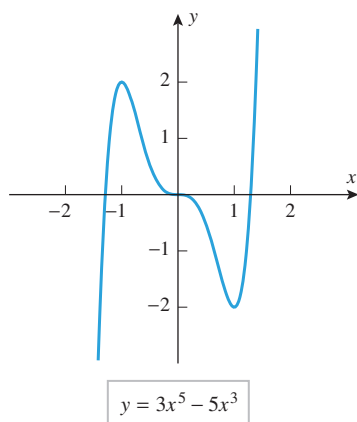
Solution. We have

$$f'(x) = 15x^4 - 15x^2 = 15x^2(x^2 - 1) = 15x^2(x + 1)(x - 1)$$

$$f''(x) = 60x^3 - 30x = 30x(2x^2 - 1)$$

Solving $f'(x) = 0$ yields the stationary points $x = 0$, $x = -1$, and $x = 1$. As shown in the following table, we can conclude from the second derivative test that f has a relative maximum at $x = -1$ and a relative minimum at $x = 1$.

STATIONARY POINT	$30x(2x^2 - 1)$	$f''(x)$	SECOND DERIVATIVE TEST
$x = -1$	-30	-	f has a relative maximum
$x = 0$	0	0	Inconclusive
$x = 1$	30	+	f has a relative minimum



▲ Figure 3.2.8

The test is inconclusive at $x = 0$, so we will try the first derivative test at that point. A sign analysis of f' is given in the following table:

INTERVAL	$15x^2(x + 1)(x - 1)$	$f'(x)$
$-1 < x < 0$	(+)(+)(-)	-
$0 < x < 1$	(+)(+)(-)	-

Since there is no sign change in f' at $x = 0$, there is neither a relative maximum nor a relative minimum at that point. All of this is consistent with the graph of f shown in Figure 3.2.8. ◀

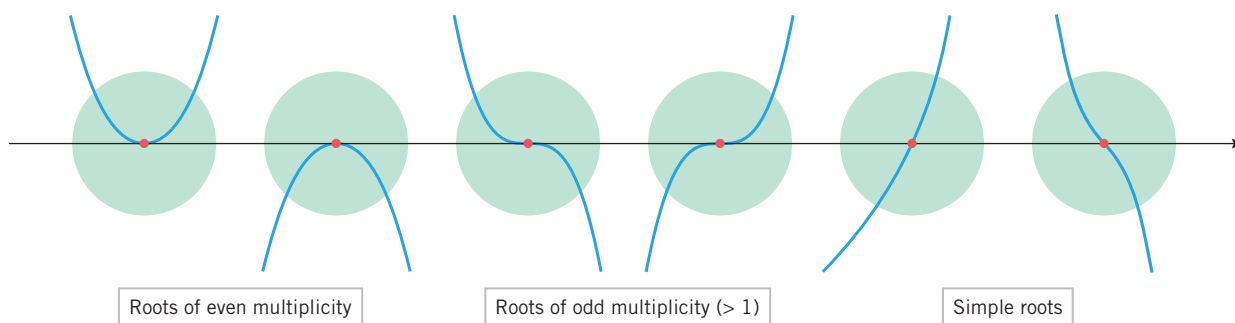
■ GEOMETRIC IMPLICATIONS OF MULTIPLICITY

Our final goal in this section is to outline a general procedure that can be used to analyze and graph polynomials. To do so, it will be helpful to understand how the graph of a polynomial behaves in the vicinity of its roots. For example, it would be nice to know what property of the polynomial in Example 5 produced the inflection point and horizontal tangent at the root $x = 0$.

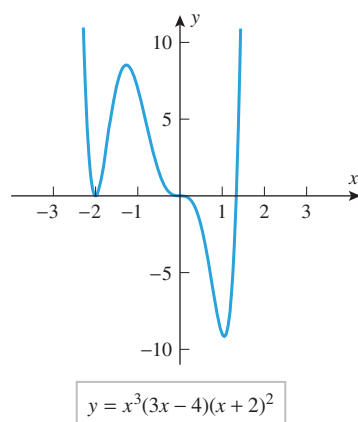
Recall that a root $x = r$ of a polynomial $p(x)$ has **multiplicity m** if $(x - r)^m$ divides $p(x)$ but $(x - r)^{m+1}$ does not. A root of multiplicity 1 is called a **simple root**. Figure 3.2.9 and the following theorem show that the behavior of a polynomial in the vicinity of a real root is determined by the multiplicity of that root (we omit the proof).

3.2.5 THE GEOMETRIC IMPLICATIONS OF MULTIPLICITY Suppose that $p(x)$ is a polynomial with a root of multiplicity m at $x = r$.

- (a) If m is even, then the graph of $y = p(x)$ is tangent to the x -axis at $x = r$, does not cross the x -axis there, and does not have an inflection point there.
- (b) If m is odd and greater than 1, then the graph is tangent to the x -axis at $x = r$, crosses the x -axis there, and also has an inflection point there.
- (c) If $m = 1$ (so that the root is simple), then the graph is not tangent to the x -axis at $x = r$, crosses the x -axis there, and may or may not have an inflection point there.



▲ Figure 3.2.9



▲ Figure 3.2.10

► **Example 6** Make a conjecture about the behavior of the graph of

$$y = x^3(3x - 4)(x + 2)^2$$

in the vicinity of its x -intercepts, and test your conjecture by generating the graph.

Solution. The x -intercepts occur at $x = 0$, $x = \frac{4}{3}$, and $x = -2$. The root $x = 0$ has multiplicity 3, which is odd, so at that point the graph should be tangent to the x -axis, cross the x -axis, and have an inflection point there. The root $x = -2$ has multiplicity 2, which is even, so the graph should be tangent to but not cross the x -axis there. The root $x = \frac{4}{3}$ is simple, so at that point the curve should cross the x -axis without being tangent to it. All of this is consistent with the graph in Figure 3.2.10. ◀

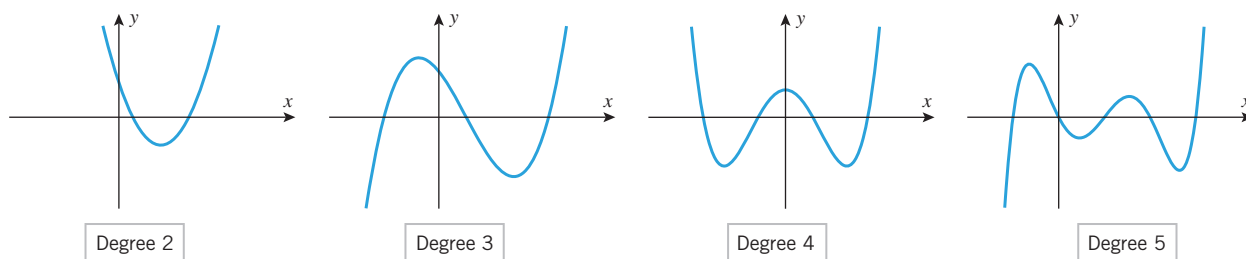
ANALYSIS OF POLYNOMIALS

Historically, the term “curve sketching” meant using calculus to help draw the graph of a function by hand—the graph was the goal. Since graphs can now be produced with great precision using calculators and computers, the purpose of curve sketching has changed. Today, we typically start with a graph produced by a calculator or computer, then use curve sketching to identify important features of the graph that the calculator or computer might have missed. Thus, the goal of curve sketching is no longer the graph itself, but rather the information it reveals about the function.

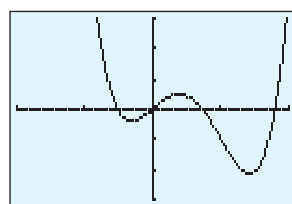
Polynomials are among the simplest functions to graph and analyze. Their significant features are symmetry, intercepts, relative extrema, inflection points, and the behavior as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$. Figure 3.2.11 shows the graphs of four polynomials in x . The graphs in Figure 3.2.11 have properties that are common to all polynomials:

- The natural domain of a polynomial is $(-\infty, +\infty)$.
- Polynomials are continuous everywhere.
- Polynomials are differentiable everywhere, so their graphs have no corners or vertical tangent lines.
- The graph of a nonconstant polynomial eventually increases or decreases without bound as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$. This is because the limit of a nonconstant polynomial as $x \rightarrow +\infty$ or as $x \rightarrow -\infty$ is $\pm\infty$, depending on the sign of the term of highest degree and whether the polynomial has even or odd degree [see Formulas (13) and (14) of Section 1.3 and the related discussion].
- The graph of a polynomial of degree n (> 2) has at most n x -intercepts, at most $n - 1$ relative extrema, and at most $n - 2$ inflection points. This is because the x -intercepts, relative extrema, and inflection points of a polynomial $p(x)$ are among the real solutions of the equations $p(x) = 0$, $p'(x) = 0$, and $p''(x) = 0$, and the polynomials in these equations have degree n , $n - 1$, and $n - 2$, respectively. Thus, for example, the graph of a quadratic polynomial has at most two x -intercepts, one relative extremum, and no inflection points; and the graph of a cubic polynomial has at most three x -intercepts, two relative extrema, and one inflection point.

For each of the graphs in Figure 3.2.11, count the number of x -intercepts, relative extrema, and inflection points, and confirm that your count is consistent with the degree of the polynomial.



▲ Figure 3.2.11



$[-2, 2] \times [-3, 3]$

$$y = 3x^4 - 6x^3 + 2x$$

▲ Figure 3.2.12

► **Example 7** Figure 3.2.12 shows the graph of

$$y = 3x^4 - 6x^3 + 2x$$

produced on a graphing calculator. Confirm that the graph is not missing any significant features.

Solution. We can be confident that the graph shows all significant features of the polynomial because the polynomial has degree 4 and we can account for four roots, three relative extrema, and two inflection points. Moreover, the graph suggests the correct behavior as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$, since

$$\begin{aligned}\lim_{x \rightarrow +\infty} (3x^4 - 6x^3 + 2x) &= \lim_{x \rightarrow +\infty} 3x^4 = +\infty \\ \lim_{x \rightarrow -\infty} (3x^4 - 6x^3 + 2x) &= \lim_{x \rightarrow -\infty} 3x^4 = +\infty \quad \blacktriangleleft\end{aligned}$$

► **Example 8** Sketch the graph of the equation

$$y = x^3 - 3x + 2$$

and identify the locations of the intercepts, relative extrema, and inflection points.

A review of polynomial factoring is given in Appendix C.

Solution. The following analysis will produce the information needed to sketch the graph:

- *x-intercepts:* Factoring the polynomial yields

$$x^3 - 3x + 2 = (x + 2)(x - 1)^2$$

which tells us that the *x*-intercepts are $x = -2$ and $x = 1$.

- *y-intercept:* Setting $x = 0$ yields $y = 2$.
- *End behavior:* We have

$$\lim_{x \rightarrow +\infty} (x^3 - 3x + 2) = \lim_{x \rightarrow +\infty} x^3 = +\infty$$

$$\lim_{x \rightarrow -\infty} (x^3 - 3x + 2) = \lim_{x \rightarrow -\infty} x^3 = -\infty$$

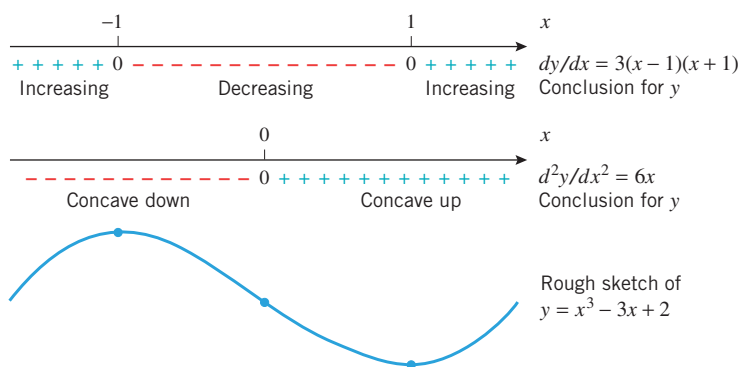
so the graph increases without bound as $x \rightarrow +\infty$ and decreases without bound as $x \rightarrow -\infty$.

- *Derivatives:*

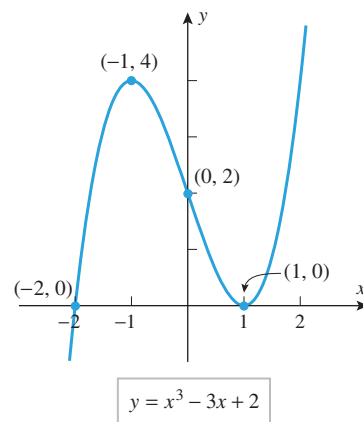
$$\frac{dy}{dx} = 3x^2 - 3 = 3(x - 1)(x + 1)$$

$$\frac{d^2y}{dx^2} = 6x$$

- *Increase, decrease, relative extrema, inflection points:* Figure 3.2.13 gives a sign analysis of the first and second derivatives and indicates its geometric significance. There are stationary points at $x = -1$ and $x = 1$. Since the sign of dy/dx changes from $+$ to $-$ at $x = -1$, there is a relative maximum there, and since it changes from $-$ to $+$ at $x = 1$, there is a relative minimum there. The sign of d^2y/dx^2 changes from $-$ to $+$ at $x = 0$, so there is an inflection point there.
- *Final sketch:* Figure 3.2.14 shows the final sketch with the coordinates of the intercepts, relative extrema, and inflection point labeled. ◀



▲ Figure 3.2.13



▲ Figure 3.2.14

✓ QUICK CHECK EXERCISES 3.2 (See page 207 for answers.)

1. A function f has a relative maximum at x_0 if there is an open interval containing x_0 on which $f(x)$ is _____ $f(x_0)$ for every x in the interval.
2. Suppose that f is defined everywhere and $x = 2, 3, 5, 7$ are critical points for f . If $f'(x)$ is positive on the intervals $(-\infty, 2)$ and $(5, 7)$, and if $f'(x)$ is negative on the intervals $(2, 3)$, $(3, 5)$, and $(7, +\infty)$, then f has relative maxima at $x =$ _____ and f has relative minima at $x =$ _____.
3. Suppose that f is defined everywhere and $x = -2$ and $x = 1$ are critical points for f . If $f''(x) = 2x + 1$, then f has a relative _____ at $x = -2$ and f has a relative _____ at $x = 1$.

4. Let $f(x) = (x^2 - 4)^2$. Then $f'(x) = 4x(x^2 - 4)$ and $f''(x) = 4(3x^2 - 4)$. Identify the locations of the (a) rela-

tive maxima, (b) relative minima, and (c) inflection points on the graph of f .

EXERCISE SET 3.2



Graphing Utility



CAS

FOCUS ON CONCEPTS

- In each part, sketch the graph of a continuous function f with the stated properties.
 - f is concave up on the interval $(-\infty, +\infty)$ and has exactly one relative extremum.
 - f is concave up on the interval $(-\infty, +\infty)$ and has no relative extrema.
 - The function f has exactly two relative extrema on the interval $(-\infty, +\infty)$, and $f(x) \rightarrow +\infty$ as $x \rightarrow +\infty$.
 - The function f has exactly two relative extrema on the interval $(-\infty, +\infty)$, and $f(x) \rightarrow -\infty$ as $x \rightarrow +\infty$.
- In each part, sketch the graph of a continuous function f with the stated properties.
 - f has exactly one relative extremum on $(-\infty, +\infty)$, and $f(x) \rightarrow 0$ as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$.
 - f has exactly two relative extrema on $(-\infty, +\infty)$, and $f(x) \rightarrow 0$ as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$.
 - f has exactly one inflection point and one relative extremum on $(-\infty, +\infty)$.
 - f has infinitely many relative extrema, and $f(x) \rightarrow 0$ as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$.
- Use both the first and second derivative tests to show that $f(x) = 3x^2 - 6x + 1$ has a relative minimum at $x = 1$.
 - Use both the first and second derivative tests to show that $f(x) = x^3 - 3x + 3$ has a relative minimum at $x = 1$ and a relative maximum at $x = -1$.
- Use both the first and second derivative tests to show that $f(x) = \sin^2 x$ has a relative minimum at $x = 0$.
 - Use both the first and second derivative tests to show that $g(x) = \tan^2 x$ has a relative minimum at $x = 0$.
 - Give an informal verbal argument to explain without calculus why the functions in parts (a) and (b) have relative minima at $x = 0$.
- Show that both of the functions $f(x) = (x - 1)^4$ and $g(x) = x^3 - 3x^2 + 3x - 2$ have stationary points at $x = 1$.
 - What does the second derivative test tell you about the nature of these stationary points?
 - What does the first derivative test tell you about the nature of these stationary points?
- Show that $f(x) = 1 - x^5$ and $g(x) = 3x^4 - 8x^3$ both have stationary points at $x = 0$.
 - What does the second derivative test tell you about the nature of these stationary points?

- (c) What does the first derivative test tell you about the nature of these stationary points?

7–14 Locate the critical points and identify which critical points are stationary points. ■

7. $f(x) = 4x^4 - 16x^2 + 17$ 8. $f(x) = 3x^4 + 12x$

9. $f(x) = \frac{x+1}{x^2+3}$ 10. $f(x) = \frac{x^2}{x^3+8}$

11. $f(x) = \sqrt[3]{x^2 - 25}$ 12. $f(x) = x^2(x-1)^{2/3}$

13. $f(x) = |\sin x|$ 14. $f(x) = \sin |x|$

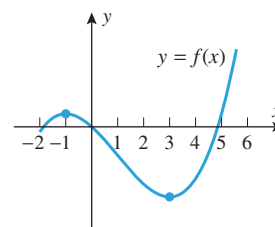
15–18 True-False Assume that f is continuous everywhere. Determine whether the statement is true or false. Explain your answer. ■

- If f has a relative maximum at $x = 1$, then $f(1) \geq f(2)$.
- If f has a relative maximum at $x = 1$, then $x = 1$ is a critical point for f .
- If $f''(x) > 0$, then f has a relative minimum at $x = 1$.
- If $p(x)$ is a polynomial such that $p'(x)$ has a simple root at $x = 1$, then p has a relative extremum at $x = 1$.

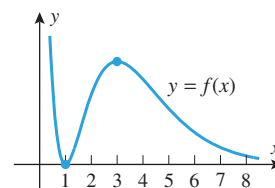
FOCUS ON CONCEPTS

19–20 The graph of a function $f(x)$ is given. Sketch graphs of $y = f'(x)$ and $y = f''(x)$. ■

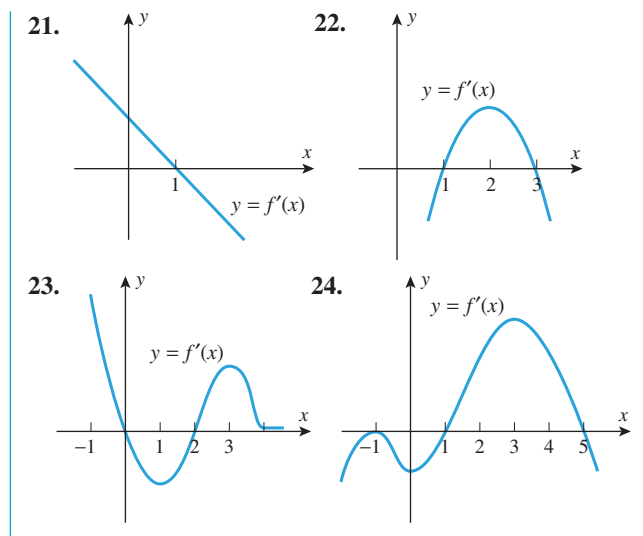
19.



20.



21–24 Use the graph of f' shown in the figure on the next page to estimate all values of x at which f has (a) relative minima, (b) relative maxima, and (c) inflection points. (d) Draw a rough sketch of the graph of a function f with the given derivative. ■



25–28 Use the given derivative to find all critical points of f , and at each critical point determine whether a relative maximum, relative minimum, or neither occurs. Assume in each case that f is continuous everywhere. ■

25. $f'(x) = x^2(x^3 - 5)$ 26. $f'(x) = 4x^3 - 9x$

27. $f'(x) = \frac{2-3x}{\sqrt[3]{x+2}}$ 28. $f'(x) = \frac{x^2-7}{\sqrt[3]{x^2+4}}$

29–32 Find the relative extrema using both first and second derivative tests. ■

29. $f(x) = 1 + 8x - 3x^2$ 30. $f(x) = x^4 - 12x^3$

31. $f(x) = \sin 2x$, $0 < x < \pi$

32. $f(x) = x + \sin 2x$, $0 < x < \pi$

33–42 Use any method to find the relative extrema of the function f . ■

33. $f(x) = x^4 - 4x^3 + 4x^2$ 34. $f(x) = x(x-4)^3$

35. $f(x) = x^3(x+1)^2$ 36. $f(x) = x^2(x+1)^3$

37. $f(x) = 2x + 3x^{2/3}$ 38. $f(x) = 2x + 3x^{1/3}$

39. $f(x) = \frac{x+3}{x-2}$ 40. $f(x) = \frac{x^2}{x^4+16}$

41. $f(x) = |3x - x^2|$ 42. $f(x) = |1 + \sqrt[3]{x}|$

43–52 Give a graph of the polynomial and label the coordinates of the intercepts, stationary points, and inflection points. Check your work with a graphing utility. ■

43. $p(x) = x^2 - 3x - 4$ 44. $p(x) = 1 + 8x - x^2$

45. $p(x) = 2x^3 - 3x^2 - 36x + 5$

46. $p(x) = 2 - x + 2x^2 - x^3$

47. $p(x) = (x+1)^2(2x-x^2)$

48. $p(x) = x^4 - 6x^2 + 5$

49. $p(x) = x^4 - 2x^3 + 2x - 1$ 50. $p(x) = 4x^3 - 9x^4$

51. $p(x) = x(x^2 - 1)^2$ 52. $p(x) = x(x^2 - 1)^3$

- 53.** In each part: (i) Make a conjecture about the behavior of the graph in the vicinity of its x -intercepts. (ii) Make a rough sketch of the graph based on your conjecture and the limits of the polynomial as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$. (iii) Compare your sketch to the graph generated with a graphing utility.
 (a) $y = x(x-1)(x+1)$ (b) $y = x^2(x-1)^2(x+1)^2$
 (c) $y = x^2(x-1)^2(x+1)^3$ (d) $y = x(x-1)^5(x+1)^4$
- 54.** Sketch the graph of $y = (x-a)^m(x-b)^n$ for the stated values of m and n , assuming that $a < b$ (six graphs in total).
 (a) $m = 1$, $n = 1, 2, 3$ (b) $m = 2$, $n = 2, 3$
 (c) $m = 3$, $n = 3$

55–58 Find the relative extrema in the interval $0 < x < 2\pi$, and confirm that your results are consistent with the graph of f generated with a graphing utility. ■

55. $f(x) = |\sin 2x|$ 56. $f(x) = \sqrt{3}x + 2 \sin x$

57. $f(x) = \cos^2 x$ 58. $f(x) = \frac{\sin x}{2 - \cos x}$

59–60 Use a graphing utility to generate the graphs of f' and f'' over the stated interval, and then use those graphs to estimate the x -coordinates of the relative extrema of f . Check that your estimates are consistent with the graph of f . ■

59. $f(x) = x^4 - 24x^2 + 12x$, $-5 \leq x \leq 5$

60. $f(x) = \sin \frac{1}{2}x \cos x$, $-\pi/2 \leq x \leq \pi/2$

61–64 Use a CAS to graph f' and f'' , and then use those graphs to estimate the x -coordinates of the relative extrema of f . Check that your estimates are consistent with the graph of f . ■

61. $f(x) = \frac{10x^3 - 3}{3x^2 - 5x + 8}$ 62. $f(x) = \frac{x^3 - x^2}{x^2 + 1}$

63. $f(x) = \sqrt{x^4 + \cos^2 x}$ 64. $f(x) = \frac{x^3 - 8x + 7}{\sqrt{x^2 + 1}}$

65. In each part, find k so that f has a relative extremum at the point where $x = 3$.

(a) $f(x) = x^2 + \frac{k}{x}$ (b) $f(x) = \frac{x}{x^2 + k}$

66. (a) Use a CAS to graph the function

$$f(x) = \frac{x^4 + 1}{x^2 + 1}$$

and use the graph to estimate the x -coordinates of the relative extrema.

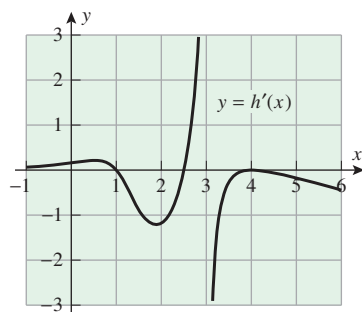
(b) Find the exact x -coordinates by using the CAS to solve the equation $f'(x) = 0$.

67. Let h and g have relative maxima at x_0 . Prove or disprove:

(a) $h + g$ has a relative maximum at x_0

(b) $h - g$ has a relative maximum at x_0 .

68. Sketch some curves that show that the three parts of the first derivative test (Theorem 3.2.3) can be false without the assumption that f is continuous at x_0 .



◀ Figure Ex-45

46. Let $f(x) = (1 - 2x)h(x)$, where $h(x)$ is as given in Exercise 45. Suppose that $x = 5$ is a critical point for $f(x)$.

- Estimate $h(5)$.
- Use the second derivative test to determine whether $f(x)$ has a relative maximum or a relative minimum at $x = 5$.

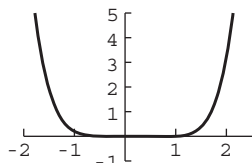
47. A rectangular plot of land is to be fenced off so that the area enclosed will be 400 ft^2 . Let L be the length of fencing needed and x the length of one side of the rectangle. Show that $L = 2x + 800/x$ for $x > 0$, and sketch the graph of L versus x for $x > 0$.

48. A box with a square base and open top is to be made from sheet metal so that its volume is 500 in^3 . Let S be the area of the surface of the box and x the length of a side of the square base. Show that $S = x^2 + 2000/x$ for $x > 0$, and sketch the graph of S versus x for $x > 0$.

49. The accompanying figure shows a computer-generated graph of the polynomial $y = 0.1x^5(x - 1)$ using a view-

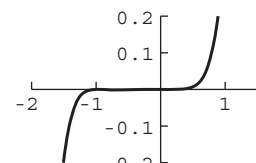
ing window of $[-2, 2.5] \times [-1, 5]$. Show that the choice of the vertical scale caused the computer to miss important features of the graph. Find the features that were missed and make your own sketch of the graph that shows the missing features.

50. The accompanying figure shows a computer-generated graph of the polynomial $y = 0.1x^5(x + 1)^2$ using a viewing window of $[-2, 1.5] \times [-0.2, 0.2]$. Show that the choice of the vertical scale caused the computer to miss important features of the graph. Find the features that were missed and make your own sketch of the graph that shows the missing features.



Generated by Mathematica

▲ Figure Ex-49



Generated by Mathematica

▲ Figure Ex-50

51. **Writing** Suppose that $x = x_0$ is a point at which a function f is continuous but not differentiable and that $f'(x)$ approaches different finite limits as x approaches x_0 from either side. Invent your own term to describe the graph of f at such a point and discuss the appropriateness of your term.
52. **Writing** Suppose that the graph of a function f is obtained using a graphing utility. Discuss the information that calculus techniques can provide about f to add to what can already be inferred about f from the graph as shown on your utility's display.

✓ QUICK CHECK ANSWERS 3.3

1. (a) $(-1, 0)$, $(3, 0)$, $(0, \frac{9}{8})$ (b) $x = -2$ and $x = 4$ (c) $y = 3$ (d) $(-\infty, -2)$, $(-1, 3)$, and $(4, +\infty)$ (e) $(-\infty, -2)$ and $(-2, 1]$ (f) $(-\infty, -2)$ and $(4, +\infty)$ (g) $(1, \frac{4}{3})$ 2. (a) $(-2, 0)$, $(2, 0)$ (b) $x = 0$ (c) $y = 0$ (d) $(-\infty, -2)$ and $(2, +\infty)$ (e) $(-\infty, -4]$ and $(0, 4]$ (f) $(-\infty, -4\sqrt{11/5})$ and $(4\sqrt{11/5}, +\infty)$ (g) $\pm 4\sqrt{11/5} \approx \pm 5.93$

3.4 ABSOLUTE MAXIMA AND MINIMA

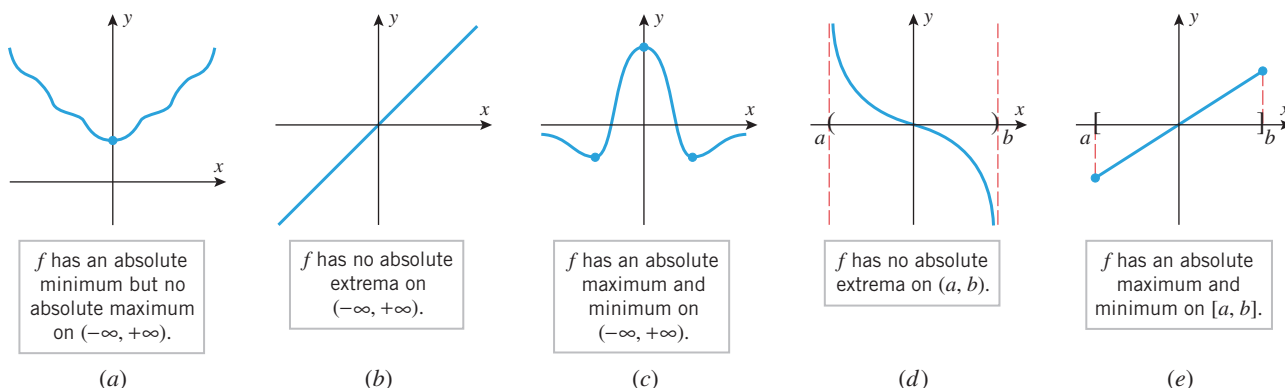
At the beginning of Section 3.2 we observed that if the graph of a function f is viewed as a two-dimensional mountain range (Figure 3.2.1), then the relative maxima and minima correspond to the tops of the hills and the bottoms of the valleys; that is, they are the high and low points in their immediate vicinity. In this section we will be concerned with the more encompassing problem of finding the highest and lowest points over the entire mountain range, that is, we will be looking for the top of the highest hill and the bottom of the deepest valley. In mathematical terms, we will be looking for the largest and smallest values of a function over an interval.

■ ABSOLUTE EXTREMA

We will begin with some terminology for describing the largest and smallest values of a function on an interval.

3.4.1 DEFINITION Consider an interval in the domain of a function f and a point x_0 in that interval. We say that f has an **absolute maximum** at x_0 if $f(x) \leq f(x_0)$ for all x in the interval, and we say that f has an **absolute minimum** at x_0 if $f(x_0) \leq f(x)$ for all x in the interval. We say that f has an **absolute extremum** at x_0 if it has either an absolute maximum or an absolute minimum at that point.

If f has an absolute maximum at the point x_0 on an interval, then $f(x_0)$ is the largest value of f on the interval, and if f has an absolute minimum at x_0 , then $f(x_0)$ is the smallest value of f on the interval. In general, there is no guarantee that a function will actually have an absolute maximum or minimum on a given interval (Figure 3.4.1).



▲ Figure 3.4.1

THE EXTREME VALUE THEOREM

Parts (a)–(d) of Figure 3.4.1 show that a continuous function may or may not have absolute maxima or minima on an infinite interval or on a finite open interval. However, the following theorem shows that a continuous function must have both an absolute maximum and an absolute minimum on every *finite closed* interval [see part (e) of Figure 3.4.1].

The hypotheses in the Extreme-Value Theorem are essential. That is, if either the interval is not closed or f is not continuous on the interval, then f need not have absolute extrema on the interval (Exercises 4–6).

3.4.2 THEOREM (Extreme-Value Theorem) If a function f is continuous on a finite closed interval $[a, b]$, then f has both an absolute maximum and an absolute minimum on $[a, b]$.

REMARK

Although the proof of this theorem is too difficult to include here, you should be able to convince yourself of its validity with a little experimentation—try graphing various continuous functions over the interval $[0, 1]$, and convince yourself that there is no way to avoid having a highest and lowest point on a graph. As a physical analogy, if you imagine the graph to be a roller-coaster track starting at $x = 0$ and ending at $x = 1$, the roller coaster will have to pass through a highest point and a lowest point during the trip.

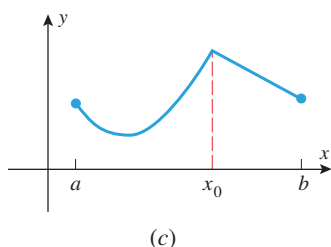
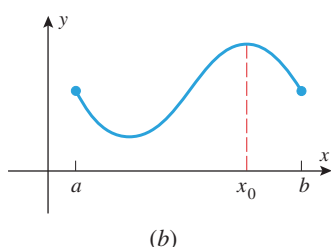
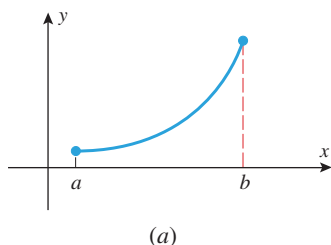
The Extreme-Value Theorem is an example of what mathematicians call an **existence theorem**. Such theorems state conditions under which certain objects exist, in this case absolute extrema. However, knowing that an object exists and finding it are two separate things. We will now address methods for determining the locations of absolute extrema under the conditions of the Extreme-Value Theorem.

If f is continuous on the finite closed interval $[a, b]$, then the absolute extrema of f occur either at the endpoints of the interval or inside on the open interval (a, b) . If the

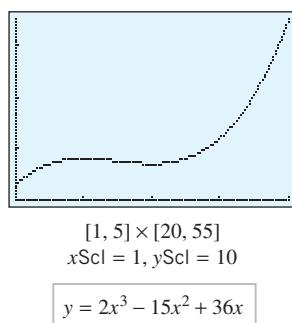
absolute extrema happen to fall inside, then the following theorem tells us that they must occur at critical points of f .

Theorem 3.4.3 is also valid on infinite open intervals, that is, intervals of the form $(-\infty, +\infty)$, $(a, +\infty)$, and $(-\infty, b)$.

3.4.3 THEOREM If f has an absolute extremum on an open interval (a, b) , then it must occur at a critical point of f .



▲ **Figure 3.4.2** In part (a) the absolute maximum occurs at an endpoint of $[a, b]$, in part (b) it occurs at a stationary point in (a, b) , and in part (c) it occurs at a critical point in (a, b) where f is not differentiable.



▲ **Figure 3.4.3**

PROOF If f has an absolute maximum on (a, b) at x_0 , then $f(x_0)$ is also a relative maximum for f ; for if $f(x_0)$ is the largest value of f on all (a, b) , then $f(x_0)$ is certainly the largest value for f in the immediate vicinity of x_0 . Thus, x_0 is a critical point of f by Theorem 3.2.2. The proof for absolute minima is similar. ■

It follows from this theorem that if f is continuous on the finite closed interval $[a, b]$, then the absolute extrema occur either at the endpoints of the interval or at critical points inside the interval (Figure 3.4.2). Thus, we can use the following procedure to find the absolute extrema of a continuous function on a finite closed interval $[a, b]$.

A Procedure for Finding the Absolute Extrema of a Continuous Function f on a Finite Closed Interval $[a, b]$

Step 1. Find the critical points of f in (a, b) .

Step 2. Evaluate f at all the critical points and at the endpoints a and b .

Step 3. The largest of the values in Step 2 is the absolute maximum value of f on $[a, b]$ and the smallest value is the absolute minimum.

► **Example 1** Find the absolute maximum and minimum values of the function $f(x) = 2x^3 - 15x^2 + 36x$ on the interval $[1, 5]$, and determine where these values occur.

Solution. Since f is continuous and differentiable everywhere, the absolute extrema must occur either at endpoints of the interval or at solutions to the equation $f'(x) = 0$ in the open interval $(1, 5)$. The equation $f'(x) = 0$ can be written as

$$6x^2 - 30x + 36 = 6(x^2 - 5x + 6) = 6(x - 2)(x - 3) = 0$$

Thus, there are stationary points at $x = 2$ and at $x = 3$. Evaluating f at the endpoints, at $x = 2$, and at $x = 3$ yields

$$f(1) = 2(1)^3 - 15(1)^2 + 36(1) = 23$$

$$f(2) = 2(2)^3 - 15(2)^2 + 36(2) = 28$$

$$f(3) = 2(3)^3 - 15(3)^2 + 36(3) = 27$$

$$f(5) = 2(5)^3 - 15(5)^2 + 36(5) = 55$$

from which we conclude that the absolute minimum of f on $[1, 5]$ is 23, occurring at $x = 1$, and the absolute maximum of f on $[1, 5]$ is 55, occurring at $x = 5$. This is consistent with the graph of f in Figure 3.4.3. ◀

► **Example 2** Find the absolute extrema of $f(x) = 6x^{4/3} - 3x^{1/3}$ on the interval $[-1, 1]$, and determine where these values occur.

Solution. Note that f is continuous everywhere and therefore the Extreme-Value Theorem guarantees that f has a maximum and a minimum value in the interval $[-1, 1]$. Differentiating, we obtain

$$f'(x) = 8x^{1/3} - x^{-2/3} = x^{-2/3}(8x - 1) = \frac{8x - 1}{x^{2/3}}$$

Thus, $f'(x) = 0$ at $x = \frac{1}{8}$, and $f'(x)$ is undefined at $x = 0$. Evaluating f at these critical points and endpoints yields Table 3.4.1, from which we conclude that an absolute minimum value of $-\frac{9}{8}$ occurs at $x = \frac{1}{8}$, and an absolute maximum value of 9 occurs at $x = -1$. ◀

Table 3.4.1

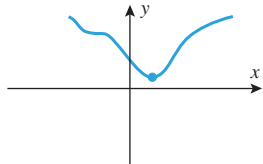
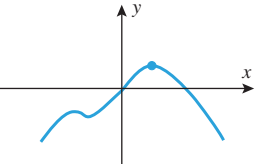
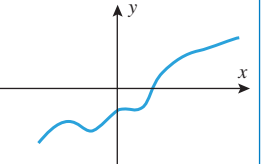
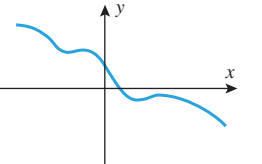
x	-1	0	$\frac{1}{8}$	1
$f(x)$	9	0	$-\frac{9}{8}$	3

■ ABSOLUTE EXTREMA ON INFINITE INTERVALS

We observed earlier that a continuous function may or may not have absolute extrema on an infinite interval (see Figure 3.4.1). However, certain conclusions about the existence of absolute extrema of a continuous function f on $(-\infty, +\infty)$ can be drawn from the behavior of $f(x)$ as $x \rightarrow -\infty$ and as $x \rightarrow +\infty$ (Table 3.4.2).

Table 3.4.2

ABSOLUTE EXTREMA ON INFINITE INTERVALS

LIMITS	$\lim_{x \rightarrow -\infty} f(x) = +\infty$ $\lim_{x \rightarrow +\infty} f(x) = +\infty$	$\lim_{x \rightarrow -\infty} f(x) = -\infty$ $\lim_{x \rightarrow +\infty} f(x) = -\infty$	$\lim_{x \rightarrow -\infty} f(x) = -\infty$ $\lim_{x \rightarrow +\infty} f(x) = +\infty$	$\lim_{x \rightarrow -\infty} f(x) = +\infty$ $\lim_{x \rightarrow +\infty} f(x) = -\infty$
CONCLUSION IF f IS CONTINUOUS EVERYWHERE	f has an absolute minimum but no absolute maximum on $(-\infty, +\infty)$.	f has an absolute maximum but no absolute minimum on $(-\infty, +\infty)$.	f has neither an absolute maximum nor an absolute minimum on $(-\infty, +\infty)$.	f has neither an absolute maximum nor an absolute minimum on $(-\infty, +\infty)$.
GRAPH				

► **Example 3** What can you say about the existence of absolute extrema on $(-\infty, +\infty)$ for polynomials?

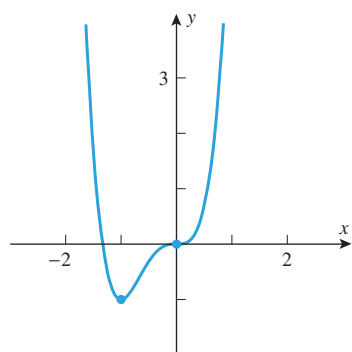
Solution. If $p(x)$ is a polynomial of odd degree, then

$$\lim_{x \rightarrow +\infty} p(x) \quad \text{and} \quad \lim_{x \rightarrow -\infty} p(x) \quad (1)$$

have opposite signs (one is $+\infty$ and the other is $-\infty$), so there are no absolute extrema. On the other hand, if $p(x)$ has even degree, then the limits in (1) have the same sign (both $+\infty$ or both $-\infty$). If the leading coefficient is positive, then both limits are $+\infty$, and there is an absolute minimum but no absolute maximum; if the leading coefficient is negative, then both limits are $-\infty$, and there is an absolute maximum but no absolute minimum. ◀

► **Example 4** Determine by inspection whether $p(x) = 3x^4 + 4x^3$ has any absolute extrema. If so, find them and state where they occur.

Solution. Since $p(x)$ has even degree and the leading coefficient is positive, $p(x) \rightarrow +\infty$ as $x \rightarrow \pm\infty$. Thus, there is an absolute minimum but no absolute maximum. From Theorem 3.4.3 [applied to the interval $(-\infty, +\infty)$], the absolute minimum must occur at a critical



$$p(x) = 3x^4 + 4x^3$$

▲ Figure 3.4.4

point of p . Since p is differentiable everywhere, we can find all critical points by solving the equation $p'(x) = 0$. This equation is

$$12x^3 + 12x^2 = 12x^2(x + 1) = 0$$

from which we conclude that the critical points are $x = 0$ and $x = -1$. Evaluating p at these critical points yields

$$p(0) = 0 \quad \text{and} \quad p(-1) = -1$$

Therefore, p has an absolute minimum of -1 at $x = -1$ (Figure 3.4.4). ◀

■ ABSOLUTE EXTREMA ON OPEN INTERVALS

We know that a continuous function may or may not have absolute extrema on an open interval. However, certain conclusions about the existence of absolute extrema of a continuous function f on a finite open interval (a, b) can be drawn from the behavior of $f(x)$ as $x \rightarrow a^+$ and as $x \rightarrow b^-$ (Table 3.4.3). Similar conclusions can be drawn for intervals of the form $(-\infty, b)$ or $(a, +\infty)$.

Table 3.4.3

ABSOLUTE EXTREMA ON OPEN INTERVALS

LIMITS	$\lim_{x \rightarrow a^+} f(x) = +\infty$ $\lim_{x \rightarrow b^-} f(x) = +\infty$	$\lim_{x \rightarrow a^+} f(x) = -\infty$ $\lim_{x \rightarrow b^-} f(x) = -\infty$	$\lim_{x \rightarrow a^+} f(x) = -\infty$ $\lim_{x \rightarrow b^-} f(x) = +\infty$	$\lim_{x \rightarrow a^+} f(x) = +\infty$ $\lim_{x \rightarrow b^-} f(x) = -\infty$
CONCLUSION IF f IS CONTINUOUS ON (a, b)	f has an absolute minimum but no absolute maximum on (a, b) .	f has an absolute maximum but no absolute minimum on (a, b) .	f has neither an absolute maximum nor an absolute minimum on (a, b) .	f has neither an absolute maximum nor an absolute minimum on (a, b) .
GRAPH				

► **Example 5** Determine whether the function

$$f(x) = \frac{1}{x^2 - x}$$

has any absolute extrema on the interval $(0, 1)$. If so, find them and state where they occur.

Solution. Since f is continuous on the interval $(0, 1)$ and

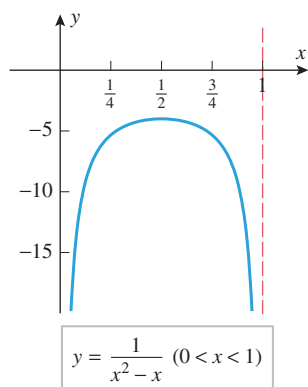
$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{1}{x^2 - x} = \lim_{x \rightarrow 0^+} \frac{1}{x(x - 1)} = -\infty$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \frac{1}{x^2 - x} = \lim_{x \rightarrow 1^-} \frac{1}{x(x - 1)} = -\infty$$

the function f has an absolute maximum but no absolute minimum on the interval $(0, 1)$. By Theorem 3.4.3 the absolute maximum must occur at a critical point of f in the interval $(0, 1)$. We have

$$f'(x) = -\frac{2x - 1}{(x^2 - x)^2}$$

so the only solution of the equation $f'(x) = 0$ is $x = \frac{1}{2}$. Although f is not differentiable at $x = 0$ or at $x = 1$, these values are doubly disqualified since they are neither in the domain



▲ Figure 3.4.5

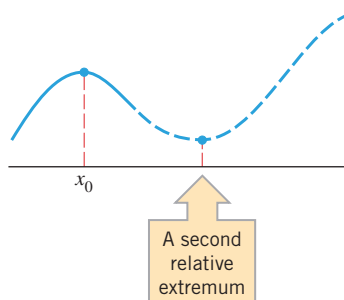
of f nor in the interval $(0, 1)$. Thus, the absolute maximum occurs at $x = \frac{1}{2}$, and this absolute maximum is

$$f\left(\frac{1}{2}\right) = \frac{1}{\left(\frac{1}{2}\right)^2 - \frac{1}{2}} = -4$$

(Figure 3.4.5). ◀

■ ABSOLUTE EXTREMA OF FUNCTIONS WITH ONE RELATIVE EXTREMUM

If a continuous function has only one relative extremum on a finite or infinite interval, then that relative extremum must of necessity also be an absolute extremum. To understand why this is so, suppose that f has a relative maximum at x_0 in an interval, and there are no other relative extrema of f on the interval. If $f(x_0)$ is *not* the absolute maximum of f on the interval, then the graph of f has to make an upward turn somewhere on the interval to rise above $f(x_0)$. However, this cannot happen because in the process of making an upward turn it would produce a second relative extremum (Figure 3.4.6). Thus, $f(x_0)$ must be the absolute maximum as well as a relative maximum. This idea is captured in the following theorem, which we state without proof.



▲ Figure 3.4.6

3.4.4 THEOREM Suppose that f is continuous and has exactly one relative extremum on an interval, say at x_0 .

- (a) If f has a relative minimum at x_0 , then $f(x_0)$ is the absolute minimum of f on the interval.
- (b) If f has a relative maximum at x_0 , then $f(x_0)$ is the absolute maximum of f on the interval.

This theorem is often helpful in situations where other methods are difficult or tedious to apply.

► **Example 6** Find the absolute extrema, if any, of the function $f(x) = x^3 - 3x^2 + 4$ on the interval $(0, +\infty)$.

Solution. We have

$$\lim_{x \rightarrow +\infty} f(x) = +\infty$$

(verify), so f does not have an absolute maximum on the interval $(0, +\infty)$. However, the continuity of f together with the fact that

$$\lim_{x \rightarrow 0^+} f(x) = f(0) = 4$$

is finite allow for the possibility that f has an absolute minimum on $(0, +\infty)$. If so, it would have to occur at a critical point of f , so we consider

$$f'(x) = 3x^2 - 6x = 3x(x - 2)$$

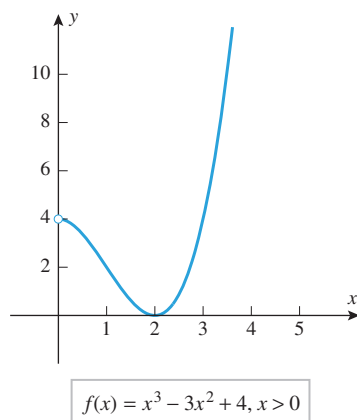
We see that $x = 0$ and $x = 2$ are the critical points of f . Of these, only $x = 2$ is in the interval $(0, +\infty)$, so this is the point at which an absolute minimum could occur. To see whether an absolute minimum actually does occur at this point, we can apply part (a) of Theorem 3.4.4. Since

$$f''(x) = 6x - 6$$

we have

$$f''(2) = 6 > 0$$

so a relative minimum occurs at $x = 2$ by the second derivative test. Thus, f has an absolute minimum at $x = 2$, and this absolute minimum is $f(2) = 0$ (Figure 3.4.7). ◀

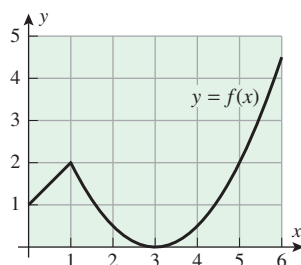


▲ Figure 3.4.7

Does the function in Example 6 have an absolute minimum on the interval $(-\infty, +\infty)$?

QUICK CHECK EXERCISES 3.4 (See page 224 for answers.)

1. Use the accompanying graph to find the x -coordinates of the relative extrema and absolute extrema of f on $[0, 6]$.



◀ Figure Ex-1

2. Suppose that a function f is continuous on $[-4, 4]$ and has critical points at $x = -3, 0, 2$. Use the accompanying table

to determine the absolute maximum and absolute minimum values, if any, for f on the indicated intervals.

- (a) $[1, 4]$ (b) $[-2, 2]$ (c) $[-4, 4]$ (d) $(-4, 4)$

x	-4	-3	-2	-1	0	1	2	3	4
$f(x)$	2224	-1333	0	1603	2096	2293	2400	2717	6064

3. Let $f(x) = x^3 - 3x^2 - 9x + 25$. Use the derivative $f'(x) = 3(x + 1)(x - 3)$ to determine the absolute maximum and absolute minimum values, if any, for f on each of the given intervals.
- (a) $[0, 4]$ (b) $[-2, 4]$ (c) $[-4, 2]$
 (d) $[-5, 10]$ (e) $(-5, 4)$

EXERCISE SET 3.4



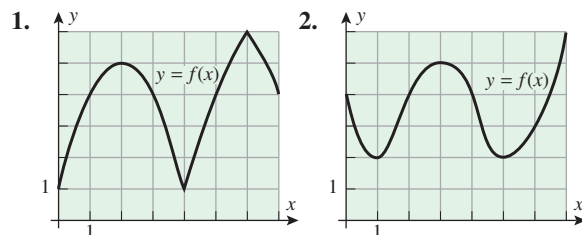
Graphing Utility



CAS

FOCUS ON CONCEPTS

1–2 Use the graph to find x -coordinates of the relative extrema and absolute extrema of f on $[0, 7]$. ■



3. In each part, sketch the graph of a continuous function f with the stated properties on the interval $[0, 10]$.

- (a) f has an absolute minimum at $x = 0$ and an absolute maximum at $x = 10$.
 (b) f has an absolute minimum at $x = 2$ and an absolute maximum at $x = 7$.
 (c) f has relative minima at $x = 1$ and $x = 8$, has relative maxima at $x = 3$ and $x = 7$, has an absolute minimum at $x = 5$, and has an absolute maximum at $x = 10$.

4. In each part, sketch the graph of a continuous function f with the stated properties on the interval $(-\infty, +\infty)$.

- (a) f has no relative extrema or absolute extrema.
 (b) f has an absolute minimum at $x = 0$ but no absolute maximum.
 (c) f has an absolute maximum at $x = -5$ and an absolute minimum at $x = 5$.

5. Let

$$f(x) = \begin{cases} \frac{1}{1-x}, & 0 \leq x < 1 \\ 0, & x = 1 \end{cases}$$

Explain why f has a minimum value but no maximum value on the closed interval $[0, 1]$.

6. Let

$$f(x) = \begin{cases} x, & 0 < x < 1 \\ \frac{1}{2}, & x = 0, 1 \end{cases}$$

Explain why f has neither a minimum value nor a maximum value on the closed interval $[0, 1]$.

7–16 Find the absolute maximum and minimum values of f on the given closed interval, and state where those values occur. ■

7. $f(x) = 4x^2 - 12x + 10$; $[1, 2]$
 8. $f(x) = 8x - x^2$; $[0, 6]$
 9. $f(x) = (x - 2)^3$; $[1, 4]$
 10. $f(x) = 2x^3 + 3x^2 - 12x$; $[-3, 2]$
 11. $f(x) = \frac{3x}{\sqrt{4x^2 + 1}}$; $[-1, 1]$
 12. $f(x) = (x^2 + x)^{2/3}$; $[-2, 3]$
 13. $f(x) = x - 2 \sin x$; $[-\pi/4, \pi/2]$
 14. $f(x) = \sin x - \cos x$; $[0, \pi]$
 15. $f(x) = 1 + |9 - x^2|$; $[-5, 1]$
 16. $f(x) = |6 - 4x|$; $[-3, 3]$

17–20 True–False Determine whether the statement is true or false. Explain your answer. ■

17. If a function f is continuous on $[a, b]$, then f has an absolute maximum on $[a, b]$.
 18. If a function f is continuous on (a, b) , then f has an absolute minimum on (a, b) .
 19. If a function f has an absolute minimum on (a, b) , then there is a critical point of f in (a, b) .
 20. If a function f is continuous on $[a, b]$ and f has no relative extreme values in (a, b) , then the absolute maximum value of f exists and occurs either at $x = a$ or at $x = b$.

21–28 Find the absolute maximum and minimum values of f , if any, on the given interval, and state where those values occur. ■

21. $f(x) = x^2 - x - 2$; $(-\infty, +\infty)$
22. $f(x) = 3 - 4x - 2x^2$; $(-\infty, +\infty)$
23. $f(x) = 4x^3 - 3x^4$; $(-\infty, +\infty)$
24. $f(x) = x^4 + 4x$; $(-\infty, +\infty)$
25. $f(x) = 2x^3 - 6x + 2$; $(-\infty, +\infty)$
26. $f(x) = x^3 - 9x + 1$; $(-\infty, +\infty)$
27. $f(x) = \frac{x^2 + 1}{x + 1}$; $(-5, -1)$
28. $f(x) = \frac{x - 2}{x + 1}$; $(-1, 5)$

 **29–38** Use a graphing utility to estimate the absolute maximum and minimum values of f , if any, on the stated interval, and then use calculus methods to find the exact values. ■

29. $f(x) = (x^2 - 2x)^2$; $(-\infty, +\infty)$
30. $f(x) = (x - 1)^2(x + 2)^2$; $(-\infty, +\infty)$
31. $f(x) = x^{2/3}(20 - x)$; $[-1, 20]$
32. $f(x) = \frac{x}{x^2 + 2}$; $[-1, 4]$
33. $f(x) = 1 + \frac{1}{x}$; $(0, +\infty)$
34. $f(x) = \frac{2x^2 - 3x + 3}{x^2 - 2x + 2}$; $[1, +\infty)$
35. $f(x) = \frac{2 - \cos x}{\sin x}$; $[\pi/4, 3\pi/4]$
36. $f(x) = \sin^2 x + \cos x$; $[-\pi, \pi]$
37. $f(x) = \sin(\cos x)$; $[0, 2\pi]$
38. $f(x) = \cos(\sin x)$; $[0, \pi]$
39. Find the absolute maximum and minimum values of

$$f(x) = \begin{cases} 4x - 2, & x < 1 \\ (x - 2)(x - 3), & x \geq 1 \end{cases}$$

on $[\frac{1}{2}, \frac{7}{2}]$.

40. Let $f(x) = x^2 + px + q$. Find the values of p and q such that $f(1) = 3$ is an extreme value of f on $[0, 2]$. Is this value a maximum or minimum?

41–42 If f is a periodic function, then the locations of all absolute extrema on the interval $(-\infty, +\infty)$ can be obtained by finding the locations of the absolute extrema for one period and using the periodicity to locate the rest. Use this idea in these exercises to find the absolute maximum and minimum values of the function, and state the x -values at which they occur. ■

41. $f(x) = 2 \cos x + \cos 2x$
42. $f(x) = 3 \cos \frac{x}{3} + 2 \cos \frac{x}{2}$

43–44 One way of proving that $f(x) \leq g(x)$ for all x in a given interval is to show that $0 \leq g(x) - f(x)$ for all x in the interval; and one way of proving the latter inequality is to show that the absolute minimum value of $g(x) - f(x)$ on the interval is nonnegative. Use this idea to prove the inequalities in these exercises. ■

43. Prove that $\sin x \leq x$ for all x in the interval $[0, 2\pi]$.
44. Prove that $\cos x \geq 1 - (x^2/2)$ for all x in the interval $[0, 2\pi]$.
45. What is the smallest possible slope for a tangent to the graph of the equation $y = x^3 - 3x^2 + 5x$?
46. (a) Show that $f(x) = \sec x + \csc x$ has a minimum value but no maximum value on the interval $(0, \pi/2)$.
(b) Find the minimum value in part (a).

-  **47.** Show that the absolute minimum value of

$$f(x) = x^2 + \frac{x^2}{(8 - x)^2}, \quad x > 8$$

occurs at $x = 10$ by using a CAS to find $f'(x)$ and to solve the equation $f'(x) = 0$.

-  **48.** The vertical displacement $f(t)$ of a cork bobbing up and down on the ocean's surface may be modeled by the function

$$f(t) = A \cos t + B \sin t$$

where $A > 0$ and $B > 0$. Use a CAS to find the maximum and minimum values of $f(t)$ in terms of A and B .

- 49.** Suppose that the equations of motion of a paper airplane during the first 12 seconds of flight are

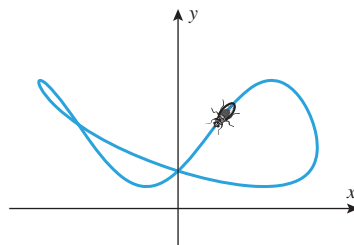
$$x = t - 2 \sin t, \quad y = 2 - 2 \cos t \quad (0 \leq t \leq 12)$$

What are the highest and lowest points in the trajectory, and when is the airplane at those points?

- 50.** The accompanying figure shows the path of a fly whose equations of motion are

$$x = \frac{\cos t}{2 + \sin t}, \quad y = 3 + \sin(2t) - 2 \sin^2 t \quad (0 \leq t \leq 2\pi)$$

- (a) How high and low does it fly?
- (b) How far left and right of the origin does it fly?



◀ Figure Ex-50

51. Let $f(x) = ax^2 + bx + c$, where $a > 0$. Prove that $f(x) \geq 0$ for all x if and only if $b^2 - 4ac \leq 0$. [Hint: Find the minimum of $f(x)$.]
52. Prove Theorem 3.4.3 in the case where the extreme value is a minimum.
53. **Writing** Suppose that f is continuous and positive-valued everywhere and that the x -axis is an asymptote for the graph

4

INTEGRATION



Jon Ferrey/Allsport/Getty Images

If a dragster moves with varying velocity over a certain time interval, it is possible to find the distance it travels during that time interval using techniques of calculus.

In this chapter we will begin with an overview of the problem of finding areas—we will discuss what the term “area” means, and we will outline two approaches to defining and calculating areas. Following this overview, we will discuss the Fundamental Theorem of Calculus, which is the theorem that relates the problems of finding tangent lines and areas, and we will discuss techniques for calculating areas. We will then use the ideas in this chapter to define the average value of a function, to continue our study of rectilinear motion, and to examine some consequences of the chain rule in integral calculus.

4.1 AN OVERVIEW OF THE AREA PROBLEM

In this introductory section we will consider the problem of calculating areas of plane regions with curvilinear boundaries. All of the results in this section will be reexamined in more detail later in this chapter. Our purpose here is simply to introduce and motivate the fundamental concepts.

■ THE AREA PROBLEM

Formulas for the areas of polygons, such as squares, rectangles, triangles, and trapezoids, were well known in many early civilizations. However, the problem of finding formulas for regions with curved boundaries (a circle being the simplest example) caused difficulties for early mathematicians.

The first real progress in dealing with the general area problem was made by the Greek mathematician Archimedes, who obtained areas of regions bounded by circular arcs, parabolas, spirals, and various other curves using an ingenious procedure that was later called the *method of exhaustion*. The method, when applied to a circle, consists of inscribing a succession of regular polygons in the circle and allowing the number of sides to increase indefinitely (Figure 4.1.1). As the number of sides increases, the polygons tend to “exhaust” the region inside the circle, and the areas of the polygons become better and better approximations of the exact area of the circle.

To see how this works numerically, let $A(n)$ denote the area of a regular n -sided polygon inscribed in a circle of radius 1. Table 4.1.1 shows the values of $A(n)$ for various choices of n . Note that for large values of n the area $A(n)$ appears to be close to π (square units),

FOCUS ON CONCEPTS

19. Explain how to use the formula for $A(x)$ found in the solution to Example 2 to determine the area between the graph of $y = x^2$ and the interval $[3, 6]$.
20. Repeat Exercise 19 for the interval $[-3, 9]$.
21. Let A denote the area between the graph of $f(x) = \sqrt{x}$ and the interval $[0, 1]$, and let B denote the area between the graph of $f(x) = x^2$ and the interval $[0, 1]$. Explain geometrically why $A + B = 1$.
22. Let A denote the area between the graph of $f(x) = 1/x$ and the interval $[1, 2]$, and let B denote the area between the graph of f and the interval $[\frac{1}{2}, 1]$. Explain geometrically why $A = B$.

23–24 The area $A(x)$ under the graph of f and over the interval $[a, x]$ is given. Find the function f and the value of a . ■

23. $A(x) = x^2 - 4$

24. $A(x) = x^2 - x$

25. Writing Compare and contrast the rectangle method and the antiderivative method.

26. Writing Suppose that f is a nonnegative continuous function on an interval $[a, b]$ and that $g(x) = f(x) + C$, where C is a positive constant. What will be the area of the region between the graphs of f and g ?

✓ QUICK CHECK ANSWERS 4.1

1. (a) $\frac{\pi}{2}$ (b) $1 + \frac{\sqrt{3}}{2}$ 2. 2 3. 9 4. $A(x) = \frac{x^2}{2}$; $A'(x) = \frac{2x}{2} = x = f(x)$ 5. $\cos x + 1$

4.2 THE INDEFINITE INTEGRAL

In the last section we saw how antidifferentiation could be used to find exact areas. In this section we will develop some fundamental results about antidifferentiation.

■ ANTIDERIVATIVES

4.2.1 DEFINITION A function F is called an **antiderivative** of a function f on a given open interval if $F'(x) = f(x)$ for all x in the interval.

For example, the function $F(x) = \frac{1}{3}x^3$ is an antiderivative of $f(x) = x^2$ on the interval $(-\infty, +\infty)$ because for each x in this interval

$$F'(x) = \frac{d}{dx} \left[\frac{1}{3}x^3 \right] = x^2 = f(x)$$

However, $F(x) = \frac{1}{3}x^3$ is not the only antiderivative of f on this interval. If we add any constant C to $\frac{1}{3}x^3$, then the function $G(x) = \frac{1}{3}x^3 + C$ is also an antiderivative of f on $(-\infty, +\infty)$, since

$$G'(x) = \frac{d}{dx} \left[\frac{1}{3}x^3 + C \right] = x^2 + 0 = f(x)$$

In general, once any single antiderivative is known, other antiderivatives can be obtained by adding constants to the known antiderivative. Thus,

$$\frac{1}{3}x^3, \quad \frac{1}{3}x^3 + 2, \quad \frac{1}{3}x^3 - 5, \quad \frac{1}{3}x^3 + \sqrt{2}$$

are all antiderivatives of $f(x) = x^2$.

It is reasonable to ask if there are antiderivatives of a function f that cannot be obtained by adding some constant to a known antiderivative F . The answer is *no*—once a single antiderivative of f on an open interval is known, all other antiderivatives on that interval are obtainable by adding constants to the known antiderivative. This is so because Theorem

3.8.3 tells us that if two functions have the same derivative on an open interval, then the functions differ by a constant on the interval. The following theorem summarizes these observations.

4.2.2 THEOREM If $F(x)$ is any antiderivative of $f(x)$ on an open interval, then for any constant C the function $F(x) + C$ is also an antiderivative on that interval. Moreover, each antiderivative of $f(x)$ on the interval can be expressed in the form $F(x) + C$ by choosing the constant C appropriately.

THE INDEFINITE INTEGRAL

The process of finding antiderivatives is called **antidifferentiation** or **integration**. Thus, if

$$\frac{d}{dx}[F(x)] = f(x) \quad (1)$$

then **integrating** (or **antidifferentiating**) the function $f(x)$ produces an antiderivative of the form $F(x) + C$. To emphasize this process, Equation (1) is recast using **integral notation**,

$$\int f(x) dx = F(x) + C \quad (2)$$

where C is understood to represent an arbitrary constant. It is important to note that (1) and (2) are just different notations to express the same fact. For example,

$$\int x^2 dx = \frac{1}{3}x^3 + C \quad \text{is equivalent to} \quad \frac{d}{dx} \left[\frac{1}{3}x^3 \right] = x^2$$

Note that if we differentiate an antiderivative of $f(x)$, we obtain $f(x)$ back again. Thus,

$$\frac{d}{dx} \left[\int f(x) dx \right] = f(x) \quad (3)$$

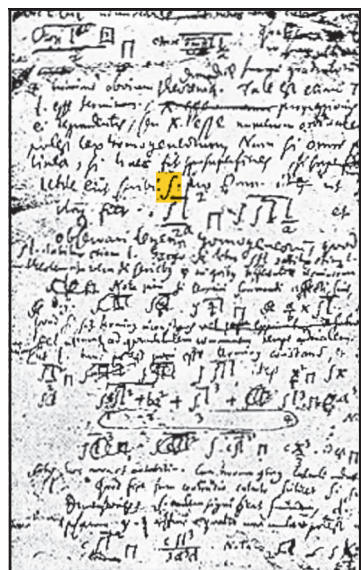
The expression $\int f(x) dx$ is called an **indefinite integral**. The adjective “indefinite” emphasizes that the result of antidifferentiation is a “generic” function, described only up to a constant term. The “elongated s” that appears on the left side of (2) is called an **integral sign**,* the function $f(x)$ is called the **integrand**, and the constant C is called the **constant of integration**. Equation (2) should be read as:

The integral of $f(x)$ with respect to x is equal to $F(x)$ plus a constant.

The differential symbol, dx , in the differentiation and antidifferentiation operations

$$\frac{d}{dx} [] \quad \text{and} \quad \int [] dx$$

* This notation was devised by Leibniz. In his early papers Leibniz used the notation “omn.” (an abbreviation for the Latin word “omnes”) to denote integration. Then on October 29, 1675 he wrote, “It will be useful to write $\int$ for omn., thus $\int l$ for omn. $l \dots$ ” Two or three weeks later he refined the notation further and wrote $\int [] dx$ rather than $\int$ alone. This notation is so useful and so powerful that its development by Leibniz must be regarded as a major milestone in the history of mathematics and science.



Reproduced from C. I. Gerhardt's "Briefwechsel von G. W. Leibniz mit Mathematikern (1899)."

Extract from the manuscript of Leibniz dated October 29, 1675 in which the integral sign first appeared (see yellow highlight).

serves to identify the independent variable. If an independent variable other than x is used, say t , then the notation must be adjusted appropriately. Thus,

$$\frac{d}{dt}[F(t)] = f(t) \quad \text{and} \quad \int f(t) dt = F(t) + C$$

are equivalent statements. Here are some examples of derivative formulas and their equivalent integration formulas:

DERIVATIVE FORMULA	EQUIVALENT INTEGRATION FORMULA
$\frac{d}{dx}[x^3] = 3x^2$	$\int 3x^2 dx = x^3 + C$
$\frac{d}{dx}[\sqrt{x}] = \frac{1}{2\sqrt{x}}$	$\int \frac{1}{2\sqrt{x}} dx = \sqrt{x} + C$
$\frac{d}{dt}[\tan t] = \sec^2 t$	$\int \sec^2 t dt = \tan t + C$
$\frac{d}{du}[u^{3/2}] = \frac{3}{2}u^{1/2}$	$\int \frac{3}{2}u^{1/2} du = u^{3/2} + C$

For simplicity, the dx is sometimes absorbed into the integrand. For example,

$$\int 1 dx \quad \text{can be written as} \quad \int dx$$

$$\int \frac{1}{x^2} dx \quad \text{can be written as} \quad \int \frac{dx}{x^2}$$

■ INTEGRATION FORMULAS

Integration is essentially educated guesswork—given the derivative f of a function F , one tries to guess what the function F is. However, many basic integration formulas can be obtained directly from their companion differentiation formulas. Some of the most important are given in Table 4.2.1.

Table 4.2.1
INTEGRATION FORMULAS

DIFFERENTIATION FORMULA	INTEGRATION FORMULA	DIFFERENTIATION FORMULA	INTEGRATION FORMULA
1. $\frac{d}{dx}[x] = 1$	$\int dx = x + C$	5. $\frac{d}{dx}[\tan x] = \sec^2 x$	$\int \sec^2 x dx = \tan x + C$
2. $\frac{d}{dx}\left[\frac{x^{r+1}}{r+1}\right] = x^r \quad (r \neq -1)$	$\int x^r dx = \frac{x^{r+1}}{r+1} + C \quad (r \neq -1)$	6. $\frac{d}{dx}[-\cot x] = \csc^2 x$	$\int \csc^2 x dx = -\cot x + C$
3. $\frac{d}{dx}[\sin x] = \cos x$	$\int \cos x dx = \sin x + C$	7. $\frac{d}{dx}[\sec x] = \sec x \tan x$	$\int \sec x \tan x dx = \sec x + C$
4. $\frac{d}{dx}[-\cos x] = \sin x$	$\int \sin x dx = -\cos x + C$	8. $\frac{d}{dx}[-\csc x] = \csc x \cot x$	$\int \csc x \cot x dx = -\csc x + C$

► **Example 1** The second integration formula in Table 4.2.1 will be easier to remember if you express it in words:

To integrate a power of x (other than -1), add 1 to the exponent and divide by the new exponent.

Formula 2 in Table 4.2.1 is not applicable to integrating x^{-1} ; we will see how to integrate this function in Chapter 6.

Here are some examples:

$$\int x^2 dx = \frac{x^3}{3} + C \quad r = 2$$

$$\int x^3 dx = \frac{x^4}{4} + C \quad r = 3$$

$$\int \frac{1}{x^5} dx = \int x^{-5} dx = \frac{x^{-5+1}}{-5+1} + C = -\frac{1}{4x^4} + C \quad r = -5$$

$$\int \sqrt{x} dx = \int x^{\frac{1}{2}} dx = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C = \frac{2}{3}x^{\frac{3}{2}} + C = \frac{2}{3}(\sqrt{x})^3 + C \quad r = \frac{1}{2} \quad \blacktriangleleft$$

■ PROPERTIES OF THE INDEFINITE INTEGRAL

Our first properties of antiderivatives follow directly from the simple constant factor, sum, and difference rules for derivatives.

4.2.3 THEOREM Suppose that $F(x)$ and $G(x)$ are antiderivatives of $f(x)$ and $g(x)$, respectively, and that c is a constant. Then:

(a) A constant factor can be moved through an integral sign; that is,

$$\int cf(x) dx = cF(x) + C$$

(b) An antiderivative of a sum is the sum of the antiderivatives; that is,

$$\int [f(x) + g(x)] dx = F(x) + G(x) + C$$

(c) An antiderivative of a difference is the difference of the antiderivatives; that is,

$$\int [f(x) - g(x)] dx = F(x) - G(x) + C$$

PROOF In general, to establish the validity of an equation of the form

$$\int h(x) dx = H(x) + C$$

one must show that

$$\frac{d}{dx}[H(x)] = h(x)$$

We are given that $F(x)$ and $G(x)$ are antiderivatives of $f(x)$ and $g(x)$, respectively, so we know that

$$\frac{d}{dx}[F(x)] = f(x) \quad \text{and} \quad \frac{d}{dx}[G(x)] = g(x)$$

Thus,

$$\frac{d}{dx}[cF(x)] = c \frac{d}{dx}[F(x)] = cf(x)$$

$$\frac{d}{dx}[F(x) + G(x)] = \frac{d}{dx}[F(x)] + \frac{d}{dx}[G(x)] = f(x) + g(x)$$

$$\frac{d}{dx}[F(x) - G(x)] = \frac{d}{dx}[F(x)] - \frac{d}{dx}[G(x)] = f(x) - g(x)$$

which proves the three statements of the theorem. ■

The statements in Theorem 4.2.3 can be summarized by the following formulas:

$$\int cf(x) dx = c \int f(x) dx \quad (4)$$

$$\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx \quad (5)$$

$$\int [f(x) - g(x)] dx = \int f(x) dx - \int g(x) dx \quad (6)$$

However, these equations must be applied carefully to avoid errors and unnecessary complexities arising from the constants of integration. For example, if you use (4) to integrate $2x$ by writing

$$\int 2x dx = 2 \int x dx = 2 \left(\frac{x^2}{2} + C \right) = x^2 + 2C$$

then you will have an unnecessarily complicated form of the arbitrary constant. This kind of problem can be avoided by inserting the constant of integration in the final result rather than in intermediate calculations. Exercises 59 and 60 explore how careless application of these formulas can lead to errors.

► **Example 2** Evaluate

$$(a) \int 4 \cos x dx \quad (b) \int (x + x^2) dx$$

Solution (a). Since $F(x) = \sin x$ is an antiderivative for $f(x) = \cos x$ (Table 4.2.1), we obtain

$$\int 4 \cos x dx = 4 \int \cos x dx = 4 \sin x + C \quad (4)$$

Solution (b). From Table 4.2.1 we obtain

$$\int (x + x^2) dx = \int x dx + \int x^2 dx = \frac{x^2}{2} + \frac{x^3}{3} + C \quad (5)$$

Parts (b) and (c) of Theorem 4.2.3 can be extended to more than two functions, which in combination with part (a) results in the following general formula:

$$\begin{aligned} \int [c_1 f_1(x) + c_2 f_2(x) + \cdots + c_n f_n(x)] dx \\ = c_1 \int f_1(x) dx + c_2 \int f_2(x) dx + \cdots + c_n \int f_n(x) dx \end{aligned} \quad (7)$$

► **Example 3**

$$\begin{aligned} \int (3x^6 - 2x^2 + 7x + 1) dx &= 3 \int x^6 dx - 2 \int x^2 dx + 7 \int x dx + \int 1 dx \\ &= \frac{3x^7}{7} - \frac{2x^3}{3} + \frac{7x^2}{2} + x + C \quad \blacktriangleleft \end{aligned}$$

Sometimes it is useful to rewrite an integrand in a different form before performing the integration. This is illustrated in the following example.

► **Example 4** Evaluate

$$(a) \int \frac{\cos x}{\sin^2 x} dx \quad (b) \int \frac{t^2 - 2t^4}{t^4} dt$$

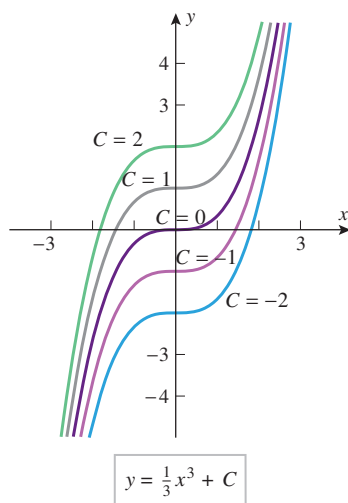
Solution (a).

$$\int \frac{\cos x}{\sin^2 x} dx = \int \frac{1}{\sin x} \frac{\cos x}{\sin x} dx = \int \csc x \cot x dx = -\csc x + C$$

Formula 8 in Table 4.2.1

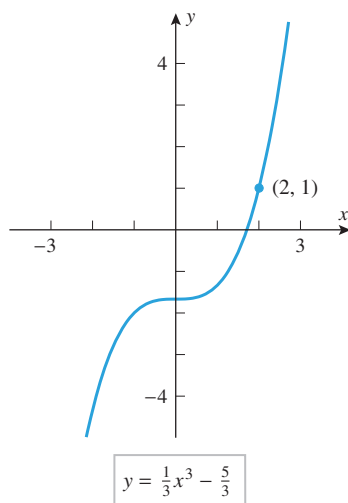
Solution (b).

$$\begin{aligned} \int \frac{t^2 - 2t^4}{t^4} dt &= \int \left(\frac{1}{t^2} - 2 \right) dt = \int (t^{-2} - 2) dt \\ &= \frac{t^{-1}}{-1} - 2t + C = -\frac{1}{t} - 2t + C \end{aligned}$$



▲ Figure 4.2.1

In Example 5, the requirement that the graph of f pass through the point $(2, 1)$ selects the single integral curve $y = \frac{1}{3}x^3 - \frac{5}{3}$ from the family of curves $y = \frac{1}{3}x^3 + C$ (Figure 4.2.2).



▲ Figure 4.2.2

INTEGRAL CURVES

Graphs of antiderivatives of a function f are called **integral curves** of f . We know from Theorem 4.2.2 that if $y = F(x)$ is any integral curve of $f(x)$, then all other integral curves are vertical translations of this curve, since they have equations of the form $y = F(x) + C$. For example, $y = \frac{1}{3}x^3$ is one integral curve for $f(x) = x^2$, so all the other integral curves have equations of the form $y = \frac{1}{3}x^3 + C$; conversely, the graph of any equation of this form is an integral curve (Figure 4.2.1).

In many problems one is interested in finding a function whose derivative satisfies specified conditions. The following example illustrates a geometric problem of this type.

► **Example 5** Suppose that a curve $y = f(x)$ in the xy -plane has the property that at each point (x, y) on the curve, the tangent line has slope x^2 . Find an equation for the curve given that it passes through the point $(2, 1)$.

Solution. Since the slope of the line tangent to $y = f(x)$ is dy/dx , we have $dy/dx = x^2$, and

$$y = \int x^2 dx = \frac{1}{3}x^3 + C$$

Since the curve passes through $(2, 1)$, a specific value for C can be found by using the fact that $y = 1$ if $x = 2$. Substituting these values in the above equation yields

$$1 = \frac{1}{3}(2^3) + C \quad \text{or} \quad C = -\frac{5}{3}$$

so an equation of the curve is

$$y = \frac{1}{3}x^3 - \frac{5}{3}$$

(Figure 4.2.2). ◀

INTEGRATION FROM THE VIEWPOINT OF DIFFERENTIAL EQUATIONS

We will now consider another way of looking at integration that will be useful in our later work. Suppose that $f(x)$ is a known function and we are interested in finding a function $F(x)$ such that $y = F(x)$ satisfies the equation

$$\frac{dy}{dx} = f(x) \tag{8}$$

The solutions of this equation are the antiderivatives of $f(x)$, and we know that these can be obtained by integrating $f(x)$. For example, the solutions of the equation

$$\frac{dy}{dx} = x^2 \tag{9}$$

are

$$y = \int x^2 dx = \frac{x^3}{3} + C$$

Equation (8) is called a **differential equation** because it involves a derivative of an unknown function. Differential equations are different from the kinds of equations we have encountered so far in that the unknown is a *function* and not a *number* as in an equation such as $x^2 + 5x - 6 = 0$.

Sometimes we will not be interested in finding all of the solutions of (8), but rather we will want only the solution whose graph passes through a specified point (x_0, y_0) . For example, in Example 5 we solved (9) for the integral curve that passed through the point $(2, 1)$.

For simplicity, it is common in the study of differential equations to denote a solution of $dy/dx = f(x)$ as $y(x)$ rather than $F(x)$, as earlier. With this notation, the problem of finding a function $y(x)$ whose derivative is $f(x)$ and whose graph passes through the point (x_0, y_0) is expressed as

$$\frac{dy}{dx} = f(x), \quad y(x_0) = y_0 \quad (10)$$

This is called an **initial-value problem**, and the requirement that $y(x_0) = y_0$ is called the **initial condition** for the problem.

► **Example 6** Solve the initial-value problem

$$\frac{dy}{dx} = \cos x, \quad y(0) = 1$$

Solution. The solution of the differential equation is

$$y = \int \cos x dx = \sin x + C \quad (11)$$

The initial condition $y(0) = 1$ implies that $y = 1$ if $x = 0$; substituting these values in (11) yields

$$1 = \sin(0) + C \quad \text{or} \quad C = 1$$

Thus, the solution of the initial-value problem is $y = \sin x + 1$. ◀

■ SLOPE FIELDS

If we interpret dy/dx as the slope of a tangent line, then at a point (x, y) on an integral curve of the equation $dy/dx = f(x)$, the slope of the tangent line is $f(x)$. What is interesting about this is that the slopes of the tangent lines to the integral curves can be obtained without actually solving the differential equation. For example, if

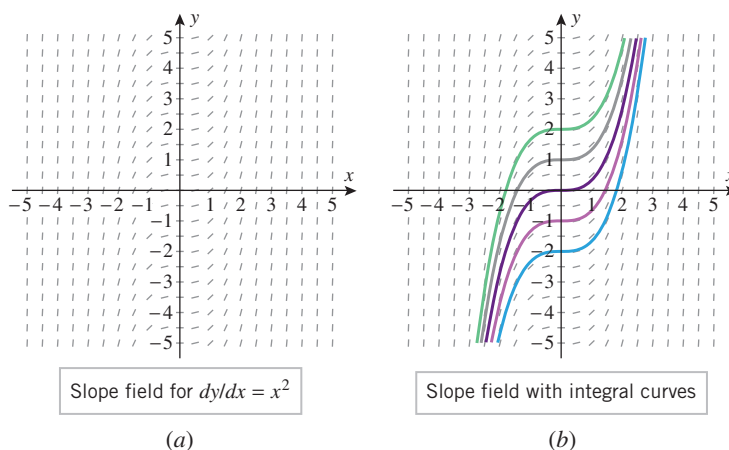
$$\frac{dy}{dx} = \sqrt{x^2 + 1}$$

then we know without solving the equation that at the point where $x = 1$ the tangent line to an integral curve has slope $\sqrt{1^2 + 1} = \sqrt{2}$; and more generally, at a point where $x = a$, the tangent line to an integral curve has slope $\sqrt{a^2 + 1}$.

A geometric description of the integral curves of a differential equation $dy/dx = f(x)$ can be obtained by choosing a rectangular grid of points in the xy -plane, calculating the slopes of the tangent lines to the integral curves at the gridpoints, and drawing small portions of the tangent lines through those points. The resulting picture, which is called a **slope field** or **direction field** for the equation, shows the “direction” of the integral curves at the gridpoints. With sufficiently many gridpoints it is often possible to visualize the integral

curves themselves; for example, Figure 4.2.3a shows a slope field for the differential equation $dy/dx = x^2$, and Figure 4.2.3b shows that same field with the integral curves imposed on it—the more gridpoints that are used, the more completely the slope field reveals the shape of the integral curves. However, the amount of computation can be considerable, so computers are usually used when slope fields with many gridpoints are needed.

Slope fields will be studied in more detail later in the text.



► Figure 4.2.3

✓ QUICK CHECK EXERCISES 4.2 (See page 281 for answers.)

- A function F is an antiderivative of a function f on an interval if _____ for all x in the interval.
- Write an equivalent integration formula for each given derivative formula.

$$(a) \frac{d}{dx}[\sqrt{x}] = \frac{1}{2\sqrt{x}}$$

$$(b) \frac{d}{dx}[\sin x] = \cos x$$
- Evaluate the integrals.

$$(a) \int [x^3 + x + 5] dx$$

$$(b) \int [\sec^2 x - \csc x \cot x] dx$$
- The graph of $y = x^2 + x$ is an integral curve for the function $f(x) = \underline{\hspace{2cm}}$. If G is a function whose graph

is also an integral curve for f , and if $G(1) = 5$, then $G(x) = \underline{\hspace{2cm}}$.

- A slope field for the differential equation

$$\frac{dy}{dx} = \frac{2x}{x^2 - 4}$$

has a line segment with slope _____ through the point $(0, 5)$ and has a line segment with slope _____ through the point $(-4, 1)$.

EXERCISE SET 4.2 Graphing Utility CAS

- In each part, confirm that the formula is correct, and state a corresponding integration formula.

$$(a) \frac{d}{dx}[\sqrt{1+x^2}] = \frac{x}{\sqrt{1+x^2}}$$

$$(b) \frac{d}{dx} \left[\frac{1}{3} \sin(1+x^3) \right] = x^2 \cos(1+x^3)$$
- In each part, confirm that the stated formula is correct by differentiating.

$$(a) \int x \sin x dx = \sin x - x \cos x + C$$

$$(b) \int \frac{dx}{(1-x^2)^{3/2}} = \frac{x}{\sqrt{1-x^2}} + C$$

FOCUS ON CONCEPTS

- What is a *constant of integration*? Why does an answer to an integration problem involve a constant of integration?
- What is an *integral curve* of a function f ? How are two integral curves of a function f related?

5–8 Find the derivative and state a corresponding integration formula. ■

- $\frac{d}{dx}[\sqrt{x^3 + 5}]$
 - $\frac{d}{dx}[\sin(2\sqrt{x})]$

- $\frac{d}{dx} \left[\frac{x}{x^2 + 3} \right]$
 - $\frac{d}{dx}[\sin x - x \cos x]$

9–10 Evaluate the integral by rewriting the integrand appropriately, if required, and applying the power rule (Formula 2 in Table 4.2.1). ■

9. (a) $\int x^8 dx$ (b) $\int x^{5/7} dx$ (c) $\int x^3 \sqrt{x} dx$
 10. (a) $\int \sqrt[3]{x^2} dx$ (b) $\int \frac{1}{x^6} dx$ (c) $\int x^{-7/8} dx$

11–14 Evaluate each integral by applying Theorem 4.2.3 and Formula 2 in Table 4.2.1 appropriately. ■

11. $\int \left[5x + \frac{2}{3x^5} \right] dx$ 12. $\int [x^{-1/2} - 3x^{7/5} + \frac{1}{9}] dx$
 13. $\int [x^{-3} - 3x^{1/4} + 8x^2] dx$
 14. $\int \left[\frac{10}{y^{3/4}} - \sqrt[3]{y} + \frac{4}{\sqrt{y}} \right] dy$

15–30 Evaluate the integral and check your answer by differentiating. ■

15. $\int x(1 + x^3) dx$ 16. $\int (2 + y^2)^2 dy$
 17. $\int x^{1/3}(2 - x)^2 dx$ 18. $\int (1 + x^2)(2 - x) dx$
 19. $\int \frac{x^5 + 2x^2 - 1}{x^4} dx$ 20. $\int \frac{1 - 2t^3}{t^3} dt$
 21. $\int [3 \sin x - 2 \sec^2 x] dx$ 22. $\int [\csc^2 t - \sec t \tan t] dt$
 23. $\int \sec x (\sec x + \tan x) dx$ 24. $\int \csc x (\sin x + \cot x) dx$
 25. $\int \frac{\sec \theta}{\cos \theta} d\theta$ 26. $\int \frac{dy}{\csc y}$
 27. $\int \frac{\sin x}{\cos^2 x} dx$ 28. $\int \left[\phi + \frac{2}{\sin^2 \phi} \right] d\phi$
 29. $\int [1 + \sin^2 \theta \csc \theta] d\theta$ 30. $\int \frac{\sec x + \cos x}{2 \cos x} dx$

31. Evaluate the integral

$$\int \frac{1}{1 + \sin x} dx$$

by multiplying the numerator and denominator by an appropriate expression.

32. Use the double-angle formula $\cos 2x = 2 \cos^2 x - 1$ to evaluate the integral

$$\int \frac{1}{1 + \cos 2x} dx$$

33–36 True–False Determine whether the statement is true or false. Explain your answer. ■

33. If $F(x)$ is an antiderivative of $f(x)$, then

$$\int f(x) dx = F(x) + C$$

34. If C denotes a constant of integration, the two formulas

$$\int \cos x dx = \sin x + C$$

$$\int \cos x dx = (\sin x + \pi) + C$$

are both correct equations.

35. The function $f(x) = \sec x + 1$ is a solution to the initial-value problem

$$\frac{dy}{dx} = \sec x \tan x, \quad y(0) = 1$$

36. Every integral curve of the slope field

$$\frac{dy}{dx} = \frac{1}{\sqrt{x^2 + 1}}$$

is the graph of an increasing function of x .

37. Use a graphing utility to generate some representative integral curves of the function $f(x) = 5x^4 - \sec^2 x$ over the interval $(-\pi/2, \pi/2)$.

38. Use a graphing utility to generate some representative integral curves of the function $f(x) = (x^2 - 1)/x^2$ over the interval $(0, 5)$.

39–40 Solve the initial-value problems. ■

39. (a) $\frac{dy}{dx} = \sqrt[3]{x}, y(1) = 2$
 (b) $\frac{dy}{dt} = \sin t + 1, y\left(\frac{\pi}{3}\right) = \frac{1}{2}$
 (c) $\frac{dy}{dx} = \frac{x+1}{\sqrt{x}}, y(1) = 0$
 40. (a) $\frac{dy}{dx} = \frac{1}{(2x)^3}, y(1) = 0$
 (b) $\frac{dy}{dt} = \sec^2 t - \sin t, y\left(\frac{\pi}{4}\right) = 1$
 (c) $\frac{dy}{dx} = x^2 \sqrt{x^3}, y(0) = 0$

41–44 A particle moves along an s -axis with position function $s = s(t)$ and velocity function $v(t) = s'(t)$. Use the given information to find $s(t)$. ■

41. $v(t) = 32t; s(0) = 20$ 42. $v(t) = \cos t; s(0) = 2$

43. $v(t) = 3\sqrt{t}; s(4) = 1$ 44. $v(t) = \sin t; s(0) = 0$

45. Find the general form of a function whose second derivative is $\sqrt{x}$. [Hint: Solve the equation $f''(x) = \sqrt{x}$ for $f(x)$ by integrating both sides twice.]

46. Find a function f such that $f''(x) = x + \cos x$ and such that $f(0) = 1$ and $f'(0) = 2$. [Hint: Integrate both sides of the equation twice.]

47–51 Find an equation of the curve that satisfies the given conditions. ■

47. At each point (x, y) on the curve the slope is $2x + 1$; the curve passes through the point $(-3, 0)$.

48. At each point (x, y) on the curve the slope is $(x + 1)^2$; the curve passes through the point $(-2, 8)$.

49. At each point (x, y) on the curve the slope is $-\sin x$; the curve passes through the point $(0, 2)$.
50. At each point (x, y) on the curve the slope equals the square of the distance between the point and the y -axis; the point $(-1, 2)$ is on the curve.
51. At each point (x, y) on the curve, y satisfies the condition $d^2y/dx^2 = 6x$; the line $y = 5 - 3x$ is tangent to the curve at the point where $x = 1$.

 52. In each part, use a CAS to solve the initial-value problem.

(a) $\frac{dy}{dx} = x^2 \cos 3x$, $y(\pi/2) = -1$

(b) $\frac{dy}{dx} = \frac{x^3}{(4 + x^2)^{3/2}}$, $y(0) = -2$

 53. (a) Use a graphing utility to generate a slope field for the differential equation $dy/dx = x$ in the region $-5 \leq x \leq 5$ and $-5 \leq y \leq 5$.

(b) Graph some representative integral curves of the function $f(x) = x$.

(c) Find an equation for the integral curve that passes through the point $(2, 1)$.

 54. (a) Use a graphing utility to generate a slope field for the differential equation $dy/dx = \sqrt{x}$ in the region $0 \leq x \leq 10$ and $-5 \leq y \leq 5$.

(b) Graph some representative integral curves of the function $f(x) = \sqrt{x}$ for $x > 0$.

(c) Find an equation for the integral curve that passes through the point $(0, 1)$.

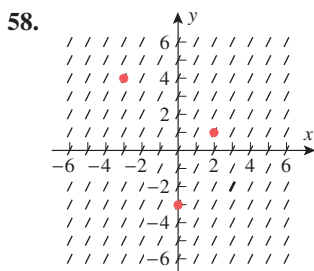
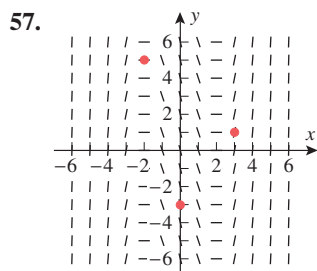
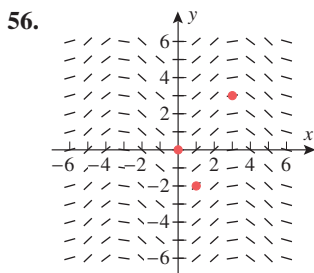
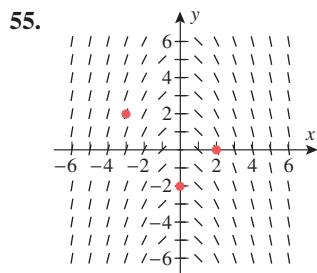
55–58 The given slope field figure corresponds to one of the differential equations below. Identify the differential equation that matches the figure, and sketch solution curves through the highlighted points.

(a) $\frac{dy}{dx} = 2$

(b) $\frac{dy}{dx} = -x$

(c) $\frac{dy}{dx} = x^2 - 4$

(d) $\frac{dy}{dx} = \sin x$ 



FOCUS ON CONCEPTS

59. Critique the following “proof” that an arbitrary constant must be zero:

$$C = \int 0 \, dx = \int 0 \cdot 0 \, dx = 0 \int 0 \, dx = 0$$

60. Critique the following “proof” that an arbitrary constant must be zero:

$$\begin{aligned} 0 &= \left(\int x \, dx \right) - \left(\int x \, dx \right) \\ &= \int (x - x) \, dx = \int 0 \, dx = C \end{aligned}$$

61. Let F and G be the functions defined by

$$F(x) = \frac{x \sin x}{x} \quad \text{and} \quad G(x) = \begin{cases} 2 + \sin x, & x > 0 \\ -1 + \sin x, & x < 0 \end{cases}$$

(a) Show that F and G have the same derivative.

(b) Show that $G(x) \neq F(x) + C$ for any constant C .

(c) Do parts (a) and (b) contradict Theorem 4.2.2? Explain.

62. Follow the directions of Exercise 61 using

$$F(x) = \frac{x^2 + 3x}{x} \quad \text{and} \quad G(x) = \begin{cases} x + 3, & x > 0 \\ x, & x < 0 \end{cases}$$

63–64 Use a trigonometric identity to evaluate the integral. 

63. $\int \tan^2 x \, dx$

64. $\int \cot^2 x \, dx$

65–66 Use the identities $\cos 2\theta = 1 - 2\sin^2 \theta = 2\cos^2 \theta - 1$ to help evaluate the integrals 

65. $\int \sin^2(x/2) \, dx$

66. $\int \cos^2(x/2) \, dx$

67. The speed of sound in air at 0°C (or 273 K on the Kelvin scale) is 1087 ft/s , but the speed v increases as the temperature T rises. Experimentation has shown that the rate of change of v with respect to T is

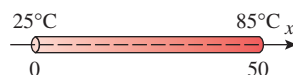
$$\frac{dv}{dT} = \frac{1087}{2\sqrt{273}} T^{-1/2}$$

where v is in feet per second and T is in kelvins (K). Find a formula that expresses v as a function of T .

68. Suppose that a uniform metal rod 50 cm long is insulated laterally, and the temperatures at the exposed ends are maintained at 25°C and 85°C , respectively. Assume that an x -axis is chosen as in the accompanying figure and that the temperature $T(x)$ satisfies the equation

$$\frac{d^2T}{dx^2} = 0$$

Find $T(x)$ for $0 \leq x \leq 50$.



◀ Figure Ex-68

69. **Writing** What is an *initial-value problem*? Describe the sequence of steps for solving an initial-value problem.

70. **Writing** What is a *slope field*? How are slope fields and integral curves related?

✓ QUICK CHECK ANSWERS 4.2

1. $F'(x) = f(x)$ 2. (a) $\int \frac{1}{2\sqrt{x}} dx = \sqrt{x} + C$ (b) $\int \cos x dx = \sin x + C$
 3. (a) $\frac{1}{4}x^4 + \frac{1}{2}x^2 + 5x + C$ (b) $\tan x + \csc x + C$ 4. $2x + 1$; $x^2 + x + 3$ 5. 0; $-\frac{2}{3}$

4.3 INTEGRATION BY SUBSTITUTION

*In this section we will study a technique, called **substitution**, that can often be used to transform complicated integration problems into simpler ones.*

■ u-SUBSTITUTION

The method of substitution can be motivated by examining the chain rule from the viewpoint of antidifferentiation. For this purpose, suppose that F is an antiderivative of f and that g is a differentiable function. The chain rule implies that the derivative of $F(g(x))$ can be expressed as

$$\frac{d}{dx}[F(g(x))] = F'(g(x))g'(x)$$

which we can write in integral form as

$$\int F'(g(x))g'(x) dx = F(g(x)) + C \quad (1)$$

or since F is an antiderivative of f ,

$$\int f(g(x))g'(x) dx = F(g(x)) + C \quad (2)$$

For our purposes it will be useful to let $u = g(x)$ and to write $du/dx = g'(x)$ in the differential form $du = g'(x) dx$. With this notation (2) can be expressed as

$$\int f(u) du = F(u) + C \quad (3)$$

The process of evaluating an integral of form (2) by converting it into form (3) with the substitution

$$u = g(x) \quad \text{and} \quad du = g'(x) dx$$

is called the **method of u-substitution**. Here our emphasis is *not* on the interpretation of the expression $du = g'(x) dx$. Rather, the differential notation serves primarily as a useful “bookkeeping” device for the method of u -substitution. The following example illustrates how the method works.

► **Example 1** Evaluate $\int (x^2 + 1)^{50} \cdot 2x dx$.

Solution. If we let $u = x^2 + 1$, then $du/dx = 2x$, which implies that $du = 2x dx$. Thus, the given integral can be written as

$$\int (x^2 + 1)^{50} \cdot 2x dx = \int u^{50} du = \frac{u^{51}}{51} + C = \frac{(x^2 + 1)^{51}}{51} + C \quad \blacktriangleleft$$

It is important to realize that in the method of u -substitution you have control over the choice of u , but once you make that choice you have no control over the resulting expression for du . Thus, in the last example we *chose* $u = x^2 + 1$ but $du = 2x dx$ was *computed*.

Fortunately, our choice of u , combined with the computed du , worked out perfectly to produce an integral involving u that was easy to evaluate. However, in general, the method of u -substitution will fail if the chosen u and the computed du cannot be used to produce an integrand in which no expressions involving x remain, or if you cannot evaluate the resulting integral. Thus, for example, the substitution $u = x^2$, $du = 2x \, dx$ will not work for the integral

$$\int 2x \sin x^4 \, dx$$

because this substitution results in the integral

$$\int \sin u^2 \, du$$

which still cannot be evaluated in terms of familiar functions.

In general, there are no hard and fast rules for choosing u , and in some problems no choice of u will work. In such cases other methods need to be used, some of which will be discussed later. Making appropriate choices for u will come with experience, but you may find the following guidelines, combined with a mastery of the basic integrals in Table 4.2.1, helpful.

Guidelines for u -Substitution

Step 1. Look for some composition $f(g(x))$ within the integrand for which the substitution

$$u = g(x), \quad du = g'(x) \, dx$$

produces an integral that is expressed entirely in terms of u and its differential du . This may or may not be possible.

Step 2. If you are successful in Step 1, then try to evaluate the resulting integral in terms of u . Again, this may or may not be possible.

Step 3. If you are successful in Step 2, then replace u by $g(x)$ to express your final answer in terms of x .

EASY TO RECOGNIZE SUBSTITUTIONS

The easiest substitutions occur when the integrand is the derivative of a known function, except for a constant added to or subtracted from the independent variable.

► Example 2

$$\int \sin(x + 9) \, dx = \int \sin u \, du = -\cos u + C = -\cos(x + 9) + C$$

$$\begin{array}{l} u = x + 9 \\ du = 1 \cdot dx = dx \end{array}$$

$$\int (x - 8)^{23} \, dx = \int u^{23} \, du = \frac{u^{24}}{24} + C = \frac{(x - 8)^{24}}{24} + C \quad \blacktriangleleft$$

$$\begin{array}{l} u = x - 8 \\ du = 1 \cdot dx = dx \end{array}$$

Another easy u -substitution occurs when the integrand is the derivative of a known function, except for a constant that multiplies or divides the independent variable. The following example illustrates two ways to evaluate such integrals.

► **Example 3** Evaluate $\int \cos 5x \, dx$.

Solution.

$$\int \cos 5x \, dx = \int (\cos u) \cdot \frac{1}{5} \, du = \frac{1}{5} \int \cos u \, du = \frac{1}{5} \sin u + C = \frac{1}{5} \sin 5x + C$$

$u = 5x$
 $du = 5 \, dx \text{ or } dx = \frac{1}{5} \, du$

Alternative Solution. There is a variation of the preceding method that some people prefer. The substitution $u = 5x$ requires $du = 5 \, dx$. If there were a factor of 5 in the integrand, then we could group the 5 and dx together to form the du required by the substitution. Since there is no factor of 5, we will insert one and compensate by putting a factor of $\frac{1}{5}$ in front of the integral. The computations are as follows:

$$\int \cos 5x \, dx = \frac{1}{5} \int \cos 5x \cdot 5 \, dx = \frac{1}{5} \int \cos u \, du = \frac{1}{5} \sin u + C = \frac{1}{5} \sin 5x + C \quad \blacktriangleleft$$

$u = 5x$
 $du = 5 \, dx$

More generally, if the integrand is a composition of the form $f(ax + b)$, where $f(x)$ is an easy to integrate function, then the substitution $u = ax + b$, $du = a \, dx$ will work.

► **Example 4**

$$\int \frac{dx}{(\frac{1}{3}x - 8)^5} = \int \frac{3 \, du}{u^5} = 3 \int u^{-5} \, du = -\frac{3}{4} u^{-4} + C = -\frac{3}{4} \left(\frac{1}{3}x - 8\right)^{-4} + C \quad \blacktriangleleft$$

$u = \frac{1}{3}x - 8$
 $du = \frac{1}{3} \, dx \text{ or } dx = 3 \, du$

With the help of Theorem 4.2.3, a complicated integral can sometimes be computed by expressing it as a sum of simpler integrals.

► **Example 5**

$$\begin{aligned} \int \left(\frac{1}{x^2} + \sec^2 \pi x \right) dx &= \int \frac{dx}{x^2} + \int \sec^2 \pi x \, dx \\ &= -\frac{1}{x} + \int \sec^2 \pi x \, dx \\ &= -\frac{1}{x} + \frac{1}{\pi} \int \sec^2 u \, du \end{aligned}$$

$u = \pi x$
 $du = \pi \, dx \text{ or } dx = \frac{1}{\pi} \, du$

$$= -\frac{1}{x} + \frac{1}{\pi} \tan u + C = -\frac{1}{x} + \frac{1}{\pi} \tan \pi x + C \quad \blacktriangleleft$$

The next three examples illustrate a substitution $u = g(x)$ where $g(x)$ is a nonlinear function.

► **Example 6** Evaluate $\int \sin^2 x \cos x \, dx$.

Solution. If we let $u = \sin x$, then

$$\frac{du}{dx} = \cos x, \quad \text{so} \quad du = \cos x \, dx$$

Thus,

$$\int \sin^2 x \cos x \, dx = \int u^2 \, du = \frac{u^3}{3} + C = \frac{\sin^3 x}{3} + C \quad \blacktriangleleft$$

► **Example 7** Evaluate $\int \frac{\cos \sqrt{x}}{\sqrt{x}} \, dx$.

Solution. If we let $u = \sqrt{x}$, then

$$\frac{du}{dx} = \frac{1}{2\sqrt{x}}, \quad \text{so} \quad du = \frac{1}{2\sqrt{x}} \, dx \quad \text{or} \quad 2 \, du = \frac{1}{\sqrt{x}} \, dx$$

Thus,

$$\int \frac{\cos \sqrt{x}}{\sqrt{x}} \, dx = \int 2 \cos u \, du = 2 \int \cos u \, du = 2 \sin u + C = 2 \sin \sqrt{x} + C \quad \blacktriangleleft$$

► **Example 8** Evaluate $\int t^4 \sqrt[3]{3 - 5t^5} \, dt$.

Solution.

$$\begin{aligned} \int t^4 \sqrt[3]{3 - 5t^5} \, dt &= -\frac{1}{25} \int \sqrt[3]{u} \, du = -\frac{1}{25} \int u^{1/3} \, du \\ &= -\frac{1}{25} \frac{u^{4/3}}{4/3} + C = -\frac{3}{100} (3 - 5t^5)^{4/3} + C \quad \blacktriangleleft \end{aligned}$$

$$\begin{aligned} u &= 3 - 5t^5 \\ du &= -25t^4 \, dt \quad \text{or} \quad -\frac{1}{25} \, du = t^4 \, dt \end{aligned}$$

LESS APPARENT SUBSTITUTIONS

The method of substitution is relatively straightforward, provided the integrand contains an easily recognized composition $f(g(x))$ and the remainder of the integrand is a constant multiple of $g'(x)$. If this is not the case, the method may still apply but may require more computation.

► **Example 9** Evaluate $\int x^2 \sqrt{x-1} \, dx$.

Solution. The composition $\sqrt{x-1}$ suggests the substitution

$$u = x - 1 \quad \text{so that} \quad du = dx \tag{4}$$

From the first equality in (4)

$$x^2 = (u + 1)^2 = u^2 + 2u + 1$$

so that

$$\begin{aligned}\int x^2 \sqrt{x-1} \, dx &= \int (u^2 + 2u + 1) \sqrt{u} \, du = \int (u^{5/2} + 2u^{3/2} + u^{1/2}) \, du \\ &= \frac{2}{7} u^{7/2} + \frac{4}{5} u^{5/2} + \frac{2}{3} u^{3/2} + C \\ &= \frac{2}{7} (x-1)^{7/2} + \frac{4}{5} (x-1)^{5/2} + \frac{2}{3} (x-1)^{3/2} + C \quad \blacktriangleleft\end{aligned}$$

► **Example 10** Evaluate $\int \cos^3 x \, dx$.

Solution. The only compositions in the integrand that suggest themselves are

$$\cos^3 x = (\cos x)^3 \quad \text{and} \quad \cos^2 x = (\cos x)^2$$

However, neither the substitution $u = \cos x$ nor the substitution $u = \cos^2 x$ work (verify). In this case, an appropriate substitution is not suggested by the composition contained in the integrand. On the other hand, note from Equation (2) that the derivative $g'(x)$ appears as a factor in the integrand. This suggests that we write

$$\int \cos^3 x \, dx = \int \cos^2 x \cos x \, dx$$

and solve the equation $du = \cos x \, dx$ for $u = \sin x$. Since $\sin^2 x + \cos^2 x = 1$, we then have

$$\begin{aligned}\int \cos^3 x \, dx &= \int \cos^2 x \cos x \, dx = \int (1 - \sin^2 x) \cos x \, dx = \int (1 - u^2) \, du \\ &= u - \frac{u^3}{3} + C = \sin x - \frac{1}{3} \sin^3 x + C \quad \blacktriangleleft\end{aligned}$$

TECHNOLOGY MASTERY

If you have a CAS, use it to calculate the integrals in the examples in this section. If your CAS produces an answer that is different from the one in the text, then confirm algebraically that the two answers agree. Also, explore the effect of using the CAS to simplify the expressions it produces for the integrals.

INTEGRATION USING COMPUTER ALGEBRA SYSTEMS

The advent of computer algebra systems has made it possible to evaluate many kinds of integrals that would be laborious to evaluate by hand. For example, a handheld calculator evaluated the integral

$$\int \frac{5x^2}{(1+x)^{1/3}} \, dx = \frac{3(x+1)^{2/3}(5x^2 - 6x + 9)}{8} + C$$

in about a second. The computer algebra system *Mathematica*, running on a personal computer, required even less time to evaluate this same integral. However, just as one would not want to rely on a calculator to compute $2 + 2$, so one would not want to use a CAS to integrate a simple function such as $f(x) = x^2$. Thus, even if you have a CAS, you will want to develop a reasonable level of competence in evaluating basic integrals. Moreover, the mathematical techniques that we will introduce for evaluating basic integrals are precisely the techniques that computer algebra systems use to evaluate more complicated integrals.

✓ QUICK CHECK EXERCISES 4.3 (See page 287 for answers.)

1. Indicate the u -substitution.

(a) $\int 3x^2(1+x^3)^{25} \, dx = \int u^{25} \, du$ if $u = \underline{\hspace{2cm}}$
and $du = \underline{\hspace{2cm}}$.

(b) $\int 2x \sin x^2 \, dx = \int \sin u \, du$ if $u = \underline{\hspace{2cm}}$ and
 $du = \underline{\hspace{2cm}}$.

(c) $\int \frac{18x}{\sqrt{1+9x^2}} \, dx = \int \frac{1}{\sqrt{u}} \, du$ if $u = \underline{\hspace{2cm}}$ and
 $du = \underline{\hspace{2cm}}$.

2. Supply the missing integrand corresponding to the indicated u -substitution.

(a) $\int 5(5x - 3)^{-1/3} dx = \int \text{_____} du; u = 5x - 3$

(b) $\int (3 - \tan x) \sec^2 x dx = \int \text{_____} du;$

$u = 3 - \tan x$

(c) $\int \frac{\sqrt[3]{8 + \sqrt{x}}}{\sqrt{x}} dx = \int \text{_____} du; u = 8 + \sqrt{x}$

EXERCISE SET 4.3



Graphing Utility



CAS

1–8 Evaluate the integrals using the indicated substitutions. ■

1. (a) $\int 2x(x^2 + 1)^{23} dx; u = x^2 + 1$

(b) $\int \cos^3 x \sin x dx; u = \cos x$

2. (a) $\int \frac{1}{\sqrt{x}} \sin \sqrt{x} dx; u = \sqrt{x}$

(b) $\int \frac{3x dx}{\sqrt{4x^2 + 5}}; u = 4x^2 + 5$

3. (a) $\int \sec^2(4x + 1) dx; u = 4x + 1$

(b) $\int y\sqrt{1 + 2y^2} dy; u = 1 + 2y^2$

4. (a) $\int \sqrt{\sin \pi \theta} \cos \pi \theta d\theta; u = \sin \pi \theta$

(b) $\int (2x + 7)(x^2 + 7x + 3)^{4/5} dx; u = x^2 + 7x + 3$

5. (a) $\int \cot x \csc^2 x dx; u = \cot x$

(b) $\int (1 + \sin t)^9 \cos t dt; u = 1 + \sin t$

6. (a) $\int \cos 2x dx; u = 2x$ (b) $\int x \sec^2 x^2 dx; u = x^2$

7. (a) $\int x^2 \sqrt{1 + x} dx; u = 1 + x$

(b) $\int [\csc(\sin x)]^2 \cos x dx; u = \sin x$

8. (a) $\int \sin(x - \pi) dx; u = x - \pi$

(b) $\int \frac{5x^4}{(x^5 + 1)^2} dx; u = x^5 + 1$

11. $\int (4x - 3)^9 dx$

13. $\int \sin 7x dx$

15. $\int \sec 4x \tan 4x dx$

17. $\int t\sqrt{7t^2 + 12} dt$

19. $\int \frac{6}{(1 - 2x)^3} dx$

21. $\int \frac{x^3}{(5x^4 + 2)^3} dx$

23. $\int \frac{\sin(5/x)}{x^2} dx$

25. $\int \cos^4 3t \sin 3t dt$

27. $\int x \sec^2(x^2) dx$

29. $\int \cos 4\theta \sqrt{2 - \sin 4\theta} d\theta$

31. $\int \sec^3 2x \tan 2x dx$

33. $\int \frac{y}{\sqrt{2y + 1}} dy$

35. $\int \sin^3 2\theta d\theta$

36. $\int \sec^4 3\theta d\theta$ [Hint: Apply a trigonometric identity.]

12. $\int x^3 \sqrt{5 + x^4} dx$

14. $\int \cos \frac{x}{3} dx$

16. $\int \sec^2 5x dx$

18. $\int \frac{x}{\sqrt{4 - 5x^2}} dx$

20. $\int \frac{x^2 + 1}{\sqrt{x^3 + 3x}} dx$

22. $\int \frac{\sin(1/x)}{3x^2} dx$

24. $\int \frac{\sec^2(\sqrt{x})}{\sqrt{x}} dx$

26. $\int \cos 2t \sin^5 2t dt$

28. $\int \frac{\cos 4\theta}{(1 + 2 \sin 4\theta)^4} d\theta$

30. $\int \tan^3 5x \sec^2 5x dx$

32. $\int [\sin(\sin \theta)] \cos \theta d\theta$

34. $\int x\sqrt{4 - x} dx$

37–39 Evaluate the integrals assuming that n is a positive integer and $b \neq 0$. ■

37. $\int (a + bx)^n dx$

38. $\int \sqrt[n]{a + bx} dx$

39. $\int \sin^n(a + bx) \cos(a + bx) dx$

- **C** 40. Use a CAS to check the answers you obtained in Exercises 37–39. If the answer produced by the CAS does not match yours, show that the two answers are equivalent. [Suggestion: *Mathematica* users may find it helpful to apply the *Simplify* command to the answer.]

FOCUS ON CONCEPTS

9. Explain the connection between the chain rule for differentiation and the method of u -substitution for integration.
10. Explain how the substitution $u = ax + b$ helps to perform an integration in which the integrand is $f(ax + b)$, where $f(x)$ is an easy to integrate function.

11–36 Evaluate the integrals using appropriate substitutions. ■

FOCUS ON CONCEPTS

41. (a) Evaluate the integral $\int \sin x \cos x \, dx$ by two methods: first by letting $u = \sin x$, and then by letting $u = \cos x$.
 (b) Explain why the two apparently different answers obtained in part (a) are really equivalent.
42. (a) Evaluate the integral $\int (5x - 1)^2 \, dx$ by two methods: first square and integrate, then let $u = 5x - 1$.
 (b) Explain why the two apparently different answers obtained in part (a) are really equivalent.

43–44 Solve the initial-value problems. ■

43. $\frac{dy}{dx} = \sqrt{5x + 1}$, $y(3) = -2$

44. $\frac{dy}{dx} = 2 + \sin 3x$, $y(\pi/3) = 0$

45. (a) Evaluate $\int [x/\sqrt{x^2 + 1}] \, dx$.
 (b) Use a graphing utility to generate some typical integral curves of $f(x) = x/\sqrt{x^2 + 1}$ over the interval $(-5, 5)$.

46. (a) Evaluate $\int 2x \sin(25 - x^2) \, dx$.
 (b) Use a graphing utility to generate some typical integral curves of $f(x) = 2x \sin(25 - x^2)$ over the interval $(-5, 5)$.

47. Find a function f such that the slope of the tangent line at a point (x, y) on the curve $y = f(x)$ is $\sqrt{3x + 1}$ and the curve passes through the point $(0, 1)$.
48. A population of minnows in a lake is estimated to be 100,000 at the beginning of the year 2010. Suppose that t years after the beginning of 2010 the rate of growth of the population $p(t)$ (in thousands) is given by $p'(t) = (3 + 0.12t)^{3/2}$. Estimate the projected population at the beginning of the year 2015.
49. Let $y(t)$ denote the number of *E. coli* cells in a container of nutrient solution t minutes after the start of an experiment. Assume that $y(t)$ is modeled by the initial-value problem

$$\frac{dy}{dt} = 0.95(0.79 + 0.024t)^{3/2}, \quad y(0) = 20$$

Use this model to estimate the number of *E. coli* cells in the container 20 minutes after the start of the experiment.

50. **Writing** If you want to evaluate an integral by u -substitution, how do you decide what part of the integrand to choose for u ?
51. **Writing** The evaluation of an integral can sometimes result in apparently different answers (Exercises 41 and 42). Explain why this occurs and give an example. How might you show that two apparently different answers are actually equivalent?

✓ QUICK CHECK ANSWERS 4.3

1. (a) $1 + x^3$; $3x^2 \, dx$ (b) x^2 ; $2x \, dx$ (c) $1 + 9x^2$; $18x \, dx$ 2. (a) $u^{-1/3}$ (b) $-u$ (c) $2\sqrt[3]{u}$

4.4 THE DEFINITION OF AREA AS A LIMIT; SIGMA NOTATION

Our main goal in this section is to use the rectangle method to give a precise mathematical definition of the “area under a curve.”

■ SIGMA NOTATION

To simplify our computations, we will begin by discussing a useful notation for expressing lengthy sums in a compact form. This notation is called **sigma notation** or **summation notation** because it uses the uppercase Greek letter Σ (sigma) to denote various kinds of sums. To illustrate how this notation works, consider the sum

$$1^2 + 2^2 + 3^2 + 4^2 + 5^2$$

in which each term is of the form k^2 , where k is one of the integers from 1 to 5. In sigma notation this sum can be written as

$$\sum_{k=1}^5 k^2$$

which is read “the summation of k^2 , where k runs from 1 to 5.” The notation tells us to form the sum of the terms that result when we substitute successive integers for k in the expression k^2 , starting with $k = 1$ and ending with $k = 5$.

6



Craig Lovell/Corbis Images

The growth and decline of animal populations and natural resources can be modeled using basic functions studied in this chapter.

EXPONENTIAL, LOGARITHMIC, AND INVERSE TRIGONOMETRIC FUNCTIONS

We begin this chapter with a review of exponential and logarithmic functions. These functions have important applications, from modeling population growth and the spread of disease, to the measurement of the magnitude of an earthquake or the perceived loudness of a sound. Logarithmic and exponential functions are best understood within the context of inverse functions and we will derive an important relationship between the derivative of a function and the derivative of its inverse. This connection will allow us to compute derivative formulas for logarithmic and exponential functions, along with their associated integration formulas. Later in the chapter we will exploit this connection again, to find the derivatives of inverse trigonometric functions, together with some related integration formulas. Along the way, we will discuss L'Hôpital's rule, a powerful tool for evaluating limits. We conclude the chapter with a study of some important combinations of exponential functions known as "hyperbolic functions."

6.1 EXPONENTIAL AND LOGARITHMIC FUNCTIONS

When logarithms were introduced in the seventeenth century as a computational tool, they provided scientists of that period computing power that was previously unimaginable. Although computers and calculators have replaced logarithm tables for numerical calculations, the logarithmic functions have wide-ranging applications in mathematics and science. In this section we will review some properties of exponents and logarithms and then develop results about exponential and logarithmic functions.

■ IRRATIONAL EXPONENTS

Recall from algebra that if b is a nonzero real number, then nonzero *integer* powers of b are defined by

$$b^n = \underbrace{b \times b \times \cdots \times b}_{n \text{ factors}} \quad \text{and} \quad b^{-n} = \frac{1}{b^n}$$

and if $n = 0$, then $b^0 = 1$. Also, if p/q is a positive *rational* number expressed in lowest terms, then

$$b^{p/q} = \sqrt[q]{b^p} = (\sqrt[q]{b})^p \quad \text{and} \quad b^{-p/q} = \frac{1}{b^{p/q}}$$

If b is negative, then some fractional powers of b will have imaginary values—the quantity $(-2)^{1/2} = \sqrt{-2}$, for example. To avoid this complication, we will assume throughout this section that $b > 0$, even if it is not stated explicitly.

50. The equation $Q = 12e^{-0.055t}$ gives the mass Q in grams of radioactive potassium-42 that will remain from some initial quantity after t hours of radioactive decay.
- How many grams were there initially?
 - How many grams remain after 4 hours?
 - How long will it take to reduce the amount of radioactive potassium-42 to half of the initial amount?

51. The acidity of a substance is measured by its pH value, which is defined by the formula

$$\text{pH} = -\log[H^+]$$

where the symbol $[H^+]$ denotes the concentration of hydrogen ions measured in moles per liter. Distilled water has a pH of 7; a substance is called *acidic* if it has $\text{pH} < 7$ and *basic* if it has $\text{pH} > 7$. Find the pH of each of the following substances and state whether it is acidic or basic.

	SUBSTANCE	$[H^+]$
(a)	Arterial blood	3.9×10^{-8} mol/L
(b)	Tomatoes	6.3×10^{-5} mol/L
(c)	Milk	4.0×10^{-7} mol/L
(d)	Coffee	1.2×10^{-6} mol/L

52. Use the definition of pH in Exercise 51 to find $[H^+]$ in a solution having a pH equal to
- 2.44
 - 8.06.
53. The perceived loudness β of a sound in decibels (dB) is related to its intensity I in watts per square meter (W/m^2) by the equation

$$\beta = 10 \log(I/I_0)$$

where $I_0 = 10^{-12} \text{ W/m}^2$. Damage to the average ear occurs at 90 dB or greater. Find the decibel level of each of the following sounds and state whether it will cause ear damage.

	SOUND	I
(a)	Jet aircraft (from 50 ft)	$1.0 \times 10^2 \text{ W/m}^2$
(b)	Amplified rock music	1.0 W/m^2
(c)	Garbage disposal	$1.0 \times 10^{-4} \text{ W/m}^2$
(d)	TV (mid volume from 10 ft)	$3.2 \times 10^{-5} \text{ W/m}^2$

54–56 Use the definition of the decibel level of a sound (see Exercise 51). ■

54. If one sound is three times as intense as another, how much greater is its decibel level?
55. According to one source, the noise inside a moving automobile is about 70 dB, whereas an electric blender generates 93 dB. Find the ratio of the intensity of the noise of the blender to that of the automobile.
56. Suppose that the intensity level of an echo is $\frac{2}{3}$ the intensity level of the original sound. If each echo results in another echo, how many echoes will be heard from a 120 dB sound given that the average human ear can hear a sound as low as 10 dB?
57. On the *Richter scale*, the magnitude M of an earthquake is related to the released energy E in joules (J) by the equation

$$\log E = 4.4 + 1.5M$$

- Find the energy E of the 1906 San Francisco earthquake that registered $M = 8.2$ on the Richter scale.
 - If the released energy of one earthquake is 10 times that of another, how much greater is its magnitude on the Richter scale?
58. Suppose that the magnitudes of two earthquakes differ by 1 on the Richter scale. Find the ratio of the released energy of the larger earthquake to that of the smaller earthquake. [Note: See Exercise 57 for terminology.]

✓ QUICK CHECK ANSWERS 6.1

1. $(-\infty, +\infty)$; $(0, +\infty)$ 2. $(-\infty, 1)$; $(-\infty, +\infty)$ 3. (a) 4^0 (b) $4^{1/2}$ (c) 4^{-2} (d) $4^{3/4}$ (e) $4^{\log_4 5}$ 4. (a) $\ln \frac{1}{2} = -\ln 2$ (b) 2 (c) $\ln 2$ 5. (a) e^3 (b) 101 (c) 2

6.2 DERIVATIVES AND INTEGRALS INVOLVING LOGARITHMIC FUNCTIONS

In this section we will obtain derivative formulas for logarithmic functions, and we will explain why the natural logarithm function is preferred in calculus over logarithms with other bases. The derivative formulas that we derive will allow us to find and use corresponding integral formulas.

■ DERIVATIVES OF LOGARITHMIC FUNCTIONS

We begin by establishing that $f(x) = \ln x$ is differentiable for $x > 0$ by using the derivative definition to find its derivative. To obtain this derivative, we need the fact that $\ln x$ is continuous for $x > 0$. Since e^x is continuous by Theorem 6.1.1(b), we know that $\ln x$ is

continuous for $x > 0$ by Theorem 1.5.7. We will also need the limit

$$\lim_{v \rightarrow 0} (1 + v)^{1/v} = e \quad (1)$$

that was given in Formula (6) of Section 6.1 (with x rather than v as the variable). Using the definition of the derivative, we obtain

$$\begin{aligned} \frac{d}{dx} [\ln x] &= \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln x}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \ln \left(\frac{x+h}{x} \right) && \text{The quotient property of} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \ln \left(1 + \frac{h}{x} \right) && \text{logarithms in Theorem 6.1.3} \\ &= \lim_{v \rightarrow 0} \frac{1}{vx} \ln(1+v) && \text{Let } v = h/x \text{ and note that} \\ &= \frac{1}{x} \lim_{v \rightarrow 0} \frac{1}{v} \ln(1+v) && \text{ } v \rightarrow 0 \text{ if and only if } h \rightarrow 0. \\ &= \frac{1}{x} \lim_{v \rightarrow 0} \ln(1+v)^{1/v} && \text{ } x \text{ is fixed in this limit computation, so } 1/x \\ &= \frac{1}{x} \ln \left[\lim_{v \rightarrow 0} (1+v)^{1/v} \right] && \text{The power property of} \\ &= \frac{1}{x} \ln e && \text{logarithms in Theorem 6.1.3} \\ &= \frac{1}{x} && \text{ } \ln x \text{ is continuous on } (0, +\infty) \text{ so we can} \\ & && \text{move the limit through the function symbol.} \\ & && \text{Since } \ln e = 1 \end{aligned}$$

Thus,

$$\frac{d}{dx} [\ln x] = \frac{1}{x}, \quad x > 0 \quad (2)$$

A derivative formula for the general logarithmic function $\log_b x$ can be obtained from (2) by using Formula (9) of Section 6.1 to write

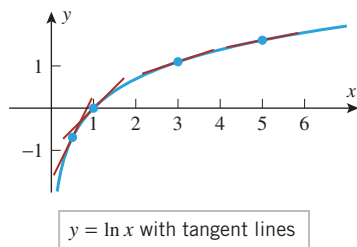
$$\frac{d}{dx} [\log_b x] = \frac{d}{dx} \left[\frac{\ln x}{\ln b} \right] = \frac{1}{\ln b} \frac{d}{dx} [\ln x]$$

It follows from this that

$$\frac{d}{dx} [\log_b x] = \frac{1}{x \ln b}, \quad x > 0 \quad (3)$$

Note that, among all possible bases, the base $b = e$ produces the simplest formula for the derivative of $\log_b x$. This is one of the reasons why the natural logarithm function is preferred over other logarithms in calculus.

► Example 1



▲ Figure 6.2.1

- (a) Figure 6.2.1 shows the graph of $y = \ln x$ and its tangent lines at the points $x = \frac{1}{2}$, 1, 3, and 5. Find the slopes of those tangent lines.
- (b) Does the graph of $y = \ln x$ have any horizontal tangent lines? Use the derivative of $\ln x$ to justify your answer.

Solution (a). From (2), the slopes of the tangent lines at the points $x = \frac{1}{2}$, 1, 3, and 5 are $1/x = 2$, 1, $\frac{1}{3}$, and $\frac{1}{5}$, respectively, which is consistent with Figure 6.2.1.

Solution (b). It does not appear from the graph of $y = \ln x$ that there are any horizontal tangent lines. This is confirmed by the fact that $dy/dx = 1/x$ is not equal to zero for any real value of x . ◀

If u is a differentiable function of x , and if $u(x) > 0$, then applying the chain rule to (2) and (3) produces the following generalized derivative formulas:

$$\frac{d}{dx}[\ln u] = \frac{1}{u} \cdot \frac{du}{dx} \quad \text{and} \quad \frac{d}{dx}[\log_b u] = \frac{1}{u \ln b} \cdot \frac{du}{dx} \quad (4-5)$$

► **Example 2** Find $\frac{d}{dx}[\ln(x^2 + 1)]$.

Solution. Using (4) with $u = x^2 + 1$ we obtain

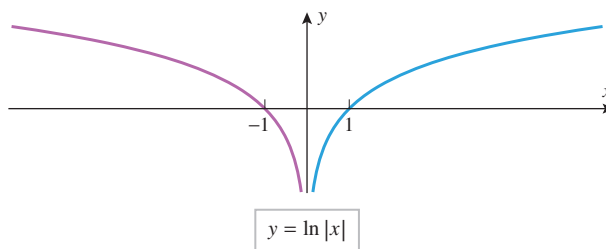
$$\frac{d}{dx}[\ln(x^2 + 1)] = \frac{1}{x^2 + 1} \cdot \frac{d}{dx}[x^2 + 1] = \frac{1}{x^2 + 1} \cdot 2x = \frac{2x}{x^2 + 1} \quad \blacktriangleleft$$

When possible, the properties of logarithms in Theorem 6.1.3 should be used to convert products, quotients, and exponents into sums, differences, and constant multiples *before* differentiating a function involving logarithms.

► **Example 3**

$$\begin{aligned} \frac{d}{dx} \left[\ln \left(\frac{x^2 \sin x}{\sqrt{1+x}} \right) \right] &= \frac{d}{dx} \left[2 \ln x + \ln(\sin x) - \frac{1}{2} \ln(1+x) \right] \\ &= \frac{2}{x} + \frac{\cos x}{\sin x} - \frac{1}{2(1+x)} \\ &= \frac{2}{x} + \cot x - \frac{1}{2+2x} \quad \blacktriangleleft \end{aligned}$$

Figure 6.2.2 shows the graph of $f(x) = \ln|x|$. This function is important because it “extends” the domain of the natural logarithm function in the sense that the values of $\ln|x|$ and $\ln x$ are the same for $x > 0$, but $\ln|x|$ is defined for all nonzero values of x , and $\ln x$ is only defined for positive values of x .



► **Figure 6.2.2**

The derivative of $\ln|x|$ for $x \neq 0$ can be obtained by considering the cases $x > 0$ and $x < 0$ separately:

Case $x > 0$. In this case $|x| = x$, so

$$\frac{d}{dx}[\ln|x|] = \frac{d}{dx}[\ln x] = \frac{1}{x}$$

Case $x < 0$. In this case $|x| = -x$, so it follows from (4) that

$$\frac{d}{dx}[\ln |x|] = \frac{d}{dx}[\ln(-x)] = \frac{1}{(-x)} \cdot \frac{d}{dx}[-x] = \frac{1}{x}$$

Since the same formula results in both cases, we have shown that

$$\frac{d}{dx}[\ln |x|] = \frac{1}{x} \quad \text{if } x \neq 0 \quad (6)$$

► **Example 4** From (6) and the chain rule,

$$\frac{d}{dx}[\ln |\sin x|] = \frac{1}{\sin x} \cdot \frac{d}{dx}[\sin x] = \frac{\cos x}{\sin x} = \cot x \quad \blacktriangleleft$$

LOGARITHMIC DIFFERENTIATION

We now consider a technique called *logarithmic differentiation* that is useful for differentiating functions that are composed of products, quotients, and powers.

► **Example 5** The derivative of

$$y = \frac{x^2 \sqrt[3]{7x-14}}{(1+x^2)^4} \quad (7)$$

is messy to calculate directly. However, if we first take the natural logarithm of both sides and then use its properties, we can write

$$\ln y = 2 \ln x + \frac{1}{3} \ln(7x-14) - 4 \ln(1+x^2)$$

Differentiating both sides with respect to x yields

$$\frac{1}{y} \frac{dy}{dx} = \frac{2}{x} + \frac{7/3}{7x-14} - \frac{8x}{1+x^2}$$

Thus, on solving for dy/dx and using (7) we obtain

$$\frac{dy}{dx} = \frac{x^2 \sqrt[3]{7x-14}}{(1+x^2)^4} \left[\frac{2}{x} + \frac{1}{3x-6} - \frac{8x}{1+x^2} \right] \quad \blacktriangleleft$$

REMARK Since $\ln y$ is only defined for $y > 0$, the computations in Example 5 are only valid for $x > 2$ (verify). However, because the derivative of $\ln y$ is the same as the derivative of $\ln |y|$, and because $\ln |y|$ is defined for $y < 0$ as well as $y > 0$, it follows that the formula obtained for dy/dx is valid for $x < 2$ as well as $x > 2$. In general, whenever a derivative dy/dx is obtained by logarithmic differentiation, the resulting derivative formula will be valid for all values of x for which $y \neq 0$. It may be valid at those points as well, but it is not guaranteed.

INTEGRALS INVOLVING $\ln x$

Formula (2) states that the function $\ln x$ is an antiderivative of $1/x$ on the interval $(0, +\infty)$, whereas Formula (6) states that the function $\ln |x|$ is an antiderivative of $1/x$ on each of the intervals $(-\infty, 0)$ and $(0, +\infty)$. Thus we have the companion integration formula to (6),

$$\int \frac{1}{u} du = \ln |u| + C \quad (8)$$

with the implicit understanding that this formula is applicable only across an interval that does not contain 0.

► **Example 6** Applying Formula (8),

$$\begin{aligned}\int_1^e \frac{1}{x} dx &= \ln |x| \Big|_1^e = \ln |e| - \ln |1| = 1 - 0 = 1 \\ \int_{-e}^{-1} \frac{1}{x} dx &= \ln |x| \Big|_{-e}^{-1} = \ln |-1| - \ln |-e| = 0 - 1 = -1 \quad \blacktriangleleft\end{aligned}$$

► **Example 7** Evaluate $\int \frac{3x^2}{x^3 + 5} dx$.

Solution. Make the substitution

$$u = x^3 + 5, \quad du = 3x^2 dx$$

so that

$$\int \frac{3x^2}{x^3 + 5} dx = \int \frac{1}{u} du = \ln |u| + C = \ln |x^3 + 5| + C \quad \blacktriangleleft$$

Formula (8)

► **Example 8** Evaluate $\int \tan x dx$.

Solution.

$$\int \tan x dx = \int \frac{\sin x}{\cos x} dx = - \int \frac{1}{u} du = -\ln |u| + C = -\ln |\cos x| + C \quad \blacktriangleleft$$

$u = \cos x$
 $du = -\sin x dx$

REMARK The last two examples illustrate an important point: any integral of the form

$$\int \frac{g'(x)}{g(x)} dx$$

(where the numerator of the integrand is the derivative of the denominator) can be evaluated by the u -substitution $u = g(x)$, $du = g'(x) dx$, since this substitution yields

$$\int \frac{g'(x)}{g(x)} dx = \int \frac{du}{u} = \ln |u| + C = \ln |g(x)| + C$$

DERIVATIVES OF REAL POWERS OF x

We know from Theorem 2.3.2 and Exercise 82 in Section 2.3 that the differentiation formula

$$\frac{d}{dx}[x^r] = rx^{r-1} \quad (9)$$

In the next section we will discuss differentiating functions that have exponents which are not constant.

holds for constant integer values of r . We will now use logarithmic differentiation to show that this formula holds if r is *any* real number (rational or irrational). In our computations we will assume that x^r is a differentiable function and that the familiar laws of exponents hold for real exponents.

Let $y = x^r$, where r is a real number. The derivative dy/dx can be obtained by logarithmic differentiation as follows:

$$\ln |y| = \ln |x^r| = r \ln |x|$$

$$\frac{d}{dx}[\ln |y|] = \frac{d}{dx}[r \ln |x|]$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{r}{x}$$

$$\frac{dy}{dx} = \frac{r}{x} y = \frac{r}{x} x^r = r x^{r-1}$$

✓ QUICK CHECK EXERCISES 6.2 (See page 427 for answers.)

- The equation of the tangent line to the graph of $y = \ln x$ at $x = e^2$ is _____.
- Find dy/dx .
 (a) $y = \ln 3x$ (b) $y = \ln \sqrt{x}$
 (c) $y = \log(1/|x|)$
- Use logarithmic differentiation to find the derivative of

$$f(x) = \frac{\sqrt{x+1}}{\sqrt[3]{x-1}}$$

- $\lim_{h \rightarrow 0} \frac{\ln(1+h)}{h} = \underline{\hspace{2cm}}$
- $\int_2^5 \frac{1}{t} dt = \underline{\hspace{2cm}}$

EXERCISE SET 6.2

1–26 Find dy/dx . ■

- | | |
|---|---|
| 1. $y = \ln 5x$ | 2. $y = \ln \frac{x}{3}$ |
| 3. $y = \ln 1+x $ | 4. $y = \ln(2 + \sqrt{x})$ |
| 5. $y = \ln x^2 - 1 $ | 6. $y = \ln x^3 - 7x^2 - 3 $ |
| 7. $y = \ln \left(\frac{x}{1+x^2} \right)$ | 8. $y = \ln \left \frac{1+x}{1-x} \right $ |
| 9. $y = \ln x^2$ | 10. $y = (\ln x)^3$ |
| 11. $y = \sqrt{\ln x}$ | 12. $y = \ln \sqrt{x}$ |
| 13. $y = x \ln x$ | 14. $y = x^3 \ln x$ |
| 15. $y = x^2 \log_2(3 - 2x)$ | 16. $y = x[\log_2(x^2 - 2x)]^3$ |
| 17. $y = \frac{x^2}{1 + \log x}$ | 18. $y = \frac{\log x}{1 + \log x}$ |
| 19. $y = \ln(\ln x)$ | 20. $y = \ln(\ln(\ln x))$ |
| 21. $y = \ln(\tan x)$ | 22. $y = \ln(\cos x)$ |
| 23. $y = \cos(\ln x)$ | 24. $y = \sin^2(\ln x)$ |
| 25. $y = \log(\sin^2 x)$ | 26. $y = \log(1 - \sin^2 x)$ |

27–30 Use the method of Example 3 to help perform the indicated differentiation. ■

- $\frac{d}{dx}[\ln((x-1)^3(x^2+1)^4)]$
- $\frac{d}{dx}[\ln((\cos^2 x)\sqrt{1+x^4})]$

- $\frac{d}{dx} \left[\ln \frac{\cos x}{\sqrt{4-3x^2}} \right]$
- $\frac{d}{dx} \left[\ln \sqrt{\frac{x-1}{x+1}} \right]$

31–34 True–False Determine whether the statement is true or false. Explain your answer. ■

- The slope of the tangent line to the graph of $y = \ln x$ at $x = a$ approaches infinity as $a \rightarrow 0^+$.
- If $\lim_{x \rightarrow +\infty} f'(x) = 0$, then the graph of $y = f(x)$ has a horizontal asymptote.
- The derivative of $\ln |x|$ is an odd function.
- We have

$$\frac{d}{dx}((\ln x)^2) = \frac{d}{dx}(2(\ln x)) = \frac{2}{x}$$

35–38 Find dy/dx using logarithmic differentiation. ■

- | | |
|---|---|
| 35. $y = x \sqrt[3]{1+x^2}$ | 36. $y = \sqrt[5]{\frac{x-1}{x+1}}$ |
| 37. $y = \frac{(x^2-8)^{1/3} \sqrt{x^3+1}}{x^6-7x+5}$ | 38. $y = \frac{\sin x \cos x \tan^3 x}{\sqrt{x}}$ |
| 39. Find | |
| (a) $\frac{d}{dx}[\log_x e]$ | (b) $\frac{d}{dx}[\log_x 2]$ |
| 40. Find | |
| (a) $\frac{d}{dx}[\log_{(1/x)} e]$ | (b) $\frac{d}{dx}[\log_{(\ln x)} e]$ |

41–44 Find the equation of the tangent line to the graph of $y = f(x)$ at $x = x_0$. ■

41. $f(x) = \ln x$; $x_0 = e^{-1}$ 42. $f(x) = \log x$; $x_0 = 10$
 43. $f(x) = \ln(-x)$; $x_0 = -e$ 44. $f(x) = \ln|x|$; $x_0 = -2$

FOCUS ON CONCEPTS

45. (a) Find the equation of a line through the origin that is tangent to the graph of $y = \ln x$.
 (b) Explain why the y -intercept of a tangent line to the curve $y = \ln x$ must be 1 unit less than the y -coordinate of the point of tangency.
46. Use logarithmic differentiation to verify the product and quotient rules. Explain what properties of $\ln x$ are important for this verification.
47. Find a formula for the area $A(w)$ of the triangle bounded by the tangent line to the graph of $y = \ln x$ at $P(w, \ln w)$, the horizontal line through P , and the y -axis.
48. Find a formula for the area $A(w)$ of the triangle bounded by the tangent line to the graph of $y = \ln x^2$ at $P(w, \ln w^2)$, the horizontal line through P , and the y -axis.
49. Verify that $y = \ln(x + e)$ satisfies $dy/dx = e^{-y}$, with $y = 1$ when $x = 0$.
50. Verify that $y = -\ln(e^2 - x)$ satisfies $dy/dx = e^y$, with $y = -2$ when $x = 0$.
51. Find a function f such that $y = f(x)$ satisfies $dy/dx = e^{-y}$, with $y = 0$ when $x = 0$.
52. Find a function f such that $y = f(x)$ satisfies $dy/dx = e^y$, with $y = -\ln 2$ when $x = 0$.
53. Let p denote the number of paramecia in a nutrient solution t days after the start of an experiment, and assume that p is defined implicitly as a function of t by the equation
- $$0 = \ln p + 0.83 - \ln(2.3 - 0.0046p) - 2.3t$$
- Use implicit differentiation to show that the rate of change of p with respect to t satisfies the equation
- $$\frac{dp}{dt} = 0.0046p(500 - p)$$
54. Let p denote the population of the United States in the year t , and assume that p is defined implicitly as a function of t by the equation
- $$0 = \ln p + 45.817 - \ln(2225 - 4.2381p) - 0.02225t$$
- Use implicit differentiation to show that the rate of change of p with respect to t satisfies the equation
- $$\frac{dp}{dt} = 10^{-5}p(2225 - 4.2381p)$$

55–57 Find the limit by interpreting the expression as an appropriate derivative. ■

55. (a) $\lim_{x \rightarrow 0} \frac{\ln(1 + 3x)}{x}$ (b) $\lim_{x \rightarrow 0} \frac{\ln(1 - 5x)}{x}$
 56. (a) $\lim_{\Delta x \rightarrow 0} \frac{\ln(e^2 + \Delta x) - 2}{\Delta x}$ (b) $\lim_{w \rightarrow 1} \frac{\ln w}{w - 1}$
 57. (a) $\lim_{x \rightarrow 0} \frac{\ln(\cos x)}{x}$ (b) $\lim_{h \rightarrow 0} \frac{(1 + h)^{\sqrt{2}} - 1}{h}$

58. Modify the derivation of Equation (2) to give another proof of Equation (3).

59–60 Evaluate the integral and check your answer by differentiating. ■

59. $\int \left[\frac{2}{x} + 3 \sin x \right] dx$ 60. $\int \left[\frac{1}{2t} + 2t \right] dt$

61–62 Evaluate the integrals using the indicated substitutions. ■

61. $\int \frac{dx}{x \ln x}$; $u = \ln x$
 62. $\int \frac{\sin 3\theta}{1 + \cos 3\theta} d\theta$; $u = 1 + \cos 3\theta$

63–64 Evaluate the integrals using appropriate substitutions. ■

63. $\int \frac{x^4}{1 + x^5} dx$ 64. $\int \frac{dx}{2x}$

65–66 Evaluate each integral by first modifying the form of the integrand and then making an appropriate substitution, if needed. ■

65. $\int \frac{t + 1}{t} dt$ 66. $\int \cot x dx$

67–68 Evaluate the integrals. ■

67. $\int_0^2 \frac{3x}{1 + x^2} dx$ 68. $\int_{1/2}^1 \frac{1}{2x} dx$

69. Evaluate the definite integral by making the indicated u -substitution.

$$\int_e^{e^2} \frac{\ln x}{x} dx; \quad u = \ln x$$

70. Evaluate the definite integral by expressing it in terms of u and evaluating the resulting integral using a formula from geometry.

$$\int_{e^{-3}}^{e^3} \frac{\sqrt{9 - (\ln x)^2}}{x} dx; \quad u = \ln x$$

71–72 Evaluate the integrals by any method. ■

71. $\int_0^e \frac{dx}{2x + e}$ 72. $\int_0^{\pi/3} \frac{\sin x}{1 + \cos x} dx$

73. Solve the initial-value problem.

$$\frac{dy}{dt} = \frac{1}{t}, \quad y(-1) = 5$$

74. **Writing** Review the derivation of the formula

$$\frac{d}{dx} [\ln x] = \frac{1}{x}$$

and then write a paragraph that discusses all the ingredients (theorems, limit properties, etc.) that are needed for this derivation.

75. **Writing** Write a paragraph that explains how logarithmic differentiation can replace a difficult differentiation computation with a simpler computation.

QUICK CHECK ANSWERS 6.2

1. $y = \frac{x}{e^2} + 1$ 2. (a) $\frac{dy}{dx} = \frac{1}{x}$ (b) $\frac{dy}{dx} = \frac{1}{2x}$ (c) $\frac{dy}{dx} = -\frac{1}{x \ln 10}$ 3. $\frac{\sqrt{x+1}}{\sqrt[3]{x-1}} \left[\frac{1}{2(x+1)} - \frac{1}{3(x-1)} \right]$ 4. 1 5. $\ln\left(\frac{5}{2}\right)$

6.3 DERIVATIVES OF INVERSE FUNCTIONS; DERIVATIVES AND INTEGRALS INVOLVING EXPONENTIAL FUNCTIONS

See Section 0.4 for a review of one-to-one functions and inverse functions.

In this section we will show how the derivative of a one-to-one function can be used to obtain the derivative of its inverse function. This will provide the tools we need to obtain derivative formulas for exponential functions from the derivative formulas for logarithmic functions. We will then find the corresponding integration formulas for exponential functions.

DIFFERENTIABILITY OF INVERSE FUNCTIONS

Our first goal in this section is to obtain a formula that relates the derivative of an invertible function f to the derivative of its inverse function.

► **Example 1** Suppose that f is a one-to-one differentiable function such that $f(2) = 1$ and $f'(2) = \frac{3}{4}$. Then the tangent line to $y = f(x)$ at the point $(2, 1)$ has equation

$$y - 1 = \frac{3}{4}(x - 2)$$

Since the graph of $y = f^{-1}(x)$ is the reflection of the graph of $y = f(x)$ about the line $y = x$, the tangent line to $y = f^{-1}(x)$ at the point $(1, 2)$ is the reflection about the line $y = x$ of the tangent line to $y = f(x)$ at the point $(2, 1)$ (Figure 6.3.1). Its equation can be obtained from that of the tangent line to the graph of f by interchanging x and y :

$$x - 1 = \frac{3}{4}(y - 2) \quad \text{or} \quad y - 2 = \frac{4}{3}(x - 1)$$

Notice that the slope of the tangent line to $y = f^{-1}(x)$ at $x = 1$ is the reciprocal of the slope of the tangent line to $y = f(x)$ at $x = 2$. That is,

$$(f^{-1})'(1) = \frac{1}{f'(2)} = \frac{4}{3} \quad \blacktriangleleft \quad (1)$$

Since $2 = f^{-1}(1)$ for the function f in Example 1, it follows that $f'(2) = f'(f^{-1}(1))$. Thus, Formula (1) can also be expressed as

$$(f^{-1})'(1) = \frac{1}{f'(f^{-1}(1))}$$

In general, if f is a differentiable and one-to-one function, then

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} \quad (2)$$

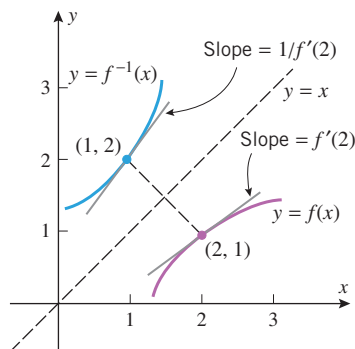
provided $f'(f^{-1}(x)) \neq 0$.

Formula (2) can be confirmed using implicit differentiation. The equation $y = f^{-1}(x)$ is equivalent to $x = f(y)$. Differentiating with respect to x we obtain

$$1 = \frac{d}{dx}[x] = \frac{d}{dx}[f(y)] = f'(y) \cdot \frac{dy}{dx}$$

so that

$$\frac{dy}{dx} = \frac{1}{f'(y)} = \frac{1}{f'(f^{-1}(x))}$$



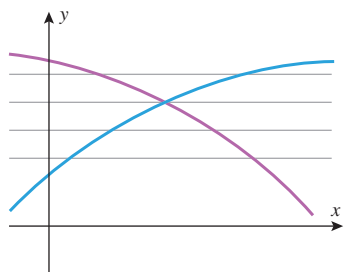
▲ Figure 6.3.1

Also from $x = f(y)$ we have $dx/dy = f'(y)$, which gives the following alternative version of Formula (2):

$$\frac{dy}{dx} = \frac{1}{dx/dy} \quad (3)$$

■ INCREASING OR DECREASING FUNCTIONS ARE ONE-TO-ONE

If the graph of a function f is always increasing or always decreasing over the domain of f , then a horizontal line will cut the graph of f in at most one point (Figure 6.3.2), so f must have an inverse function (see Section 0.4). In Theorem 3.1.2 we saw that f must be increasing on any interval on which $f'(x) > 0$ and must be decreasing on any interval on which $f'(x) < 0$. These observations, together with Formula (2), suggest the following theorem, which we state without formal proof.



▲ **Figure 6.3.2** The graph of an increasing function (blue) or a decreasing function (purple) is cut at most once by any horizontal line.

6.3.1 THEOREM Suppose that the domain of a function f is an open interval on which $f'(x) > 0$ or on which $f'(x) < 0$. Then f is one-to-one, $f^{-1}(x)$ is differentiable at all values of x in the range of f , and the derivative of $f^{-1}(x)$ is given by Formula (2).

In general, once it is established that f^{-1} is differentiable, one has the option of calculating the derivative of f^{-1} using Formula (2) or (3), or by differentiating implicitly.

► **Example 2** Consider the function $f(x) = x^5 + x + 1$.

- Show that f has a differentiable inverse function.
- Use implicit differentiation to find a formula for the derivative of f^{-1} .
- Compute $(f^{-1})'(1)$.

Solution (a). Since $f'(x) = 5x^4 + 1 > 0$

for all real values of x , it follows from Theorem 6.3.1 that f is one-to-one on the interval $(-\infty, +\infty)$ and has a differentiable inverse function.

Solution (b). Let $y = f^{-1}(x)$. Differentiating $x = f(y) = y^5 + y + 1$ implicitly with respect to x yields

$$\begin{aligned} \frac{d}{dx}[x] &= \frac{d}{dx}[y^5 + y + 1] \\ 1 &= (5y^4 + 1) \frac{dy}{dx} \\ \frac{dy}{dx} &= \frac{1}{5y^4 + 1} \end{aligned} \quad (4)$$

We cannot solve $x = y^5 + y + 1$ for y in terms of x , so we leave the expression for dy/dx in Equation (4) in terms of y .

Solution (c). From Equation (4),

$$(f^{-1})'(1) = \left. \frac{dy}{dx} \right|_{x=1} = \left. \frac{1}{5y^4 + 1} \right|_{x=1}$$

Thus, we need to know the value of $y = f^{-1}(x)$ at $x = 1$, which we can obtain by solving the equation $f(y) = 1$ for y . This equation is $y^5 + y + 1 = 1$, which, by inspection, is satisfied by $y = 0$. Thus,

$$(f^{-1})'(1) = \left. \frac{1}{5y^4 + 1} \right|_{y=0} = 1 \quad \blacktriangleleft$$

DERIVATIVES OF EXPONENTIAL FUNCTIONS

Our next objective is to show that the general exponential function b^x ($b > 0, b \neq 1$) is differentiable everywhere and to find its derivative. To do this, we will use the fact that b^x is the inverse of the function $f(x) = \log_b x$. We will assume that $b > 1$. With this assumption we have $\ln b > 0$, so

$$f'(x) = \frac{d}{dx}[\log_b x] = \frac{1}{x \ln b} > 0 \quad \text{for all } x \text{ in the interval } (0, +\infty)$$

It now follows from Theorem 6.3.1 that $f^{-1}(x) = b^x$ is differentiable for all x in the range of $f(x) = \log_b x$. But we know from Table 6.1.3 that the range of $\log_b x$ is $(-\infty, +\infty)$, so we have established that b^x is differentiable everywhere.

To obtain a derivative formula for b^x we rewrite $y = b^x$ as

$$x = \log_b y$$

and differentiate implicitly using Formula (5) of Section 6.2 to obtain

$$1 = \frac{1}{y \ln b} \cdot \frac{dy}{dx}$$

Solving for dy/dx and replacing y by b^x we have

$$\frac{dy}{dx} = y \ln b = b^x \ln b$$

Thus, we have shown that

$$\frac{d}{dx}[b^x] = b^x \ln b \quad (5)$$

In the special case where $b = e$ we have $\ln e = 1$, so that (5) becomes

$$\frac{d}{dx}[e^x] = e^x \quad (6)$$

Moreover, if u is a differentiable function of x , then it follows from (5) and (6) that

$$\frac{d}{dx}[b^u] = b^u \ln b \cdot \frac{du}{dx} \quad \text{and} \quad \frac{d}{dx}[e^u] = e^u \cdot \frac{du}{dx} \quad (7-8)$$

How does the derivation of Formula (5) change if $0 < b < 1$?

In Section 6.1 we stated that $b = e$ is the only base for which the slope of the tangent line to the curve $y = b^x$ at any point P on the curve is the y -coordinate at P (see page 412). Verify this statement.

It is important to distinguish between differentiating an exponential function b^x (variable exponent and constant base) and a power function x^b (variable base and constant exponent). For example, compare the derivative

$$\frac{d}{dx}[x^2] = 2x$$

to the derivative of 2^x in Example 3.

► **Example 3** The following computations use Formulas (5), (7) and (8).

$$\frac{d}{dx}[2^x] = 2^x \ln 2$$

$$\frac{d}{dx}[e^{-2x}] = e^{-2x} \cdot \frac{d}{dx}[-2x] = -2e^{-2x}$$

$$\frac{d}{dx}[e^{x^3}] = e^{x^3} \cdot \frac{d}{dx}[x^3] = 3x^2 e^{x^3}$$

$$\frac{d}{dx}[e^{\cos x}] = e^{\cos x} \cdot \frac{d}{dx}[\cos x] = -(\sin x)e^{\cos x} \quad \blacktriangleleft$$

Functions of the form $f(x) = u^v$ in which u and v are *nonconstant* functions of x are neither exponential functions nor power functions. Functions of this form can be differentiated using logarithmic differentiation.

► **Example 4** Use logarithmic differentiation to find $\frac{d}{dx}[(x^2 + 1)^{\sin x}]$.

Solution. Setting $y = (x^2 + 1)^{\sin x}$ we have

$$\ln y = \ln[(x^2 + 1)^{\sin x}] = (\sin x) \ln(x^2 + 1)$$

Differentiating both sides with respect to x yields

$$\begin{aligned} \frac{1}{y} \frac{dy}{dx} &= \frac{d}{dx}[(\sin x) \ln(x^2 + 1)] \\ &= (\sin x) \frac{1}{x^2 + 1} (2x) + (\cos x) \ln(x^2 + 1) \end{aligned}$$

Thus,

$$\begin{aligned} \frac{dy}{dx} &= y \left[\frac{2x \sin x}{x^2 + 1} + (\cos x) \ln(x^2 + 1) \right] \\ &= (x^2 + 1)^{\sin x} \left[\frac{2x \sin x}{x^2 + 1} + (\cos x) \ln(x^2 + 1) \right] \quad \blacktriangleleft \end{aligned}$$

■ INTEGRALS INVOLVING EXPONENTIAL FUNCTIONS

Associated with derivatives (7) and (8) are the companion integration formulas

$$\int b^u du = \frac{b^u}{\ln b} + C \quad \text{and} \quad \int e^u du = e^u + C \quad (9-10)$$

► **Example 5**

$$\int 2^x dx = \frac{2^x}{\ln 2} + C \quad \blacktriangleleft$$

► **Example 6** Evaluate $\int e^{5x} dx$.

Solution. Let $u = 5x$ so that $du = 5 dx$ or $dx = \frac{1}{5} du$, which yields

$$\int e^{5x} dx = \frac{1}{5} \int e^u du = \frac{1}{5} e^u + C = \frac{1}{5} e^{5x} + C \quad \blacktriangleleft$$

► **Example 7** The following computations use Formula (10).

$$\int e^{-x} dx = - \int e^u du = -e^u + C = -e^{-x} + C$$

$$\begin{array}{l} u = -x \\ du = -dx \end{array}$$

$$\int x^2 e^{x^3} dx = \frac{1}{3} \int e^u du = \frac{1}{3} e^u + C = \frac{1}{3} e^{x^3} + C$$

$$\begin{array}{l} u = x^3 \\ du = 3x^2 dx \end{array}$$

$$\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx = 2 \int e^u du = 2e^u + C = 2e^{\sqrt{x}} + C \quad \blacktriangleleft$$

$$\begin{array}{l} u = \sqrt{x} \\ du = \frac{1}{2\sqrt{x}} dx \end{array}$$

► **Example 8** Evaluate $\int_0^{\ln 3} e^x (1 + e^x)^{1/2} dx$.

Solution. Make the u -substitution

$$u = 1 + e^x, \quad du = e^x dx$$

and change the x -limits of integration ($x = 0$, $x = \ln 3$) to the u -limits

$$u = 1 + e^0 = 2, \quad u = 1 + e^{\ln 3} = 1 + 3 = 4$$

This yields

$$\int_0^{\ln 3} e^x (1 + e^x)^{1/2} dx = \int_2^4 u^{1/2} du = \left. \frac{2}{3} u^{3/2} \right|_2^4 = \frac{2}{3} [4^{3/2} - 2^{3/2}] = \frac{16 - 4\sqrt{2}}{3} \quad \blacktriangleleft$$

✓ QUICK CHECK EXERCISES 6.3 (See page 434 for answers.)

- Suppose that a one-to-one function f has tangent line $y = 5x + 3$ at the point $(1, 8)$. Evaluate $(f^{-1})'(8)$.
- In each case, from the given derivative, determine whether the function f is invertible.
 - $f'(x) = x^2 + 1$
 - $f'(x) = x^2 - 1$
 - $f'(x) = \sin x$
 - $f'(x) = \frac{1}{2} - e^{x^2}$
- Evaluate the derivative.
 - $\frac{d}{dx}[e^x]$
 - $\frac{d}{dx}[7^x]$
 - $\frac{d}{dx}[\cos(e^x + 1)]$
 - $\frac{d}{dx}[e^{3x-2}]$
- Let $f(x) = e^{x^3+x}$. Use $f'(x)$ to verify that f is one-to-one.
- $\int_0^{\frac{1}{2}\ln 5} e^x dx = \underline{\hspace{2cm}}$

EXERCISE SET 6.3 Graphing Utility

FOCUS ON CONCEPTS

- Let $f(x) = x^5 + x^3 + x$.
 - Show that f is one-to-one and confirm that $f(1) = 3$.
 - Find $(f^{-1})'(3)$.
- Let $f(x) = x^3 + 2e^x$.
 - Show that f is one-to-one and confirm that $f(0) = 2$.
 - Find $(f^{-1})'(2)$.

3–4 Find $(f^{-1})'(x)$ using Formula (2), and check your answer by differentiating f^{-1} directly. ■

- $f(x) = 2/(x+3)$
- $f(x) = \ln(2x+1)$

5–6 Determine whether the function f is one-to-one by examining the sign of $f'(x)$. ■

- $f(x) = x^2 + 8x + 1$
 - $f(x) = 2x^5 + x^3 + 3x + 2$
 - $f(x) = 2x + \sin x$
 - $f(x) = \left(\frac{1}{2}\right)^x$
- $f(x) = x^3 + 3x^2 - 8$
 - $f(x) = x^5 + 8x^3 + 2x - 1$
 - $f(x) = \frac{x}{x+1}$
 - $f(x) = \log_b x, \quad 0 < b < 1$

7–10 Find the derivative of f^{-1} by using Formula (3), and check your result by differentiating implicitly. ■

- $f(x) = 5x^3 + x - 7$
- $f(x) = 1/x^2, \quad x > 0$
- $f(x) = 2x^5 + x^3 + 1$
- $f(x) = 5x - \sin 2x, \quad -\frac{\pi}{4} < x < \frac{\pi}{4}$

FOCUS ON CONCEPTS

- Figure 0.4.8 is a “proof by picture” that the reflection of a point $P(a, b)$ about the line $y = x$ is the point $Q(b, a)$. Establish this result rigorously by completing each part.
 - Prove that if P is not on the line $y = x$, then P and Q are distinct, and the line $\overleftrightarrow{PQ}$ is perpendicular to the line $y = x$.
 - Prove that if P is not on the line $y = x$, the midpoint of segment PQ is on the line $y = x$.
 - Carefully explain what it means geometrically to reflect P about the line $y = x$.
 - Use the results of parts (a)–(c) to prove that Q is the reflection of P about the line $y = x$.
- Prove that the reflection about the line $y = x$ of a line with slope m , $m \neq 0$, is a line with slope $1/m$. [Hint: Apply the result of the previous exercise to a pair of points on the line of slope m and to a corresponding

pair of points on the reflection of this line about the line $y = x$.]

13. Suppose that f and g are increasing functions. Determine which of the functions $f(x) + g(x)$, $f(x)g(x)$, and $f(g(x))$ must also be increasing.
14. Suppose that f and g are one-to-one functions. Determine which of the functions $f(x) + g(x)$, $f(x)g(x)$, and $f(g(x))$ must also be one-to-one.

15–26 Find dy/dx . ■

15. $y = e^{7x}$ 16. $y = e^{-5x^2}$
 17. $y = x^3 e^x$ 18. $y = e^{1/x}$
 19. $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ 20. $y = \sin(e^x)$
 21. $y = e^{x \tan x}$ 22. $y = \frac{e^x}{\ln x}$
 23. $y = e^{(x - e^{3x})}$ 24. $y = \exp(\sqrt{1 + 5x^3})$
 25. $y = \ln(1 - xe^{-x})$ 26. $y = \ln(\cos e^x)$

27–30 Find $f'(x)$ by Formula (7) and then by logarithmic differentiation. ■

27. $f(x) = 2^{x^2}$ 28. $f(x) = 3^{-x}$
 29. $f(x) = \pi^{\sin x}$ 30. $f(x) = \pi^{x \tan x}$

31–35 Find dy/dx using the method of logarithmic differentiation. ■

31. $y = (x^3 - 2x)^{\ln x}$ 32. $y = x^{\sin x}$
 33. $y = (\ln x)^{\tan x}$ 34. $y = (x^2 + 3)^{\ln x}$
 35. $y = (\ln x)^{\ln x}$

36. (a) Explain why Formula (5) cannot be used to find $(d/dx)[x^x]$.
 (b) Find this derivative by logarithmic differentiation.

37–42 Find dy/dx using any method. ■

37. $y = (x^3 - 2x^2 + 1)e^x$ 38. $y = (2x^2 - 2x + 1)e^{2x}$
 39. $y = (x^2 + \sqrt{x})3^x$ 40. $y = (x^3 + \sqrt[3]{x})5^x$
 41. $y = 4^{3 \sin x - e^x}$ 42. $y = 2^{\cos x + \ln x}$

43–46 True-False Determine whether the statement is true or false. Explain your answer. ■

43. If a function $y = f(x)$ satisfies $dy/dx = y$, then $y = e^x$.
 44. If $y = f(x)$ is a function such that dy/dx is a rational function, then $f(x)$ is also a rational function.
 45. $\frac{d}{dx}(\log_b |x|) = \frac{1}{x \ln b}$
 46. If the tangent line to the graph of $f(x) = b^x$ has slope 1 at the point $(0, 1)$, then f is the natural exponential function.
 47. (a) Show that $f(x) = x^3 - 3x^2 + 2x$ is not one-to-one on $(-\infty, +\infty)$.
 (b) Find the largest value of k such that f is one-to-one on the interval $(-k, k)$.

48. (a) Show that the function $f(x) = x^4 - 2x^3$ is not one-to-one on $(-\infty, +\infty)$.
 (b) Find the smallest value of k such that f is one-to-one on the interval $[k, +\infty)$.
 49. Let $f(x) = x^4 + x^3 + 1$, $0 \leq x \leq 2$.
 (a) Show that f is one-to-one.
 (b) Let $g(x) = f^{-1}(x)$ and define $F(x) = f(2g(x))$. Find an equation for the tangent line to $y = F(x)$ at $x = 3$.

50. Let $f(x) = \frac{\exp(4 - x^2)}{x}$, $x > 0$.
 (a) Show that f is one-to-one.
 (b) Let $g(x) = f^{-1}(x)$ and define $F(x) = f([g(x)]^2)$. Find $F'(\frac{1}{2})$.

51. Show that for any constants A and k , the function $y = Ae^{kt}$ satisfies the equation $dy/dt = ky$.

52. Show that for any constants A and B , the function

$$y = Ae^{2x} + Be^{-4x}$$

satisfies the equation

$$y'' + 2y' - 8y = 0$$

53. Show that

- (a) $y = xe^{-x}$ satisfies the equation $xy' = (1 - x)y$
 (b) $y = xe^{-x^2/2}$ satisfies the equation $xy' = (1 - x^2)y$.

54. Show that the rate of change of $y = 100e^{-0.2x}$ with respect to x is proportional to y .

55. Suppose that the percentage of U.S. households with broadband Internet access is modeled by the equation

$$P(t) = \frac{5300}{53 + 47e^{-0.182t}}$$

where $P(t)$ is the percentage t years after an initial survey result made in the year 2007.

- (a) Use a graphing utility to graph the function $P(t)$.
 (b) In words, explain what happens to the percentage over time. Check your conclusion by finding $\lim_{t \rightarrow +\infty} P(t)$.
 (c) In words, what happens to the *rate* of population growth over time? Check your conclusion by graphing $P'(t)$.

56. Suppose that the population of oxygen-dependent bacteria in a pond is modeled by the equation

$$P(t) = \frac{60}{5 + 7e^{-t}}$$

where $P(t)$ is the population (in billions) t days after an initial observation at time $t = 0$.

- (a) Use a graphing utility to graph the function $P(t)$.
 (b) In words, explain what happens to the population over time. Check your conclusion by finding $\lim_{t \rightarrow +\infty} P(t)$.
 (c) In words, what happens to the *rate* of population growth over time? Check your conclusion by graphing $P'(t)$.

57–59 Find the limit by interpreting the expression as an appropriate derivative. ■

57. $\lim_{x \rightarrow 0} \frac{e^{3x} - 1}{x}$ 58. $\lim_{x \rightarrow 0} \frac{\exp(x^2) - 1}{x}$

59. $\lim_{h \rightarrow 0} \frac{10^h - 1}{h}$

60. Suppose that a steel ball bearing is released within a vat of fluid and begins to sink. According to one model, the speed $v(t)$ (in m/s) of the ball bearing t seconds after its release is given by the formula

$$v(t) = \frac{9.8(1 - e^{-kt})}{k}$$

where k is a positive constant that corresponds to the resistance the fluid offers against the motion of the bearing. (The smaller the value of k , the weaker will be the resistance.) For t fixed, determine the limiting value of the speed as $k \rightarrow 0^+$, and give a physical interpretation of the limit. [Hint: Interpret the limit as an appropriate derivative.]

- 61–62 Evaluate the integral and check your answer by differentiating. ■

61. $\int \left[\frac{2}{x} + 3e^x \right] dx$ 62. $\int \left[\frac{1}{2t} - \sqrt{2}e^t \right] dt$

- 63–64 Evaluate the integrals using the indicated substitutions. ■

63. $\int e^{-5x} dx; u = -5x$ 64. $\int \frac{e^x}{1 + e^x} dx; u = 1 + e^x$

- 65–72 Evaluate the integrals using appropriate substitutions. ■

65. $\int e^{2x} dx$ 66. $\int e^{-x/2} dx$ 67. $\int e^{\sin x} \cos x dx$

68. $\int x^3 e^{x^4} dx$ 69. $\int x^2 e^{-2x^3} dx$ 70. $\int \frac{e^x + e^{-x}}{e^x - e^{-x}} dx$

71. $\int \frac{dx}{e^x}$ 72. $\int \sqrt{e^x} dx$

- 73–74 Evaluate each integral by first modifying the form of the integrand. ■

73. $\int [\ln(e^x) + \ln(e^{-x})] dx$ 74. $\int e^{2 \ln x} dx$

- 75–76 Evaluate the integrals. ■

75. $\int_{\ln 2}^3 5e^x dx$ 76. $\int_0^1 (e^x - x) dx$

77. Evaluate the definite integral by making the indicated u -substitution.

$$\int_0^1 e^{2x-1} dx; \quad u = 2x - 1$$

- 78–80 Evaluate the integrals by any method. ■

78. $\int_0^{\ln 5} e^x (3 - 4e^x) dx$ 79. $\int_{-\ln 3}^{\ln 3} \frac{e^x}{e^x + 4} dx$

80. $\int_1^{\sqrt{2}} x e^{-x^2} dx$

81–84 Medication can be administered to a patient in different ways. For a given method, let $c(t)$ denote the concentration of medication in the patient's bloodstream (measured in mg/L) t hours after the dose is given. Over the time interval $0 \leq t \leq b$, the area between the graph of $c = c(t)$ and the interval $[0, b]$ indicates the “availability” of the medication for the patient's body over that time period. Determine which method provides the greater availability over the given interval. ■

81. Method 1: $c(t) = 5(e^{-0.2t} - e^{-t})$,
Method 2: $c(t) = 4(e^{-0.2t} - e^{-3t})$; $[0, 4]$

82. Method 1: $c(t) = 5(e^{-0.2t} - e^{-t})$,
Method 2: $c(t) = 4(e^{-0.2t} - e^{-3t})$; $[0, 24]$

83. Method 1: $c(t) = 5.78(e^{-0.4t} - e^{-1.3t})$,
Method 2: $c(t) = 4.15(e^{-0.4t} - e^{-3t})$; $[0, 4]$

84. Method 1: $c(t) = 5.78(e^{-0.4t} - e^{-1.3t})$,
Method 2: $c(t) = 4.15(e^{-0.4t} - e^{-3t})$; $[0, 24]$

85. Suppose that at time $t = 0$ there are 750 bacteria in a growth medium and the bacteria population $y(t)$ grows at the rate $y'(t) = 802.137e^{1.528t}$ bacteria per hour. How many bacteria will there be in 12 hours?

86. Suppose that a particle moving along a coordinate line has velocity $v(t) = 25 + 10e^{-0.05t}$ ft/s.

(a) What is the distance traveled by the particle from time $t = 0$ to time $t = 10$?

(b) Does the term $10e^{-0.05t}$ have much effect on the distance traveled by the particle over that time interval? Explain your reasoning.

87. Find a positive value of k such that the area under the graph of $y = e^{2x}$ over the interval $[0, k]$ is 3 square units.

88. Solve the initial-value problem.

$$\frac{dy}{dt} = -e^{2t}, \quad y(0) = 6$$

89. Let $y(t)$ denote the number of *E. coli* cells in a container of nutrient solution t minutes after the start of an experiment. Assume that $y(t)$ is modeled by the initial-value problem

$$\frac{dy}{dt} = (\ln 2)2^{t/20}, \quad y(0) = 20$$

Use this model to estimate the number of *E. coli* cells in the container 2 hours after the start of the experiment.

90. **Writing** Let G denote the graph of an invertible function f and consider G as a fixed set of points in the plane. Suppose we relabel the coordinate axes so that the x -axis becomes the y -axis and vice versa. Carefully explain why now the same set of points G becomes the graph of f^{-1} (with the coordinate axes in a nonstandard position). Use this result to explain Formula (2).

91. **Writing** Suppose that f has an inverse function. Carefully explain the connection between Formula (2) and implicit differentiation of the equation $x = f(y)$.